STUDY GUIDE

Equity & Capital Markets

A complete, concept-first reference — prepared for Saikumar Kaleru.

certifications / study / Finance • 2026

Overview of Equity & Capital Markets

The Problem / Why this matters

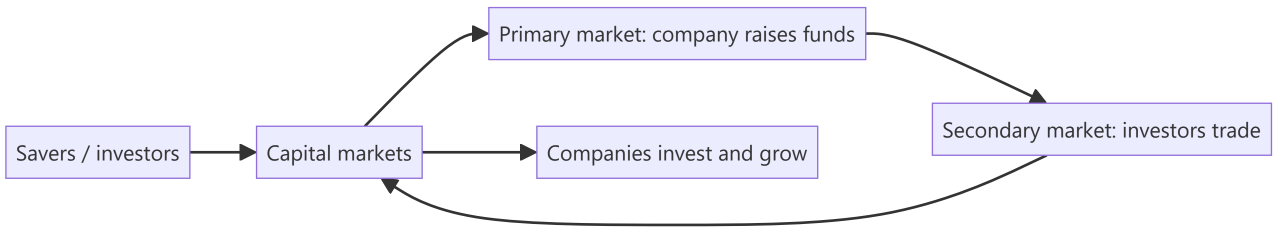
Companies need long-term capital to grow; savers need somewhere to put their money to earn a return. **Capital markets are the machinery that connects the two** — channelling household and institutional savings into productive businesses, and giving investors tradable claims on those businesses. Understanding how this system is structured — primary vs secondary, equity vs debt, the players and the plumbing — is the foundation for every equity, research, and markets role.

Core Idea

Capital markets are where **long-term funds** are raised and traded. The **primary market** is where new securities are issued and the company actually raises money; the **secondary market** is where those securities then trade among investors, providing the liquidity that makes the primary market work. Equity gives ownership; debt gives a creditor claim.

Why it works this way

Investors will only fund companies in the primary market if they believe they can **exit** later by selling to someone else — that exit is the secondary market. Liquidity in the secondary market lowers the return investors demand, which lowers the company's cost of capital. So the two markets are symbiotic: primary raises capital, secondary provides the liquidity that makes raising capital cheap.



Full technical content

Money markets vs capital markets:

	MONEY MARKET	CAPITAL MARKET
Maturity	Short-term (< 1 yr)	Long-term (> 1 yr)
Instruments	T-bills, CP, CDs, repo	Equities, bonds
Purpose	Liquidity management	Long-term funding/investment

Equity vs debt:

	EQUITY	DEBT
Claim	Ownership, residual	Creditor, fixed
Return	Dividends + capital gains, uncapped	Interest, capped
Risk	Last in line, higher risk	Senior, lower risk
Control	Voting rights	Usually none

Primary vs secondary market: - **Primary** — first issuance: IPO, follow-on (FPO), rights issue, QIP, private placement, bond issuance. *The company receives the cash.* - **Secondary** — subsequent trading on an exchange between investors. *The company gets nothing; it provides price discovery and liquidity.*

Functions of capital markets: 1. **Capital formation** — channelling savings into investment. 2. **Price discovery** — continuous pricing of companies via trading. 3. **Liquidity** — the ability to convert holdings to cash. 4. **Risk transfer & sharing** — spreading risk across many investors. 5. **Corporate governance** — market discipline and monitoring by investors.

The ecosystem (previewing later chapters): issuers (companies), investors (retail, mutual funds, insurers, FPIs, pension funds), intermediaries (investment banks, brokers, exchanges, depositories, clearing houses), and the regulator (SEBI in India).

Worked examples

Example 1 — primary vs secondary cash flow. Company Z raises ₹1,000 cr in an IPO (primary) — that ₹1,000 cr goes to the company (or selling shareholders in an OFS). The next day, an investor buys ₹5 cr of Z shares on the NSE (secondary) — that ₹5 cr goes to the *seller*, not to Z. *Z only receives money at issuance.*

Example 2 — why liquidity lowers cost of capital. Two identical companies raise equity. Company A's shares will list on a liquid exchange (easy exit); Company B's won't trade at all. Investors demand a higher return from B for locking up their money, so B's cost of equity is higher and its shares are worth less. Liquidity is valuable.

Example 3 — equity vs debt in a good year and a bad year. A firm earns ₹100 cr. Bondholders get their fixed ₹20 cr interest either way. In a great year (₹200 cr), equity keeps the extra upside; in a terrible year (₹10 cr), equity is wiped out first while bondholders still rank ahead. Equity = uncapped upside, first-loss downside.

How it is tested in interviews

- **"Difference between primary and secondary markets?"** — "Primary is where the company issues new securities and actually raises money; secondary is where investors trade those securities among themselves, giving liquidity and price discovery. The company only receives cash in the primary market."
- **"When Infosys shares trade on the NSE, does Infosys get the money?"** — "No — that's the secondary market. Infosys only raised money at its IPO/issuances."
- **"Equity vs debt?"** — "Equity is ownership with uncapped, residual, higher-risk returns; debt is a senior, fixed, capped, lower-risk claim."
- **"What are the functions of capital markets?"** — Capital formation, price discovery, liquidity, risk sharing, governance.

Traps & common mistakes

- Thinking the company receives money from **secondary** trading.
- Confusing **money markets** (short-term) with **capital markets** (long-term).
- Treating equity and debt returns as symmetric — equity's downside is first-loss.
- Underrating the role of **liquidity** in lowering cost of capital.

First-principles recap

- Capital markets connect savers to companies needing long-term funds.
- **Primary** raises capital (company gets cash); **secondary** provides liquidity (investors trade).
- Equity = ownership/residual/uncapped; debt = senior/fixed/capped.
- Functions: capital formation, price discovery, liquidity, risk sharing, governance.
- Secondary-market liquidity lowers the primary-market cost of capital.

Quick-reference

CONCEPT	ONE-LINER
Primary market	New issue; company raises cash
Secondary market	Investors trade; liquidity & price discovery
Money vs capital market	Short-term vs long-term
Equity	Ownership, residual, uncapped
Debt	Creditor, fixed, senior

Q&A — Overview of Equity & Capital Markets

Foundational theory questions and worked scenarios on the primary/secondary distinction and capital-market functions.

Q1. When Infosys shares trade on the NSE today, does Infosys receive any of that money? Explain precisely.

Model answer. No. That is secondary-market trading — an exchange of shares between two investors, with the proceeds going to the selling investor, not the company. Infosys only received cash from the market at the point of an actual issuance (its original IPO, or any subsequent follow-on/QIP/rights issue). This is one of the most common conceptual gaps to close early: the vast majority of daily trading volume on any exchange involves no cash flow to the underlying company at all.

Q2. Why does secondary-market liquidity lower a company's cost of capital, even though the company never touches secondary-market trading proceeds?

Model answer. Investors price in the ability to exit an investment when they need to. If a security is illiquid (hard to sell without moving the price significantly), investors demand a higher expected return to compensate for that liquidity risk — raising the company's cost of equity. A liquid secondary market lowers that required return, which directly lowers the price the company pays (in the form of a higher achievable valuation) to raise capital in the primary market. The two markets are economically linked even though only the primary market involves direct cash flow to the issuer.

Q3. Distinguish money markets from capital markets along two dimensions.

Model answer. Maturity: money-market instruments (T-bills, commercial paper, certificates of deposit, repo) are short-term, generally under one year; capital-market instruments (equities, bonds) are long-term. Purpose: money markets exist primarily for liquidity management (parking or raising cash briefly), while capital markets exist for long-term funding and investment (financing growth, capital projects). A company issuing commercial paper to cover working-capital timing gaps is a money-market transaction; the same company issuing a 10-year bond or new equity is a capital-market transaction.

Q4. Worked — equity vs debt payoff in three scenarios.

A firm has ₹500 cr of debt requiring ₹40 cr annual interest. Compare what debt and equity holders receive in a bad year (₹30 cr operating profit), an average year (₹80 cr), and a great year (₹250 cr).

Model answer. Bad year (₹30 cr): debtholders are owed ₹40 cr but only ₹30 cr is available — this is a default/distress scenario; debtholders receive what's available (₹30 cr, likely triggering restructuring), and equity holders receive **nothing**. Average year (₹80 cr): debtholders receive their fixed ₹40 cr; equity holders receive the residual ₹40 cr. Great year (₹250 cr): debtholders still receive only their fixed ₹40 cr (no upside participation); equity holders capture the full residual ₹210 cr. This illustrates the core asymmetry: debt has a capped, senior claim; equity has an uncapped, residual, first-loss claim — debt's payoff resembles a short put option, equity's resembles a long call.

Q5. List and briefly explain the five core functions of capital markets.

Model answer. (1) Capital formation — channelling household/institutional savings into productive investment. (2) Price discovery — continuous trading aggregates information into a market-clearing price. (3) Liquidity — letting investors convert holdings to cash without large value loss. (4) Risk transfer and sharing — spreading a company's risk across many investors rather than concentrating it with a few. (5) Corporate governance — market discipline (share-price reaction, activist investors, the threat of takeover) monitors and pressures management to act in shareholders' interest.

Q6. A private (unlisted) company and a listed company have identical financials and growth prospects. Would you expect them to trade/be valued at the same multiple? Why or why not?

Model answer. No — the private company would typically warrant a lower valuation, reflecting an illiquidity discount: its equity cannot be readily converted to cash the way a listed company's shares can (no continuous secondary market, no daily price discovery, exit typically requires a full sale process or IPO). Investors demand additional compensation for bearing this illiquidity risk, all else being equal — a direct application of the liquidity-lowers-cost-of-capital logic from Q2, run in reverse.

Q7. What's the difference between "the company raises capital" and "the market provides liquidity," and why do interviewers test this distinction so directly?

Model answer. Raising capital is specifically a primary-market function (new securities issued, cash flows to the issuer). Providing liquidity is specifically a secondary-market function (existing securities change hands between investors, cash flows between investors, not to the issuer). Interviewers test this because it's foundational — a candidate who conflates the two will make downstream errors (e.g. assuming a company benefits financially every time its stock trades, or misunderstanding why a company doesn't "lose money" when its stock price falls on the secondary market post-issuance).

Q8. Why is "risk transfer and sharing" listed as a distinct function from liquidity, when they sound related?

Model answer. Liquidity is about the *ease* of converting a holding to cash. Risk transfer/sharing is about *distributing* a company's business risk across many investors with different risk appetites and diversification needs, rather than concentrating it entirely with a single owner or a small group of founders — a founder who sells 70% of their company via an IPO isn't just gaining liquidity for their own stake, they're also transferring the associated risk of the business's fortunes to a broad, diversified investor base, each of whom bears only a small fraction of the total risk. The two functions are related but conceptually distinct: one is about convertibility, the other about risk distribution across the market.

Primary Markets & the IPO Process

The Problem / Why this matters

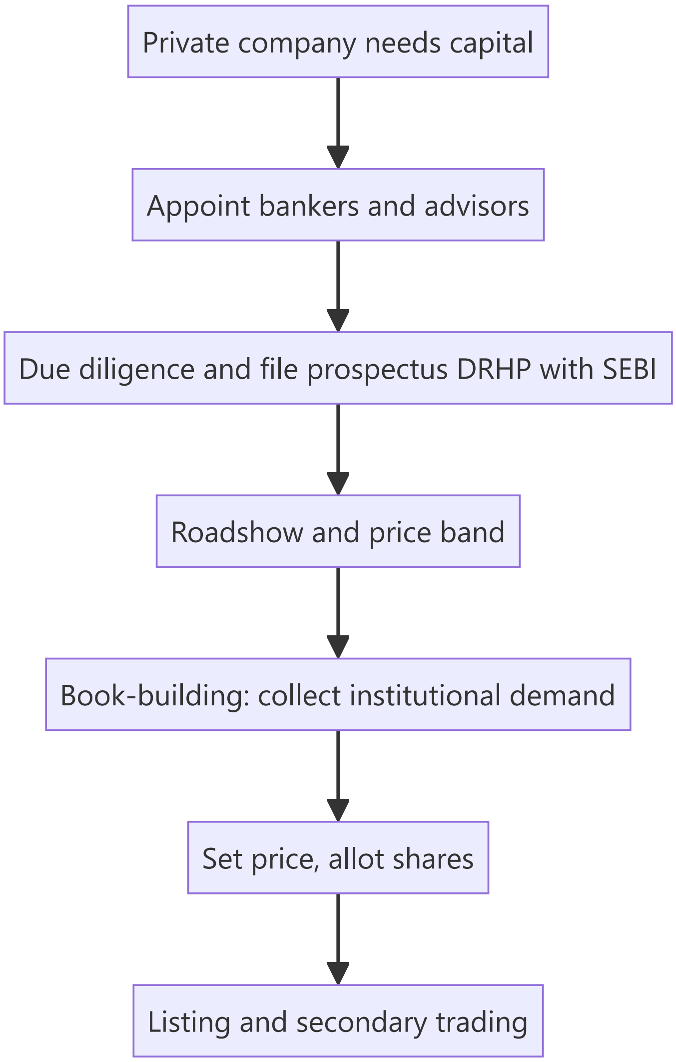
The most visible event in capital markets is a company **going public** — the IPO. It's how a private company raises large-scale equity, gives early investors an exit, and joins the public market. Understanding the IPO process — why companies list, how the price is set (book-building), who does what (the bankers, the regulator), and the risks (underpricing, lock-ups) — is essential for equity capital markets, investment banking, and research roles, and it comes up constantly in interviews.

Core Idea

The **primary market** is where companies issue securities to raise capital for the first time or afresh. An **IPO (Initial Public Offering)** is a private company's first sale of shares to the public, priced via **book-building**, intermediated by investment banks, and regulated (by SEBI in India). Beyond IPOs, primary issuance includes follow-ons, rights issues, QIPs and private placements.

Why it works this way

A company needs a mechanism to (i) discover a fair price when there's no market price yet, and (ii) distribute shares to many investors while managing risk. Book-building solves price discovery by collecting demand from institutions across a price band; underwriters manage distribution and (sometimes) guarantee the raise; the regulator protects investors through disclosure (the prospectus).



Full technical content

Why companies go public: raise growth capital, give founders/PE/VC an exit (offer for sale), acquire currency (listed shares) for M&A, raise profile/credibility, and enable employee stock liquidity. Costs: disclosure, scrutiny, short-term market pressure, listing expense, dilution.

Types of primary issuance:

TYPE	WHAT IT IS
IPO	First public sale of shares
FPO / follow-on	Further public issue by an already-listed firm
Rights issue	Offer of new shares to existing holders, usually at a discount
QIP	Qualified Institutional Placement — quick raise from institutions (India)
Private placement / preferential	Shares sold to select investors
OFS	Offer for Sale — existing holders sell (company gets no cash)

Fresh issue vs offer for sale. A **fresh issue** creates new shares — the *company* receives the money (dilutes existing holders). An **OFS** sells *existing* shares — the selling shareholders receive the money (no dilution, no new company capital). Most IPOs are a mix.

The IPO process (India): 1. Appoint **merchant bankers (BRLMs)**, legal counsel, auditors, registrar. 2. Due diligence; draft the **DRHP** (Draft Red Herring Prospectus) and file with **SEBI**. 3. SEBI review and observations; finalize the RHP. 4. **Roadshow** — market to institutions; set a **price band**. 5. **Book-building** — collect bids across the band from QIBs, non-institutional, and retail categories (India reserves quotas for each). 6. Determine the **cut-off price** from the demand book; allot shares. 7. **Listing** on NSE/BSE; secondary trading begins.

Pricing methods: book-building (price discovered from demand within a band — the norm) vs **fixed price** (set upfront). **Anchor investors** (large institutions) may be allotted a day before to signal quality.

Key phenomena: - **Underpricing** — IPOs often "pop" on day one (priced below where they trade), leaving money on the table but rewarding investors and ensuring a successful listing. - **Lock-up period** — insiders/pre-IPO holders can't sell for a set period after listing, preventing an immediate flood of stock. - **Greenshoe (over-allotment) option** — lets underwriters stabilize the price post-listing by selling up to ~15% extra and buying back if it falls. - **Winner's curse / oversubscription** — hot IPOs are heavily oversubscribed; retail allotment is scaled/lottery-based.

Worked examples

Example 1 — fresh issue vs OFS. An IPO raises ₹2,000 cr: ₹1,200 cr fresh issue (new shares → goes to the *company* for expansion) and ₹800 cr OFS (existing PE investor sells → goes to the *PE fund*, not the company). Only the ₹1,200 cr strengthens the company's balance sheet.

Example 2 — book-building price discovery. Price band ₹95–100. Institutional demand is strong at ₹100 (book covered 8× at the top), so the cut-off is set at **₹100**. Weak demand would have set it lower in the band. The market's bids, not the company's wish, set the price.

Example 3 — underpricing pop. Shares priced at ₹100 open at ₹135 on listing (+35%). Investors who got allotment gain; the company "left money on the table" (could have priced higher) but secured a strong, well-received listing and goodwill for future raises.

Example 4 — a full worked IPO subscription and allotment case study. A company issues 2 crore shares (fresh issue only, no OFS) at a price band of ₹450-₹475, reserved as: 50% QIB (Qualified Institutional Buyers), 15% NII (Non-Institutional Investors, i.e. HNIs), and 35% retail. At close: QIB portion subscribed 18x, NII portion subscribed 42x, retail portion subscribed 3.2x. **Cut-off price**: given the QIB book (the category whose demand carries the most weight in price discovery, since institutions do the deepest diligence) is subscribed nearly 20x even at the top of the band, the cut-off is set at ₹475 — the top of the band. **Retail allotment mechanics**: with the retail portion 3.2x oversubscribed, not every retail applicant gets shares — allotment is via a lottery/proportionate mechanism (for retail, typically each applicant either gets one full lot or nothing, decided by a computerised draw, rather than a proportionate partial allotment, since retail lot sizes are already small) — so a retail investor applying for the minimum lot has roughly a 1-in-3.2 (≈31%) chance of allotment in this scenario, a probability an analyst should be able to explain when asked "how does retail allotment actually work in an oversubscribed IPO." **NII allotment**, by contrast, is typically proportionate (not lottery-based) — an NII applicant subscribed 42x receives roughly 1/42nd of what they applied for, scaled across all NII applicants, reflecting the different allotment methodology SEBI mandates for the NII category versus retail.

Example 5 — a down-round / weak-demand IPO, contrasted with Example 4. A different company, same structure, sees QIB subscription of only 0.8x at the top of the band (i.e. the QIB book is *undersubscribed*) — under SEBI rules, if the QIB portion isn't at least fully subscribed, the issue may not proceed at all (a minimum-subscription threshold), forcing the company to either withdraw the IPO, extend the bidding period, or in some structures reduce the price band and re-open bidding. This scenario — contrasted directly against Example 4's strong 18x QIB demand — is what an analyst means by "the IPO market read this company's valuation as too aggressive": weak institutional demand at the top of the band is the market's most direct, unambiguous pricing signal, well before the stock ever lists and trades.

How it is tested in interviews

- **"Walk me through an IPO."** — Appoint bankers → due diligence and file DRHP with SEBI → roadshow and price band → book-building → set cut-off and allot → list and trade.
- **"Fresh issue vs OFS?"** — "Fresh issue creates new shares and the company gets the cash; OFS sells existing shares and the selling holders get the cash — no new capital for the company."
- **"What is book-building?"** — "Price discovery by collecting institutional demand across a price band and setting the cut-off from the demand book."
- **"Why are IPOs often underpriced?"** — "To ensure a successful, well-subscribed listing and reward investors for the uncertainty; it 'pops' but the issuer leaves some money on the table."
- **"What is a greenshoe / lock-up?"** — Greenshoe stabilizes the price via over-allotment; lock-up stops insiders dumping stock right after listing.

Traps & common mistakes

- Confusing **fresh issue** (company gets cash, dilution) with **OFS** (holders get cash, no dilution).
- Thinking the **company** sets the price — book-building lets **demand** set it.
- Forgetting **SEBI/DRHP** disclosure as the investor-protection backbone.
- Missing the purpose of **lock-ups and greenshoe** (orderly aftermarket).

First-principles recap

- Primary market = companies issue securities to raise capital.
- IPO = first public sale, priced by **book-building**, filed via **DRHP** with SEBI.
- **Fresh issue** funds the company; **OFS** cashes out existing holders.
- Underpricing, lock-ups, greenshoe, and anchor investors shape a successful listing.
- Beyond IPOs: FPO, rights, QIP, private placement.

Quick-reference

TERM	NOTE
IPO	First public share sale
DRHP	Draft prospectus filed with SEBI
Book-building	Demand-based price discovery in a band
Fresh issue vs OFS	Company cash vs selling-holder cash
Greenshoe	Over-allotment price stabilization
Lock-up	Insiders can't sell for a set period

Q&A — Primary Markets & the IPO Process

Theory and worked scenarios on issuance, book-building, and listing mechanics.

Q1. Walk me through the IPO process end to end.

Model answer. Appoint merchant bankers (BRLMs), legal counsel, auditors, and a registrar → due diligence and file the Draft Red Herring Prospectus (DRHP) with SEBI → SEBI review and observations, finalise the RHP → roadshow to market to institutions and set a price band → book-building to collect bids across the band from QIBs, non-institutional, and retail categories → determine the cut-off price from the demand book and allot shares → list on NSE/BSE and begin secondary trading.

Q2. Distinguish a fresh issue from an offer for sale (OFS), and why the distinction matters to a research analyst valuing the deal.

Model answer. A fresh issue creates new shares — the company receives the proceeds, which dilutes existing holders but strengthens the balance sheet (usable for growth capex, debt paydown, etc.). An OFS sells existing shares — the selling shareholder (often a PE/VC investor or promoter) receives the proceeds, with no new capital reaching the company and no incremental dilution beyond the ownership transfer itself. An analyst assessing "what does this IPO do for the company" must check the fresh-issue/OFS split in the prospectus — an IPO that's 90% OFS raises little growth capital for the business despite a large headline deal size.

Q3. Worked — book-building price discovery.

Price band ₹340-360. At ₹360 the book is covered 12x by QIBs; at ₹340 it would be covered 25x. What does this tell you, and roughly where would the cut-off likely land?

Model answer. Strong oversubscription even at the top of the band (12x at ₹360) signals demand comfortably clears the highest price, so the cut-off is very likely to be set at or near the top of the band (₹360) — book-building sets price from where the demand actually clears, not an average; a company with this level of demand has no reason to leave the band's ceiling unclaimed.

Q4. What is the greenshoe (over-allotment) option, and how does it stabilise the post-listing price?

Model answer. Underwriters are permitted to allot up to roughly 15% more shares than the base offer, funded by a share loan from a promoter/existing holder, effectively creating a short position. If the stock falls below the issue price after listing, underwriters buy shares in the open market to cover that short, which supports the price; if the stock trades well above the issue price, the option isn't exercised for buybacks. This is the primary open-market price-stabilisation mechanism during the immediate post-listing window.

Q5. Why do IPOs commonly "pop" on listing day, and is that necessarily a mistake by the company/bankers?

Model answer. Underpricing (setting the offer price modestly below where secondary-market demand will actually clear) is common and partly deliberate — it rewards investors for bearing IPO uncertainty and helps ensure the issue is fully subscribed and lists strongly, building credibility for the issuer's future capital-raising. It is a genuine trade-off, not simply an error: pricing "perfectly" to zero pop risks an undersubscribed, poorly-received listing, which can be more costly to the issuer's reputation and future access to capital than leaving some money on the table once.

Q6. What's the difference between a rights issue and a QIP, and when would a company use each?

Model answer. A rights issue offers new shares to *existing* shareholders (typically at a discount to market price), preserving their proportional ownership if they subscribe — used when a company wants to raise capital while giving existing holders first refusal and avoiding dilution of those who participate. A QIP (Qualified Institutional Placement) is a fast, India-specific route to raise capital directly from institutional investors without the extended regulatory process of a public offer — used when a listed company needs to raise capital quickly and is comfortable diluting existing (especially retail) shareholders who don't get first access, in exchange for speed and lower issuance cost.

Q7. A pre-IPO investor's shares are subject to a lock-up. What happens, mechanically and to the stock price, when the lock-up expires?

Model answer. During the lock-up, those shares cannot be sold on the open market, artificially constraining the freely-tradable float. At expiry, a large block of previously-locked shares becomes sellable at once — if the holder (often a PE/VC fund or promoter group) sells a meaningful portion, the sudden increase in sell-side supply can pressure the price downward even with no change in the company's fundamentals; sophisticated investors and analysts track lock-up expiry dates as a known, mechanical (not fundamentals-driven) source of near-term stock-price risk.

Q8. Worked — anchor investor allotment and its signalling effect.

A company allots 30% of its IPO book to anchor investors (large institutions) the day before the issue opens, at the top of the price band, with strong-name global and domestic funds participating.

Model answer. Anchor allotment serves as a costless signal to the broader market ahead of the public book-building: strong, well-known institutional names committing at the top of the band ahead of retail/other-institutional bidding suggests sophisticated investors, who presumably did deeper diligence, are confident in the pricing — this often improves subscription momentum in the subsequent book-building for the rest of the issue, though an analyst should still independently verify the fundamentals rather than relying on anchor participation alone as a substitute for their own diligence.

Secondary Markets & Trading Mechanics

The Problem / Why this matters

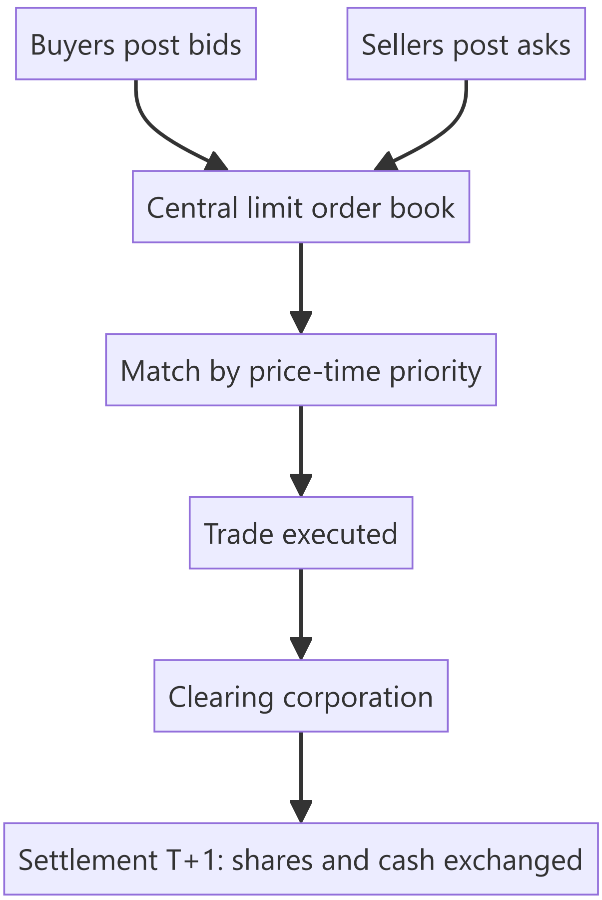
Once shares list, they trade — millions of times a day — on exchanges. How that trading actually works (order types, the order book, bid-ask spreads, matching, settlement) determines the price you get, the cost you pay, and the liquidity you rely on. Anyone touching markets — research, trading, operations, risk — must understand the plumbing. Interviewers test order types, the bid-ask spread, and settlement (T+1) directly.

Core Idea

The secondary market is a **continuous auction**: buyers post bids, sellers post asks, and the exchange's matching engine pairs them by price-time priority. The **bid-ask spread** is the cost of immediacy; **liquidity** is how easily you trade without moving the price; and every trade must **clear and settle** through depositories and clearing corporations.

Why it works this way

Price discovery needs a mechanism that continuously aggregates supply and demand. An **order-driven** market does this via a central limit order book where anyone can post orders and the best-priced ones execute first. Spreads exist because liquidity providers must be compensated for the risk of quoting; settlement infrastructure exists so buyers and sellers who never meet can trust that shares and cash actually change hands.



Full technical content

Market structures: order-driven (a central limit order book, as on NSE/BSE) vs **quote-driven** (dealers/market-makers quote two-way prices). Most modern equity markets are order-driven with electronic matching by **price-time priority** (best price first; at equal price, earliest order first).

The order book & bid-ask spread. The book shows all resting buy orders (bids) and sell orders (asks) at each price. The **best bid** and **best ask** define the spread; a **narrow spread = liquid**, a wide spread = illiquid and expensive to trade. **Depth** (quantity available near the top) shows how much you can trade without moving the price (**market impact**).

Order types:

ORDER	BEHAVIOUR
Market	Execute now at the best available price (certainty of execution, not price)
Limit	Execute only at your price or better (certainty of price, not execution)
Stop-loss	Becomes a market/limit order once a trigger price is hit
Stop-limit	Stop that becomes a limit order
Others	IOC, GTC, iceberg, day orders

Liquidity providers. Market makers and high-frequency traders continuously quote both sides, earning the spread and supplying immediacy; they narrow spreads and add depth (see market microstructure).

Long vs short. Going **long** = buy, profit if price rises (max loss = amount invested). **Short selling** = sell borrowed shares to buy back cheaper; profit if price falls, but loss is theoretically unlimited (price can rise without bound). Shorting requires a stock-borrow and is regulated.

Clearing & settlement. After a trade: the **clearing corporation** (NSE Clearing / Indian Clearing) becomes central counterparty and nets obligations; **depositories** (NSDL, CDSL) hold shares in **demat** form and transfer them; cash and securities settle on **T+1** in India (among the fastest globally; India is moving toward T+0/instant). **Rolling settlement**, margining, and a settlement guarantee fund underpin trust.

Circuit breakers & price bands limit extreme moves (index-level halts and stock-level bands) to curb panic.

Worked examples

Example 1 — market vs limit order. Best bid ₹99.8, best ask ₹100.2 (spread 0.4). A **market buy** executes immediately at ₹100.2 (pays the spread for certainty). A **limit buy at ₹100.0** sits in the book and only fills if a seller drops to ₹100.0 — better price, but it might never execute. Choose by whether you value price or certainty.

Example 2 — spread and liquidity. Stock A (large-cap) shows spread ₹0.05 with 50,000 shares at the top — deep and liquid; you trade large size cheaply. Stock B (small-cap) shows spread ₹2.00 with 200 shares — illiquid; a modest order moves the price sharply (high market impact). Same rupee value order, very different execution cost.

Example 3 — settlement. You buy 100 shares on Monday (T). Under T+1, the shares are credited to your demat and cash debited on Tuesday (T+1). The clearing corporation guarantees the trade so you don't bear your counterparty's default risk.

How it is tested in interviews

- **"Market order vs limit order?"** — "Market executes now at the best price (certain execution, uncertain price); limit executes only at your price or better (certain price, uncertain execution)."
- **"What is the bid-ask spread and what does a wide spread mean?"** — "The gap between best bid and best ask — the cost of immediacy. A wide spread means illiquidity and expensive trading."
- **"What's the max loss on a short?"** — "Theoretically unlimited, because the price can rise without bound; a long's max loss is what you invested."
- **"How does settlement work in India?"** — "T+1 rolling settlement: shares (via NSDL/CDSL demat) and cash exchange one day after the trade, guaranteed by the clearing corporation as central counterparty."
- **"What does market depth tell you?"** — "How much you can trade near the top of the book without moving the price — your likely market impact."

Traps & common mistakes

- Confusing **market** (execution certainty) and **limit** (price certainty) orders.
- Forgetting a short's loss is **unlimited**.
- Ignoring **market impact/depth** — a big order in a thin book moves the price against you.
- Not knowing India settles at **T+1** (and is moving faster).
- Overlooking the **clearing corporation** as central counterparty removing settlement risk.

First-principles recap

- Secondary trading is a continuous auction matched by **price-time priority**.
- The **bid-ask spread** is the cost of immediacy; narrow = liquid.
- **Market** = certain execution; **limit** = certain price; **stop-loss** triggers on a level.
- Short selling has **unlimited** loss potential.
- Trades **clear and settle** via clearing corporations and depositories, T+1 in India.

Quick-reference

ITEM	NOTE
Order book	Resting bids/asks by price-time priority
Spread	Best ask − best bid (cost of immediacy)
Market order	Now, best price
Limit order	Your price or better
Short loss	Unlimited
Settlement	T+1 (NSDL/CDSL demat, clearing corp CCP)

Q&A — Secondary Markets & Trading Mechanics

Theory and worked scenarios on order types, spreads, and settlement.

Q1. Market order vs limit order — what does each guarantee, and what does each not guarantee?

Model answer. A market order guarantees execution (it fills immediately against the best available price on the other side of the book) but not the price you'll pay — in a fast-moving or thin market you can be filled well away from the last traded price. A limit order guarantees the price (it will only execute at your specified price or better) but not execution — if the market never trades at your price, the order simply sits unfilled. The choice is a direct trade-off between price certainty and execution certainty.

Q2. Worked — computing market impact from the order book.

Best bid ₹248.50 (2,000 shares), next bid ₹248.20 (5,000 shares); best ask ₹249.00 (1,500 shares), next ask ₹249.40 (4,000 shares). You need to sell 5,000 shares immediately as a market order.

Model answer. Selling into the bid side: the first 2,000 shares execute at ₹248.50, the remaining 3,000 shares execute at ₹248.20 (the next price level down), since the top level's depth is exhausted. Volume-weighted average execution price = (2,000×248.50 + 3,000×248.20) / 5,000 = (497,000 + 744,600) / 5,000 = ₹248.32 — worse than the ₹248.50 best bid, illustrating market impact: a large order "walks the book" and gets a worse average price than the top-of-book quote suggests.

Q3. Why is a short seller's maximum loss theoretically unlimited, while a long position's maximum loss is capped?

Model answer. A long position's downside is capped at the amount invested — the price can only fall to zero. A short position involves selling borrowed shares with the obligation to buy them back later to return them; since a stock's price has no theoretical ceiling, the cost to buy back and close a short position can rise without bound, making the potential loss theoretically unlimited. This asymmetry is why short positions require margin, active risk management, and often stop-loss discipline that a comparable long position may not need as urgently.

Q4. What roles do the clearing corporation and the depository each play in a single trade, and why are they separate entities?

Model answer. The clearing corporation (e.g. NSE Clearing) becomes the central counterparty to every matched trade — it guarantees settlement to both sides, so neither party bears the other's default risk. The depository (NSDL/CDSL) holds securities in dematerialised electronic form and executes the actual transfer of shares between demat accounts on settlement. They're separate because they solve different frictions: the clearing corp manages counterparty/credit risk across the whole market, while the depository manages custody and the mechanical transfer of ownership records — conflating them (a common interview trap) misses that one guarantees the trade and the other physically moves the asset.

Q5. What does "market depth" measure, and why does an analyst or trader care about it beyond the headline bid-ask spread?

Model answer. Depth measures the quantity available at each price level near the top of the order book, not just the best bid/ask. A narrow spread can be misleading if depth is thin — a seemingly liquid-looking stock with a tight spread but only 200 shares at the best bid will show significant market impact (Q2) on any order larger than that, even though the quoted spread alone looks attractive. Depth, not spread alone, determines how much size can actually be traded near the current price.

Q6. Explain price-time priority in order matching with a worked example.

Two limit buy orders both at ₹100: Order A for 500 shares placed at 9:15:02am, Order B for 800 shares placed at 9:15:05am. A sell market order for 600 shares arrives.

Model answer. Price-time priority matches the best price first, and among orders at the same price, the earliest-placed order first. Order A (earlier, same price) fills completely (500 shares), and the remaining 100 shares of the incoming sell order fill against Order B, leaving Order B with 700 shares still resting in the book. Time priority at a given price level rewards being first in the queue, not just posting a competitive price.

Q7. What's the difference between an order-driven and a quote-driven market structure?

Model answer. An order-driven market (NSE/BSE's central limit order book) lets any participant post bids/asks directly, with the exchange's matching engine pairing them by price-time priority — price discovery emerges from aggregated public orders. A quote-driven market relies on designated dealers/market-makers who continuously post two-way (buy and sell) quotes that others trade against — price discovery is mediated through the dealer's quoted spread rather than a fully open book. Most modern equity markets, including Indian exchanges, are primarily order-driven, though market-makers/liquidity providers still operate within that order-driven structure to supply depth.

Q8. Why would a trader use a stop-limit order instead of a plain stop-loss order?

Model answer. A plain stop-loss becomes a *market* order once the trigger price is hit, guaranteeing execution but not the price — in a fast-falling/gapping market, the actual fill can be significantly worse than the trigger level. A stop-limit becomes a *limit* order once triggered, capping the worst acceptable execution price — but risks not executing at all if the price gaps straight through the limit without trading at an acceptable level. The choice reflects the same execution-certainty-vs-price-certainty trade-off as Q1, applied specifically to risk-management/exit orders.

Market Participants & Intermediaries

The Problem / Why this matters

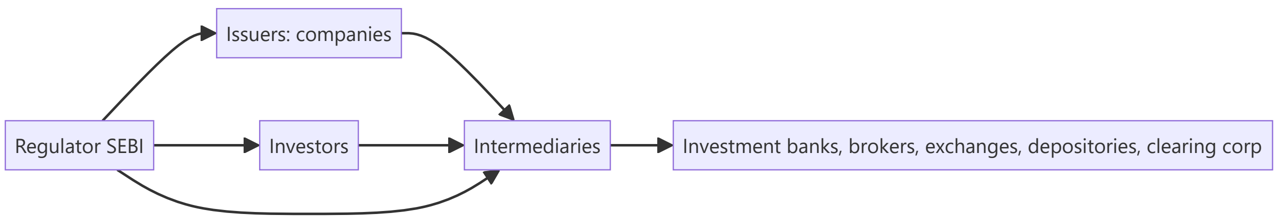
Markets don't run themselves — a web of participants and intermediaries makes issuance, trading, clearing and custody happen safely. Knowing **who does what** — issuers, the different investor types and their motivations, and the intermediaries (banks, brokers, exchanges, depositories, clearing corporations) — tells you how order flow moves, why prices react to certain investors, and where you'd fit in the ecosystem. Interviewers expect you to map the players cleanly.

Core Idea

The capital-market ecosystem has three groups: **issuers** who need capital, **investors** who supply it (each with different horizons and motivations), and **intermediaries** who connect and service them. Overlaying all of it is the **regulator** (SEBI) ensuring fairness, disclosure and investor protection.

Why it works this way

Issuers and investors can't transact directly at scale — they need someone to underwrite issues, match trades, hold securities, guarantee settlement and enforce rules. Each intermediary exists to solve a specific friction (distribution, execution, custody, counterparty risk, oversight), and the mix of investors determines how "smart," patient or reactive the money in a stock is.



Full technical content

Issuers: companies (private and public), governments (bonds), and other entities raising capital.

Investors (by type and motivation):

INVESTOR	HORIZON / MOTIVATION
Retail	Individuals; varied horizons; price-takers
Mutual funds / AMCs	Pooled retail money; benchmark-driven
Insurance & pension funds	Long-term, liability-matching, stable
FPIs / FIIs	Foreign institutional flows; can be large and fast
Domestic institutions (DIIs)	Banks, insurers, MFs domestically
Hedge funds / PMS / AIFs	Absolute-return, flexible, can short/leverage
Promoters / strategic	Control-oriented, long-term holders

Institutional flows (FPI/DII) often drive index moves; "smart money" positioning is watched closely.

Intermediaries:

INTERMEDIARY	ROLE
Investment bank / merchant banker (BRLM)	Advise on and underwrite issues; M&A; capital raising
Brokers / trading members	Execute investor orders on the exchange
Stock exchanges (NSE, BSE)	Provide the trading platform and price discovery
Depositories (NSDL, CDSL)	Hold securities in demat; effect transfer
Clearing corporations	Central counterparty; net and guarantee settlement
Custodians	Safekeep assets for institutions; settlement support
Registrars & Transfer Agents (RTAs)	Maintain shareholder records, process corporate actions
Rating agencies / research	Independent credit/equity opinions
Investment advisers / distributors	Advise and distribute to end investors

The regulator — SEBI. Sets rules for issuers (disclosure), intermediaries (registration, conduct), and markets (surveillance, insider-trading and manipulation enforcement); protects investors and develops the market. RBI oversees banking/monetary aspects; IRDAI insurance.

Sell-side vs buy-side (preview). *Sell-side* (banks, brokers) creates products, executes and publishes research to win client business. *Buy-side* (asset managers, funds) invests client money and consumes that research to make decisions.

Worked examples

Example 1 — the chain of a single trade. A mutual fund (investor) instructs its **broker** (intermediary) to buy 1 lakh shares on the **NSE** (exchange); the trade matches, the **clearing corporation** guarantees it, the **depository (CDSL/NSDL)** moves the shares into the fund's **custodian's** demat account on T+1. Five different intermediaries touched one trade, each solving a distinct friction.

Example 2 — why the investor mix matters. Two similar mid-caps: Stock A is 70% held by long-term insurers and index funds (stable, patient); Stock B is 40% held by fast FPIs and traders (volatile, flow-driven). A negative headline moves B far more, because its holder base reacts and rotates quickly. *Knowing the holder base predicts volatility.*

Example 3 — SEBI's role. A promoter tries to sell shares on undisclosed bad news (insider trading). SEBI's surveillance flags the trades, investigates, and penalizes — protecting other investors and preserving trust that keeps the market liquid.

Example 4 — a hedge fund/PMS using leverage and shorting that a mutual fund can't. A long-only mutual fund identifies an overvalued stock but has no mandate to short it — it can only underweight or avoid holding it, a comparatively weak way to express a negative view. A hedge fund or PMS with a flexible mandate can short the same stock directly, and can additionally use leverage to size a high-conviction long position beyond what its raw capital would allow. This mandate difference — not skill or information — is often the real reason a hedge fund and a mutual fund reach different position sizes or even opposite trades on the same stock, an important nuance when a candidate is asked to compare institutional investor types.

Example 5 — custodian vs broker in an actual settlement failure scenario. An institutional buy order executes on the exchange via the fund's broker, but on settlement date the fund's custodian reports a shortfall — the securities weren't available to deliver from the counterparty side. The clearing corporation's settlement-guarantee mechanism (Part on secondary markets) steps in to make the buyer whole (via the exchange's default/shortfall-handling process, typically an auction to source the shares or a cash equivalent), while the custodian's role throughout is purely to report and reconcile the fund's actual holdings against what was expected — illustrating concretely why an institution needs both a broker (who executed a trade that, from the fund's side, looked completely normal) and an independent custodian (whose job is precisely to catch and report a settlement discrepancy the broker relationship alone wouldn't surface).

How it is tested in interviews

- **"Who are the main market participants?"** — Issuers; investors (retail, mutual funds, insurers/pensions, FPIs, DIIs, hedge funds/PMS, promoters); intermediaries (banks, brokers, exchanges, depositories, clearing corporations, custodians, RTAs); regulator (SEBI).
- **"What does a depository do?"** — "Holds securities in dematerialized form and effects their transfer — NSDL and CDSL in India."
- **"What's the difference between sell-side and buy-side?"** — "Sell-side creates, executes and researches to serve clients; buy-side invests client money and consumes that research."
- **"Why do FPI flows matter?"** — "They're large and can move quickly, so they often drive index-level moves and volatility."
- **"What does SEBI do?"** — "Regulates issuers, intermediaries and markets — disclosure, conduct, surveillance, and investor protection."

Traps & common mistakes

- Confusing **depository** (holds shares) with **clearing corporation** (guarantees settlement) with **exchange** (matches trades) — three different roles.
- Lumping all investors together — horizons and behaviour differ sharply.
- Mixing up **sell-side** and **buy-side**.
- Forgetting the **regulator** as the backbone of trust.

First-principles recap

- Three groups: issuers, investors, intermediaries — plus the regulator.
- Each intermediary solves a friction: distribution, execution, custody, settlement, records.
- The **investor mix** shapes a stock's volatility and behaviour.
- Exchange (match) ≠ clearing corp (guarantee) ≠ depository (hold).
- SEBI ensures disclosure, conduct and investor protection.

Quick-reference

PLAYER	ROLE
Investment bank / BRLM	Underwrite/advise issues
Broker	Execute orders
Exchange (NSE/BSE)	Match trades, price discovery
Depository (NSDL/CDSL)	Hold & transfer demat shares
Clearing corporation	CCP, guarantee settlement
SEBI	Regulate & protect investors

Q&A — Market Participants & Intermediaries

Theory and worked scenarios on the ecosystem of issuers, investors, and intermediaries.

Q1. Map a single institutional trade through every intermediary it touches, and state each one's distinct role.

Model answer. A mutual fund (investor) instructs its broker (execution intermediary) to buy shares on the NSE (exchange, provides the matching venue and price discovery); the trade is guaranteed by the clearing corporation (central counterparty); the depository (NSDL/CDSL) transfers the shares in demat form into the fund's custodian's account (safekeeping/settlement support for institutions) on T+1. Five distinct intermediaries, each solving a different friction: distribution/execution, matching, counterparty guarantee, custody/transfer, and institutional safekeeping.

Q2. Why does the composition of a stock's investor base (retail vs FPI vs DII vs promoter) matter to an analyst forecasting volatility?

Model answer. Different investor types have different horizons and reaction speeds: FPIs can move large sums quickly in response to global risk sentiment or currency moves, amplifying volatility; DIIs (insurers, pension funds) tend to be longer-horizon and more stable, often acting as a stabilising counterweight; promoters are typically long-term, control-oriented holders. A stock heavily held by fast-moving FPIs will react more sharply to negative headlines than an otherwise-similar stock dominated by patient domestic institutional holders — the holder-base composition is itself a volatility signal, independent of the company's fundamentals.

Q3. Distinguish the roles of an exchange, a clearing corporation, and a depository — a classic three-way confusion trap.

Model answer. The exchange (NSE/BSE) provides the trading platform and matches buy/sell orders — price discovery happens here. The clearing corporation becomes the central counterparty to every matched trade, guaranteeing settlement so neither party bears the other's default risk. The depository (NSDL/CDSL) holds securities in dematerialised form and physically executes the transfer of ownership. A common interview trap is treating any two of these as interchangeable — each solves a genuinely distinct friction (matching vs guaranteeing vs custody/transfer).

Q4. What's the difference between sell-side and buy-side, and where would a candidate coming from a trading/derivatives background naturally fit?

Model answer. Sell-side firms (investment banks, brokers) create products, execute trades, and publish research to win and service client business — their research and trading desks are ultimately commercial functions serving external clients. Buy-side firms (asset managers, mutual funds, hedge funds, pension funds) invest their own or client capital directly, consuming sell-side research as one input among many to make investment decisions. A candidate with direct trading/derivatives desk experience (as opposed to client-facing sales or research-publishing experience) often has skills that map most naturally to buy-side execution/trading roles or to sell-side trading desks, as distinct from sell-side equity research roles, which screen more for the research/writing/client-communication skill set.

Q5. What does SEBI actually regulate, concretely, across issuers, intermediaries, and markets?

Model answer. For issuers: disclosure requirements (prospectus content, ongoing reporting, related-party transaction rules). For intermediaries: registration and conduct standards (brokers, merchant bankers, investment advisers must be SEBI-registered and follow conduct codes). For markets: surveillance against insider trading and price manipulation, oversight of exchanges and clearing corporations, and setting rules like circuit breakers and margin requirements. The unifying purpose across all three is investor protection and market integrity — a useful one-line answer is that SEBI's job is to make sure information is disclosed fairly, intermediaries behave honestly, and market mechanics can be trusted.

Q6. Worked scenario — how would surveillance likely detect the insider-trading pattern in this case, and what happens next?

A promoter sells a large block of shares three days before the company announces disappointing earnings, having had no prior pattern of similar sales.

Model answer. SEBI's surveillance systems flag unusual trading patterns ahead of price-sensitive announcements — specifically, a departure from an individual's established trading behaviour (no prior pattern of large sales) combined with timing that precedes a material negative disclosure is a classic red flag pattern. This typically triggers an investigation (examining trading records, communications, and access to the pre-announcement information), and if insider trading is substantiated, results in penalties (fines, trading bans, potential criminal referral) — the broader function this serves is preserving the market's trust that prices reflect fairly available information, which is what keeps outside investors willing to participate.

Q7. What's the difference between a custodian and a broker, and why might an institutional investor use both?

Model answer. A broker executes trading instructions (buy/sell orders) on the investor's behalf. A custodian safekeeps the institution's assets, handles settlement mechanics on the institution's side, and often provides additional services (corporate action processing, reporting, fund administration support). An institutional investor typically uses a broker for execution and a separate custodian for safekeeping and back-office settlement — this separation (rather than one entity doing both) is itself a risk-management practice, reducing the concentration of control over both trade execution and asset custody in a single counterparty.

Q8. Why do RTAs (Registrars & Transfer Agents) matter to a retail shareholder, even though most investors never interact with one directly?

Model answer. RTAs maintain the official record of shareholding and process corporate actions (dividend payments, rights/bonus share credits, name/address changes, dematerialisation requests) on behalf of the issuing company — even though a retail investor's day-to-day interaction is with their broker/demat account, the RTA is the entity that ultimately ensures corporate-action entitlements (a declared dividend, a bonus share) are correctly credited to the right shareholder of record, making it a quietly essential piece of infrastructure behind every corporate action an investor experiences.

Equity Instruments

The Problem / Why this matters

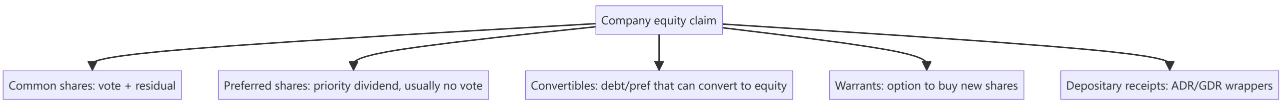
"Equity" isn't a single thing. Companies issue common shares, preferred shares, depositary receipts, warrants, and convertibles — each with different rights to cash flow, control and risk. Knowing the instruments, their rights, and where they sit in the capital structure lets you understand what you actually own, how it's valued, and why one class trades differently from another. It's foundational for equity research and capital markets roles.

Core Idea

Equity instruments are claims on a company's **residual value and control**, ranging from plain common shares (full ownership and residual risk) through preferred shares (a hybrid with priority) to derivative-like claims (warrants, convertibles) and cross-border wrappers (ADRs/GDRs). Each differs in cash-flow priority, voting rights, and payoff.

Why it works this way

Different investors want different risk-return-control packages, and companies want to tailor securities to tap the widest, cheapest capital. So the market slices the equity claim into instruments: some trade voting rights for a fixed dividend (preferred), some bundle upside options (warrants, convertibles), and some repackage foreign shares for local investors (depositary receipts).



Full technical content

Common (equity/ordinary) shares: full ownership — voting rights, residual claim on profits (dividends, discretionary) and on assets in liquidation (last in line). Uncapped upside, first-loss downside. The default meaning of "a share."

Preferred shares: a hybrid. Priority over common for **dividends** (often fixed) and in **liquidation**, but usually **no vote**. Variants: **cumulative** (missed dividends accrue), **participating** (share in extra upside), **convertible** (can convert to common), **redeemable/callable**. Sits between debt and common equity in the capital structure.

Depositary receipts: let investors hold foreign shares locally. - **ADR** (American Depositary Receipt) — a US-listed certificate representing shares of a non-US company (e.g., Infosys ADR on NYSE). - **GDR** (Global Depositary Receipt) — similar, listed outside the home and US markets. - Purpose: access foreign capital and investors without a full foreign listing.

Warrants: a company-issued option giving the holder the right to buy **new** shares at a set price for a period. Exercise creates new shares (dilutive). Often attached to bonds as a sweetener.

Convertible securities: bonds or preferred shares that can convert into common equity at a set ratio. Give the holder debt-like downside protection plus equity upside; give the issuer a lower coupon. Convert when the equity is worth more than the bond.

Rights & entitlements: existing shareholders may receive **rights** (to buy new shares at a discount — see corporate actions) or **bonus** shares.

Key rights of a shareholder: vote (elect directors, major decisions), dividends (if declared), residual assets in liquidation, information/disclosure, pre-emption (rights issues), and to transfer shares.

Where each sits (priority in liquidation): secured debt → unsecured debt → **preferred equity** → **common equity** (last). Higher priority = lower risk = lower expected return.

Worked examples

Example 1 — common vs preferred in a bad year. A firm can pay ₹10 cr of dividends. Preferred holders are owed a fixed ₹6 cr (cumulative) — they're paid first; common gets the remaining ₹4 cr (discretionary). In a terrible year with ₹4 cr available, preferred takes it all and common gets nothing (and cumulative preferred arrears accrue). Preferred = priority but capped; common = residual and variable.

Example 2 — convertible economics. A ₹1,000 convertible bond converts into 8 shares (conversion price ₹125). If the stock is ₹100, the holder keeps the bond (worth more than 8 × ₹100 = ₹800) and enjoys downside protection. If the stock rises to ₹180, converting gives 8 × ₹180 = ₹1,440 — the holder converts and captures the equity upside. The issuer paid a low coupon for embedding that option.

Example 3 — ADR arbitrage link. Infosys trades in Mumbai and as an ADR in New York. If the ADR (adjusted for the ratio and FX) diverges from the Mumbai price, arbitrageurs trade both until they realign — so the ADR essentially tracks the home share.

Example 4 — a worked ADR ratio and arbitrage-band calculation. An Indian company's ADR represents 2 underlying home-market shares (an ADR ratio of 2:1). The home share trades at ₹840, and USD/INR is 83.5. The ADR's "fair" USD price = (840 × 2) / 83.5 ≈ **$20.12**. If the actual ADR price is $20.60, it's trading at a premium of roughly ($20.60 − $20.12)/$20.12 ≈ 2.4% to fair value — small enough that transaction costs (brokerage, FX conversion spread, the operational cost of the deposit/withdrawal mechanism connecting the ADR to the underlying share) likely exceed the arbitrage profit, which is exactly why small ADR premiums/discounts can persist despite the theoretical arbitrage link; a wider gap (say 8-10%) would be a much more actionable, and more surprising, signal worth investigating for a data or execution error before assuming a genuine mispricing.

Example 5 — why a company chooses preferred shares over a straight bond to raise capital. A company wants to raise ₹200 cr without adding to its formal debt covenants (which would trip financial-ratio limits already set with existing lenders) and without immediately diluting common shareholders' voting control. Issuing cumulative redeemable preferred shares achieves both: preferred shares typically aren't counted as "debt" for covenant purposes (they're equity on the balance sheet, even though they behave economically like a hybrid), and standard preferred structures carry no voting rights, leaving common shareholders' control unchanged. The cost: preferred dividends aren't tax-deductible the way interest expense is, making preferred capital more expensive after-tax than equivalent debt — a trade-off (covenant/control flexibility vs after-tax cost) a candidate should be able to state explicitly when asked "why would a company use preferred stock instead of just borrowing."

How it is tested in interviews

- **"Common vs preferred shares?"** — "Common has voting rights and a residual, variable claim (uncapped upside, first-loss). Preferred usually has no vote but priority on a fixed dividend and in liquidation — a hybrid between debt and common."
- **"What is an ADR?"** — "A US-listed certificate representing shares of a foreign company, letting US investors hold it in dollars without a foreign listing."

- **"How does a convertible bond work?"** — "A bond that can convert into equity at a set ratio; the holder gets debt downside protection plus equity upside, and the issuer pays a lower coupon for that option."
- **"Where does preferred sit in the capital structure?"** — "Below debt, above common — priority over common for dividends and liquidation, but behind all creditors."

Traps & common mistakes

- Thinking preferred is "safer equity" without noting it's still **below all debt**.
- Confusing **warrants** (company issues new shares, dilutive) with exchange-traded **options** (no new shares).
- Forgetting convertibles are **dilutive** on conversion.
- Treating ADRs/GDRs as different companies rather than **wrappers** on the same shares.

First-principles recap

- Equity instruments slice the ownership claim by cash-flow priority, control and payoff.
- **Common** = vote + residual (uncapped/first-loss); **preferred** = priority dividend, usually no vote.
- **Convertibles** = debt/pref with an equity option; **warrants** = right to buy new shares (dilutive).
- **ADR/GDR** = wrappers letting foreign investors hold local shares.
- Priority: debt → preferred → common.

Quick-reference

INSTRUMENT	KEY FEATURE
Common share	Vote + residual claim
Preferred share	Priority dividend, usually no vote
Convertible	Converts debt/pref → equity
Warrant	Right to buy new shares (dilutive)
ADR / GDR	Foreign-share wrapper
Priority	Debt > preferred > common

Q&A — Equity Instruments

Theory and worked scenarios on common, preferred, convertible, and depositary-receipt instruments.

Q1. Common vs preferred shares — what's given up and what's gained by holding preferred instead of common?

Model answer. Preferred shareholders give up voting rights (in most standard structures) and uncapped upside participation, in exchange for priority over common holders on dividends (often a fixed rate) and on residual assets in liquidation. Common shareholders retain full voting rights and an uncapped, residual claim on profits and assets, but sit last in the priority stack — first to lose in a downturn, last to be paid even in a good year if the board doesn't declare a dividend. The trade is control-and-upside versus priority-and-a-capped-return.

Q2. Worked — cumulative preferred dividends in a bad year.

A company owes ₹8 cr/year in cumulative preferred dividends. In Year 1 it can only pay ₹3 cr total. In Year 2, profits recover and ₹15 cr is available for distribution. How much does each class receive in each year?

Model answer. Year 1: preferred is owed ₹8 cr but only ₹3 cr is available — preferred receives the full ₹3 cr (still short by ₹5 cr, which accrues as arrears since it's cumulative); common receives ₹0. Year 2: preferred is owed the current year's ₹8 cr *plus* the ₹5 cr arrears from Year 1 = ₹13 cr, paid first from the ₹15 cr available; common receives the remaining ₹15 − 13 = ₹2 cr. This illustrates why "cumulative" matters: a missed preferred dividend doesn't disappear, it compounds as a prior claim against future distributable profit before common sees anything.

Q3. Worked — convertible bond conversion decision at two different stock prices.

A ₹1,000 face-value convertible bond converts into 10 shares (conversion price ₹100). At maturity the bond can be redeemed at face value (₹1,000) or converted.

Model answer. If the stock is at ₹80: conversion value = 10 × 80 = ₹800, less than the ₹1,000 redemption value — the holder redeems for cash, the debt-like floor protects them. If the stock is at ₹150: conversion value = 10 × 150 = ₹1,500, more than the ₹1,000 redemption value — the holder converts to capture the equity upside. The convertible's value is effectively the greater of its bond floor and its as-converted equity value, which is exactly why issuers can offer a lower coupon than a plain bond — investors are compensated with this embedded option instead.

Q4. What is an ADR, and why would the ADR price of an Indian company track its NSE/BSE price closely rather than diverge freely?

Model answer. An ADR (American Depositary Receipt) is a US-listed certificate representing shares of a non-US company held by a depositary bank — it lets US investors hold and trade the company in dollars without a full separate US listing. Because the ADR and the home-market shares represent the same underlying economic claim (adjusted for the ADR ratio and the USD/INR exchange rate), any meaningful price divergence creates a pure arbitrage opportunity: arbitrageurs would buy the cheaper instrument and sell the more expensive one (or convert between them via the depositary mechanism), which pulls the two prices back into alignment — the arbitrage mechanism itself is what keeps the ADR "tracking" the home share, not any inherent property of the ADR.

Q5. Distinguish a warrant from an exchange-traded call option — same payoff shape, different mechanics and consequences.

Model answer. Both give the right to buy shares at a set price by a set date, and both have a similar payoff diagram. The key difference: exercising a warrant creates *new* shares issued by the company (dilutive to existing shareholders, and the exercise proceeds go to the company), while exercising an exchange-traded call option transfers *existing* shares between two market participants (no dilution, no cash to the company — the option writer delivers shares they already own or must acquire in the market). A candidate who conflates the two in an interview signals they haven't thought through where the shares actually come from.

Q6. Where do preferred shares sit in the capital-structure priority stack, and what does that imply about their risk relative to bonds and common equity?

Model answer. Priority order (highest to lowest claim): secured debt → unsecured debt → preferred equity → common equity. Preferred sits below all debt (both secured and unsecured), meaning in a liquidation or severe distress scenario, all debtholders must be paid in full before preferred holders receive anything — so preferred is meaningfully riskier than any debt instrument despite sometimes being marketed as a "safer" or "income" instrument. It is, however, senior to common, so it carries less risk than common equity. This full ordering — not just "preferred is safer than common" — is the complete, correct answer.

Q7. What's the difference between a rights issue entitlement and a bonus share, from the shareholder's perspective?

Model answer. A rights entitlement gives an existing shareholder the option to *buy* additional shares (usually at a discount to market price) — it requires the shareholder to pay in additional capital to exercise it, and if they don't, their proportional ownership is diluted. A bonus share is issued to existing shareholders *for free*, in proportion to their existing holding, funded by capitalising the company's reserves — it requires no payment and, because more shares now represent the same underlying company value, the share price mechanically adjusts downward (similar to a stock split) with no real change in the shareholder's proportional wealth or ownership percentage.

Q8. Why might a company issue convertible preferred shares rather than either straight debt or a straight equity issue?

Model answer. Convertible preferred lets a company raise capital at a lower cost than straight preferred/debt (investors accept a lower fixed return in exchange for the embedded equity-upside option) while deferring dilution — dilution only occurs if and when the instrument converts, typically once the stock has appreciated enough to make conversion attractive to holders. This is a common structure in growth-stage and PE/VC financing rounds specifically because it balances the investor's downside protection (priority, fixed return if it doesn't convert) against the company's desire to avoid selling common equity outright at what may be a depressed current valuation.

Market Indices

The Problem / Why this matters

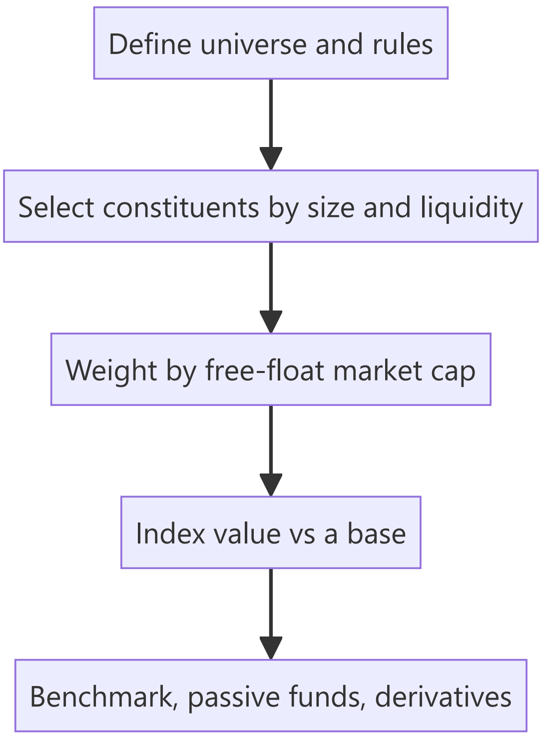
"The market was up 1% today" — but what *is* "the market"? An **index** is the number that represents it. Indices are how performance is measured, how passive funds are built, how benchmarks are set, and how derivatives (index futures/options) are priced. Understanding how they're constructed (and their quirks) matters for research, portfolio, and derivatives roles — and interviewers ask how the Nifty/Sensex are built and what "free-float market-cap weighted" means.

Core Idea

A market index is a **single number tracking the value of a defined basket of stocks**, used as a barometer of the market or a segment. Most modern indices are **free-float market-capitalization weighted**: each stock's weight is proportional to the market value of its publicly tradable shares.

Why it works this way

Investors need a comparable, investable benchmark. Weighting by free-float market cap makes the index reflect where investable money actually sits (bigger, more freely-traded companies matter more) and makes it **replicable** by index funds. It also self-adjusts: as a stock rises, its weight rises, mirroring a real buy-and-hold portfolio.



Full technical content

Construction methods (weighting):

METHOD	WEIGHT BY	EXAMPLES
Free-float market-cap	Value of publicly tradable shares	Nifty 50, Sensex, S&P 500
Full market-cap	Total shares × price	older indices
Price-weighted	Share price (not size)	Dow Jones, Nikkei
Equal-weighted	Same weight each	equal-weight indices

Free-float excludes promoter/strategic/locked holdings — only shares actually available to trade count, so the index reflects investable value.

Indian benchmarks: - **Nifty 50** (NSE) — 50 large-caps, free-float market-cap weighted. - **Sensex** (BSE) — 30 large-caps, free-float weighted. - Broader: Nifty Next 50, Nifty 100/500, Nifty Midcap/Smallcap, and sectoral indices (Bank Nifty, Nifty IT, etc.).

The divisor & continuity. An index uses a **divisor** so that mechanical changes (constituent additions/deletions, corporate actions, new share issuance) don't create artificial jumps — the divisor is adjusted to keep the index continuous. This is why a stock split doesn't move the index.

Rebalancing. Indices are reviewed periodically (e.g., semi-annually for Nifty) to add/drop constituents by size and liquidity rules. Index inclusion/exclusion drives real flows (passive funds must buy the added stock), so it moves prices.

Uses of indices: 1. **Benchmark** — measure a portfolio's relative performance (alpha). 2. **Passive investing** — index funds and ETFs replicate the index cheaply. 3. **Derivatives** — index futures and options (Nifty/Bank Nifty are among the world's most traded). 4. **Market sentiment** — a barometer of the economy/segment.

Price return vs total return index. A **price index** ignores dividends; a **total return index (TRI)** reinvests dividends — TRI is the fair benchmark for a fund that also earns dividends.

Worked examples

Example 1 — free-float weighting. Company X has ₹1,000 cr total market cap but the promoter holds 60% (locked), so free-float = 40% = ₹400 cr. Company Y has ₹600 cr market cap, 100% free-float = ₹600 cr. Despite X being "bigger," Y gets a larger index weight (₹600 cr vs ₹400 cr of investable value). *Free-float, not total size, drives weight.*

Example 2 — why a split doesn't move the index. A constituent does a 1:1 split — price halves, shares double, market cap unchanged. The index divisor absorbs the mechanical change so the index value is unaffected. Corporate actions don't distort the index.

Example 3 — index inclusion flow. A stock is added to the Nifty 50 at the next rebalance. Every passive fund tracking the Nifty must buy it to match the index, creating real buying pressure — the stock often rises into and on inclusion. Index membership has real cash-flow consequences.

Example 4 — sizing the actual passive-flow impact of an index change. Estimated total AUM tracking the Nifty 50 (index funds + ETFs + closet-indexers benchmarked to it) is roughly ₹4,50,000 cr. A stock is added to the index with an assigned weight of 0.6%. The mechanical passive buying this triggers is approximately 0.6% × 4,50,000 cr ≈ **₹2,700 cr** of forced buying across the tracking-fund universe, concentrated around the effective date of the change. If the stock's free-float market cap is, say, ₹40,000 cr and its average daily traded value is ₹150 cr, then ₹2,700 cr of buying represents roughly **18 days' worth of average trading volume** that needs to clear near the rebalance date — explaining why index-inclusion candidates often see meaningfully outsized price moves and volume spikes specifically in the days around the effective date, and why some active managers explicitly trade *ahead* of an anticipated inclusion (once a stock's growing market cap makes inclusion likely at the next scheduled review) to front-run this predictable flow.

Example 5 — why a stock's index weight can change even with no rebalance and no price move. A promoter sells a 10% stake via an open-market block deal, reducing their holding from 55% to 45% — free-float rises from 45% to 55% of total shares. Even with the share price completely unchanged, the stock's free-float market cap (and therefore its index weight) rises proportionally, since free-float weighting (Section 6.1) is a function of *tradable* value, not just price. This is why an index's divisor and constituent weights are reviewed and adjusted not only at scheduled rebalance dates, but also on ad hoc "specified events" like a material shareholding change — a detail that separates a candidate who understands index mechanics from one who only knows the "market-cap weighted" headline.

How it is tested in interviews

- **"How is the Nifty/Sensex constructed?"** — "Free-float market-capitalization weighted baskets of large-caps (Nifty 50 on NSE, Sensex 30 on BSE); weights reflect the value of publicly tradable shares."
- **"What does free-float mean and why use it?"** — "Only publicly tradable shares (excluding promoter/locked holdings). It makes the index reflect investable value and be replicable by index funds."
- **"Why doesn't a stock split move the index?"** — "The divisor adjusts for mechanical changes so the index stays continuous; market cap is unchanged by a split."
- **"Price-weighted vs market-cap-weighted?"** — "Price-weighted (Dow) weights by share price regardless of size; market-cap weighted (Nifty/S&P) weights by company value — the latter reflects the real market."
- **"Price return vs total return index?"** — "TRI reinvests dividends; it's the fair benchmark for a fund that also collects dividends."

Traps & common mistakes

- Saying "market-cap weighted" without the **free-float** nuance for Nifty/Sensex.
- Thinking a **split or bonus** moves the index (the divisor absorbs it).
- Confusing **price-weighted** (Dow) with **market-cap-weighted** (S&P/Nifty).
- Benchmarking a fund against a **price index** instead of the **total-return** index.

First-principles recap

- An index = a single number tracking a defined basket; the market's barometer.
- Nifty/Sensex are **free-float market-cap weighted** — investable value drives weight.
- A **divisor** keeps the index continuous through corporate actions and rebalances.
- Indices power benchmarking, passive funds, and derivatives.
- Use the **total-return** index to fairly benchmark dividend-earning funds.

Quick-reference

ITEM	NOTE
Nifty 50 / Sensex	Free-float market-cap weighted (50 / 30 stocks)
Free-float	Publicly tradable shares only
Divisor	Keeps index continuous through changes
Price-weighted	Dow, Nikkei (by share price)
TRI	Reinvests dividends (fair benchmark)
Inclusion	Forces passive-fund buying (real flow)

Q&A — Market Indices

Theory and worked scenarios on index construction, weighting, and mechanics.

Q1. How are the Nifty 50 and Sensex constructed, precisely — not just "market-cap weighted"?

Model answer. Both are **free-float market-capitalisation weighted** indices of large-cap stocks (50 for Nifty on NSE, 30 for Sensex on BSE) — weight is proportional to the value of each company's *publicly tradable* shares, specifically excluding promoter, government, and other strategic/locked holdings. Saying only "market-cap weighted" without the free-float qualifier is an incomplete answer and a common interview gap, since free-float is the specific design choice that makes the index reflect genuinely investable value rather than raw company size.

Q2. Worked — free-float weight calculation.

Company P: total market cap ₹5,000 cr, promoter holding 55% (locked/strategic). Company Q: total market cap ₹3,200 cr, promoter holding 15%. Which gets a larger index weight, and by how much (in free-float value terms)?

Model answer. Company P free-float = 45% × 5,000 = ₹2,250 cr. Company Q free-float = 85% × 3,200 = ₹2,720 cr. Despite Company P having a larger total market cap (₹5,000 cr vs ₹3,200 cr), Company Q gets the larger index weight because its free-float value (₹2,720 cr) exceeds P's (₹2,250 cr) — a direct illustration that total company size and index weight are not the same thing once free-float is applied.

Q3. Why doesn't a stock split or bonus issue move the index level, even though it changes the share price and share count?

Model answer. The index uses a **divisor** — a normalising constant recalculated whenever a mechanical, non-value change occurs (splits, bonus issues, constituent additions/deletions, buybacks). A split halves the price and doubles the share count with market cap unchanged; the divisor is adjusted so the index's calculated value doesn't move from this purely mechanical change. The divisor is what allows the index to be a continuous, comparable series over time despite constant structural changes to its constituents.

Q4. What's the practical, real-money consequence of a stock being added to (or removed from) a major index like the Nifty 50?

Model answer. Passive funds and ETFs that track the index are mandated to hold constituents in index-matching proportions — when a stock is added at a scheduled rebalance, every fund tracking that index must buy it (and sell any removed stock), creating real, mechanical buying (or selling) pressure independent of the company's fundamentals. This is why stocks often see a price bump running into a confirmed index-inclusion date, and why "will this stock get added to the index" is itself a research question with a trading implication, separate from fundamental valuation.

Q5. Price-weighted vs market-cap-weighted indices — what's the practical flaw of price-weighting that market-cap weighting avoids?

Model answer. A price-weighted index (like the Dow Jones) gives a higher-priced stock more influence on the index level regardless of the company's actual size — a ₹5,000 stock in a small company would move a price-weighted index more than a ₹200 stock in a much larger company, which is economically arbitrary (share price level is just a function of how many shares are outstanding, not company value). Market-cap weighting (Nifty, Sensex, S&P 500) ties influence to actual company/free-float value, which is both more economically meaningful and matches how a real buy-and-hold portfolio would naturally be weighted.

Q6. What is a total return index (TRI), and why does using a price index instead understate a long-term equity fund's true relative performance?

Model answer. A price index tracks only price changes, ignoring dividends paid by constituents. A total return index reinvests those dividends back into the index calculation. Since an actively or passively managed equity fund also earns and (usually) reinvests dividends from its holdings, benchmarking the fund's total return against a price-only index systematically understates the index's true comparable performance — over long periods, the dividend-reinvestment gap between a price index and its TRI equivalent compounds to a meaningful difference, which is why professional performance reporting should always use the TRI, not the more commonly quoted headline price index.

Q7. Worked — divisor mechanics illustrated with numbers.

A simplified 2-stock index: Stock A, 100 shares free-float at ₹50 (₹5,000), Stock B, 200 shares free-float at ₹40 (₹8,000). Total free-float value = ₹13,000. If the divisor is 130, the index value = 13,000/130 = 100. Stock A now does a 2-for-1 split: 200 shares at ₹25 (still ₹5,000 total). What must the new divisor be to keep the index at 100, and why?

Model answer. Post-split total free-float value is unchanged (200 × 25 = ₹5,000 for A, plus B's unchanged ₹8,000 = ₹13,000 total) — since the underlying value hasn't changed, the divisor also doesn't need to change here (it stays at 130, index value = 13,000/130 = 100). The divisor only needs to be *recalculated* when a change alters the *aggregate free-float value* itself without a corresponding real value change — e.g. when a constituent is swapped for a different one, or free-float percentage changes (a promoter selling down increases free-float value without a price change) — a split alone, since it's price × share-count neutral, is actually one of the simpler cases, though it's still routinely cited alongside divisor mechanics because the underlying principle (isolating real value changes from mechanical ones) is the same.

Q8. Beyond Nifty 50 and Sensex, name and briefly describe two other categories of Indian indices and what each is used for.

Model answer. Broader market-cap indices (Nifty 100/500, Nifty Midcap, Nifty Smallcap) — used to benchmark funds and strategies focused outside just the largest 50 companies, giving a fuller picture of mid- and small-cap segment performance. Sectoral/thematic indices (Bank Nifty, Nifty IT, Nifty Auto, Nifty Pharma, etc.) — used to benchmark sector-focused funds, to trade sector-specific derivatives (Bank Nifty options are among the most actively traded derivatives contracts in India), and by analysts to gauge relative sector performance and rotation trends within the broader market.

The Equity Research Process

The Problem / Why this matters

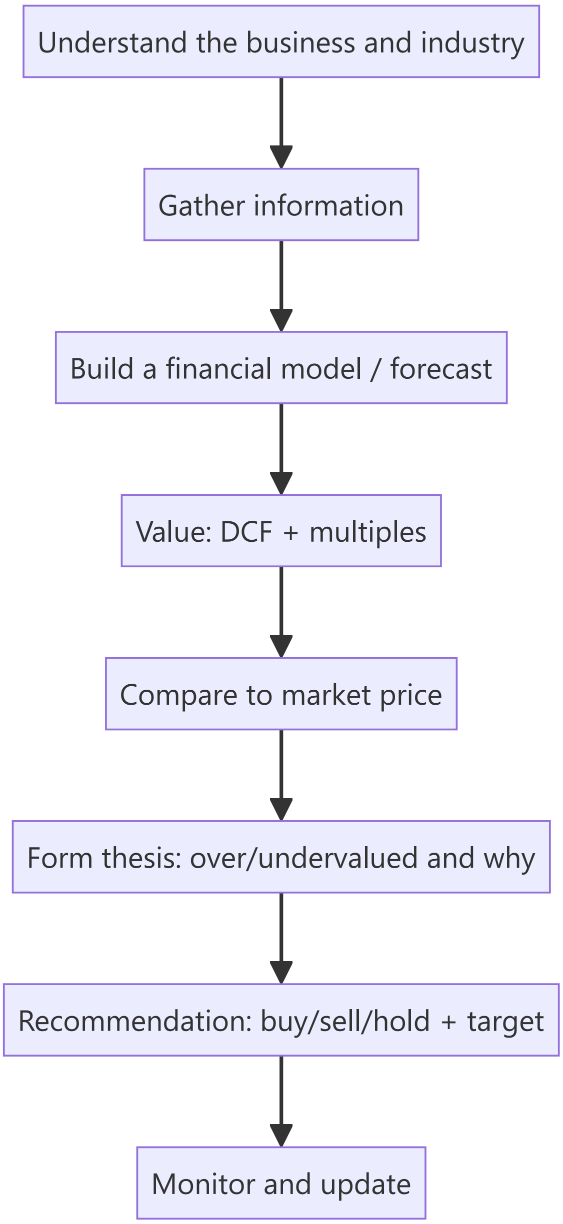
An equity research analyst's job is to form and defend a **view on a stock** — is it worth more or less than the market says, and why? That view drives buy/sell/hold recommendations that clients act on. Knowing the end-to-end process — coverage, information gathering, modelling, valuation, thesis, and the recommendation — is exactly what an equity research or investment-analyst interview tests, often as "walk me through how you'd analyse a company."

Core Idea

Equity research is a repeatable process: **understand the business → model its financials → value it → compare to the market price → form a thesis and recommendation → monitor.** The output is an investment view supported by a model and a written note.

Why it works this way

Markets are mostly efficient, so to add value an analyst must either know something the market underappreciates or interpret known facts better. That requires deeply understanding the business and its drivers, building a defensible forecast, and translating it into a value that can be compared to the price — then articulating *why* the market is wrong.



Full technical content

Step 1 — Understand the business & industry. What does the company do, how does it make money (revenue model, unit economics), what's its competitive position and moat, who are its customers and competitors, and what drives the industry (see fundamental analysis).

Step 2 — Gather information. Annual reports/10-Ks, quarterly results and concalls, management guidance, industry data, channel checks, expert networks, competitor filings, and news. Rigorous, primary-source-driven.

Step 3 — Build the model. A three-statement model with a **driver-based forecast** (revenue drivers → costs → the three linked statements), producing the free cash flows and metrics valuation needs (see the modeling chapter).

Step 4 — Value. Intrinsic (DCF) and relative (comps/multiples), triangulated to a value range and a target price (see applied valuation).

Step 5 — Form the thesis. The core argument: *why* the stock is mispriced. A good thesis is **differentiated** (a view the market doesn't fully hold), **specific**, and **falsifiable** (you can say what would prove you wrong). Identify the **catalysts** that will close the value gap and the **risks**.

Step 6 — Recommendation. Buy / Hold / Sell (or Overweight/Neutral/Underweight) with a **target price** and time horizon, and the expected return vs the current price.

Step 7 — Write and communicate. The research note (initiation or update) and verbal pitch to clients/PMs; then **monitor** results, news and thesis, updating the view as facts change.

The analyst's edge — three sources of alpha:

EDGE	DESCRIPTION
Informational	Knowing a fact the market hasn't priced (rare, and insider info is illegal)
Analytical	Interpreting known facts better (better model/insight)
Behavioural/time-horizon	Exploiting others' biases or short-termism

Most legitimate edge is **analytical** and **behavioural**.

Quality markers of good research: a clear, differentiated thesis; a defensible model with sensible assumptions; explicit catalysts and risks; and intellectual honesty (state what would make you wrong).

Channel checks and primary research in equity analysis

Public filings and models only go so far — the highest-conviction theses are usually backed by **channel checks**: primary research into a company's own ecosystem, run with the same rigour as market research (Part 3-4 of the market-research literature, directly transferable). - **Supplier checks** — calling/meeting key suppliers to gauge order volumes, pricing trends, and capacity utilisation ahead of the company's own disclosure (a leading indicator of revenue). - **Distributor/dealer checks** — for consumer and auto companies, calling a sample of dealers/distributors across regions to triangulate real sell-through (not just sell-in, which a company can inflate near quarter-end by stuffing the channel). - **Customer/end-user surveys** — for B2C names, small structured surveys on brand awareness, repeat-purchase intent, or switching behaviour (identical methodology to Part 5 of market research: sample size, question design, bias avoidance all apply directly). - **Expert networks** — paid platforms connecting analysts to former employees, industry consultants, or specialists for one-off calls; must be used within strict compliance boundaries (no MNPI — material non-public information — can be solicited or used). - **Job-posting and hiring-trend analysis** — a company rapidly hiring in a new geography/function is a public, legal signal of expansion plans, cross-checked against management commentary. - **Store-check/footfall studies** — for retail/QSR names, physically observing footfall and average transaction patterns at a sample of outlets — the equity-research analogue of a market-research "intercept" study (see Market Research, Part 5.6).

Compliance boundary (tested in every legitimate research role's onboarding): any of the above crosses a hard legal line the moment it solicits or knowingly uses **material non-public information** — a supplier confirming "volumes are up" is a legitimate industry read; a supplier disclosing an unreleased quarterly number is insider information, and using it is a criminal offence under India's SEBI (Prohibition of Insider Trading) Regulations. A research analyst must be able to draw this line without hesitation.

A full worked research note — "TechCo India Ltd" (illustrative)

Walking one company through the entire pipeline end to end is the single best interview-prep exercise for this chapter.

Business understanding: TechCo is a mid-cap Indian IT-services company, ₹8,500 crore revenue, historically 14% EBITDA margin, growing 12% p.a., trading at 18x forward P/E vs a peer group average of

22x.

Model (driver-based, simplified): Revenue = existing client base (95% retention) × average contract growth (8%) + new client wins (4% incremental) = ~12% revenue CAGR over the forecast window. Margin: assume 100bps expansion over 3 years from offshore-mix shift and utilisation improvement, taking EBITDA margin from 14% to 15% (this single assumption is the crux of the thesis — see below).

Valuation: DCF using WACC 11.5% (cost of equity via CAPM ~13%, pre-tax cost of debt ~8.5%, D/E modest) and a terminal growth rate of 5% (nominal, in line with long-run Indian GDP-growth assumptions for a mature IT-services business) yields an intrinsic value of ₹1,150/share. Cross-check via relative valuation: applying the peer-average 22x forward P/E (rather than the stock's own 18x) to next-year EPS implies ₹1,190/share — the two methods converge within ~3.5%, a strong triangulation (Part 7.2 of the Market Research handbook's triangulation principle applies identically here).

Thesis: the market is pricing TechCo at a discount to peers (18x vs 22x) because of a perceived margin ceiling; the analyst's differentiated view is that the offshore-mix shift already underway (visible in headcount-location disclosures) will close 100bps of the gap over 3 years, which the market is not yet modelling — this, not revenue growth (already well understood and priced), is the source of edge.

Catalysts: next two quarters' margin prints beating consensus; management commentary on offshore-mix % at the next earnings call; a peer re-rating event that draws attention to the sector's margin trajectory.

Risks: wage inflation offsetting the offshore-mix benefit; a large client loss (client-concentration risk — check top-5-client revenue % in the filings); rupee appreciation compressing margins (a genuine India-specific IT-services risk worth naming explicitly, since revenue is often dollar-denominated while costs are rupee-denominated).

Recommendation: Buy, target ₹1,150-1,190 (DCF/comps range), current price ₹1,000, implying ~15-19% upside over a 12-month horizon; position sized moderately given single-driver (margin) thesis concentration.

What would falsify this thesis: two consecutive quarters of flat or declining offshore-mix % in management disclosures, or a margin miss attributed to wage inflation exceeding the offshore benefit — the analyst commits, in writing, to downgrading on either signal.

Worked examples

Example 1 — the process end to end. Analyst covers an FMCG firm. Understands the business (rural distribution moat); models revenue off volume × price with margin expansion from premiumization; DCFs it and cross-checks EV/EBITDA vs peers → intrinsic ₹1,200 vs market ₹1,000. Thesis: the market underrates margin expansion from premiumization (differentiated view); catalyst: next two quarters of margin beats; risk: rural slowdown. Recommendation: **Buy, target ₹1,200, +20%.**

Example 2 — differentiated vs consensus thesis. Two analysts value the same stock at ₹1,000, exactly the market price, agreeing with consensus. Neither adds value — a research view is only useful when it *differs* from the price and has a reason. The job is to find and defend the gap.

Example 3 — thesis falsification. An analyst's Buy rests on a new product driving 15% volume growth. She states: "If the first two quarters show under 8% volume growth, the thesis is broken and I'll downgrade." That falsifiability is what separates research from cheerleading.

How it is tested in interviews

- **"Walk me through how you'd analyse a company / research a stock."** — Recite: understand the business → gather info → model → value (DCF + comps) → compare to price → form a differentiated thesis with catalysts and risks → recommend with a target → monitor.

- **"What makes a good investment thesis?"** — "Differentiated from consensus, specific, with clear catalysts and risks, and falsifiable — you can state what would prove you wrong."
- **"Where does an analyst's edge come from?"** — "Mostly analytical (better interpretation) and behavioural (exploiting short-termism/biases); informational edge is rare and insider info is illegal."
- **"How do you decide buy vs sell?"** — "Compare intrinsic value/target to the market price; the gap and catalysts drive the rating and expected return."

Traps & common mistakes

- Producing a view that **matches consensus/price** — it adds no value.
- A model with no **thesis** — data without a differentiated argument.
- No **catalysts** (why does the gap close?) or **risks** (what if you're wrong?).
- Not stating what would **falsify** the thesis — a recipe for holding losers.
- Chasing **informational edge** into insider-trading territory.

First-principles recap

- Research = understand → model → value → compare to price → thesis → recommend → monitor.
- Value is only added when your view **differs from the market**, with a reason.
- A good thesis is differentiated, specific, catalyst-driven, and **falsifiable**.
- Edge is mostly analytical and behavioural.
- The output is a rating, a target price, and a defensible note.

Quick-reference

STEP	OUTPUT
Understand business	Revenue model, moat, drivers
Model	Driver-based 3-statement forecast
Value	DCF + multiples → target
Thesis	Why mispriced + catalysts + risks
Recommendation	Buy/Hold/Sell + target + horizon
Edge	Analytical + behavioural

Q&A — The Equity Research Process

A mixed bank of theory and worked scenarios on how an equity analyst actually builds and defends a view.

Q1. Walk me through how you'd research a stock, start to finish.

Model answer. Understand the business and industry (revenue model, moat, competitors) → gather information (filings, concalls, channel checks) → build a driver-based three-statement model → value it via DCF and comps → compare intrinsic value to the market price → form a differentiated, falsifiable thesis with named catalysts and risks → issue a recommendation with a target price and horizon → monitor and update as facts change.

Q2. What are the three sources of an analyst's "edge," and which are legitimate?

Model answer. Informational (knowing a fact the market hasn't priced — rare, and using material non-public information is illegal), analytical (interpreting known, public facts better than consensus), and behavioural/time-horizon (exploiting other investors' biases or short-termism). Legitimate, sustainable edge in a compliant research role is almost entirely analytical and behavioural.

Q3. Two analysts independently value a stock at exactly the current market price. Has either added value?

Model answer. No — a view that matches the price and consensus adds nothing actionable; research only creates value when the analyst's view *differs* from the price, for a stated, defensible reason. Agreement with consensus is not itself wrong, but it isn't a research output worth publishing as a rated call.

Q4. What makes an investment thesis "falsifiable," and why does it matter?

Model answer. A falsifiable thesis states, in advance, the specific evidence that would prove it wrong (e.g. "if offshore-mix % doesn't rise for two consecutive quarters, the margin-expansion thesis is broken and I'll downgrade"). It matters because it forces intellectual honesty and prevents an analyst from silently rationalising every new data point to fit an existing Buy/Sell call — a well-known behavioural trap (confirmation bias) that falsifiability is a direct discipline against.

Q5. What is a channel check, and where is the legal line?

Model answer. A channel check is primary research into a company's own ecosystem — calling suppliers, distributors, or customers to independently gauge demand/pricing/volume trends ahead of official disclosure. The legal line is material non-public information (MNPI): a supplier saying "orders feel stronger this quarter" is a legitimate industry read; a supplier disclosing an actual unreleased number is MNPI, and knowingly trading or advising on it is a criminal offence under SEBI's insider-trading regulations. An analyst must be able to draw this distinction instantly, not as an afterthought.

Q6. Worked scenario — build a one-paragraph thesis from these facts.

Facts: a listed paints company is growing volumes 8% p.a., but raw-material (crude-linked) costs have fallen 12% over two quarters while the company's average selling price has been flat; consensus models flat gross margin for the next year.

Model answer. The market appears to be under-modelling gross-margin expansion: with input costs down 12% and pricing held flat rather than passed through to consumers, gross margin should expand roughly in line with the input-cost decline, which consensus's flat-margin assumption doesn't capture — a differentiated, testable view (falsified if the next two quarters show gross margin flat or down despite the input-cost tailwind, which would suggest the company is instead cutting price to defend volume/share).

Q7. Why is "monitor and update" a formal step in the process, not just an afterthought?

Model answer. A thesis is a hypothesis, not a fact — new quarterly results, management commentary, and competitor moves are continuous tests of it. An analyst who publishes a rating and doesn't systematically re-check it against new data is not doing research, just issuing a one-time opinion; the discipline of monitoring against the pre-stated falsification criteria (Q4) is what makes ongoing coverage credible.

Q8. What's the difference between "informational edge" and insider trading, precisely?

Model answer. Informational edge from *legally obtained, non-material, or already-public-but-underappreciated* information (e.g. noticing a disclosed but overlooked footnote, or aggregating many small legal public data points into an insight nobody else has bothered to compile) is legitimate. The line is crossed the moment the information is both **material** (would move the stock price if disclosed) and **non-public** (not available to the market generally) — using or passing on information meeting both criteria is insider trading regardless of intent or how it was obtained.

Fundamental Analysis: Top-Down & Bottom-Up

The Problem / Why this matters

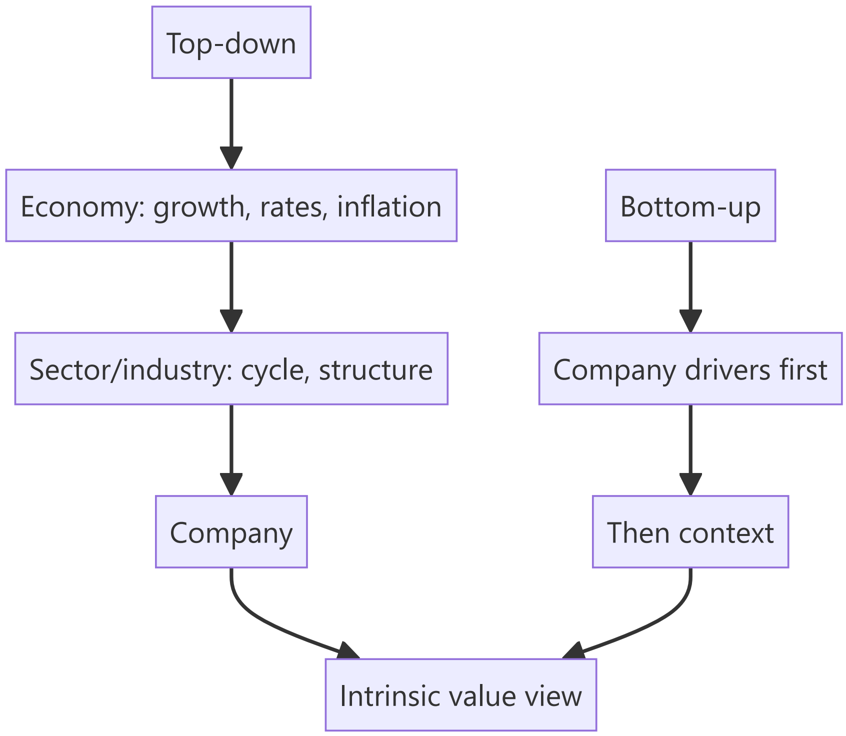
To decide whether a stock is cheap or dear, you must first understand the *fundamentals* — the real economic drivers of the business and its environment. **Fundamental analysis** is the discipline of assessing a company's intrinsic worth from the economy, industry and company up (or down). It's the core skill of equity research and long-term investing, and interviewers probe whether you can structure an analysis rather than just quote a P/E.

Core Idea

Fundamental analysis estimates a company's intrinsic value from its underlying economics. It's done **top-down** (economy → sector → company) and **bottom-up** (company-specific analysis first), usually combining both: understand the macro and industry context, then dig into the specific company's drivers, quality and valuation.

Why it works this way

A company's results are shaped by forces at three levels — the macro economy (growth, rates, inflation), the industry (structure, cycle, competition), and the company itself (strategy, execution, moat). Ignoring any level gives a wrong answer: a great company in a collapsing industry, or a mediocre one riding a boom, will disappoint if you only looked at one layer.



Full technical content

Top-down analysis (economy → sector → stock): 1. **Macro** — GDP growth, interest rates, inflation, currency, policy. Sets the tailwind or headwind (e.g., rate cuts favour rate-sensitives). 2. **Sector/industry**

— pick sectors positioned to benefit; assess industry structure and cycle. 3. **Stock** — pick the best companies within the chosen sectors.

Bottom-up analysis (company first): start with the individual company's fundamentals — competitive position, financials, management, valuation — largely regardless of the macro call, on the belief that a great business at a fair price wins over time. Most research combines both.

Analysing the company — the toolkit:

DIMENSION	WHAT TO ASSESS
Revenue model & unit economics	How it makes money; per-unit profitability
Competitive position / moat	Brand, cost, switching costs, network, scale, IP (Porter's five forces)
Growth drivers	Volume, price, mix, new products, markets
Margins & operating leverage	Profitability trajectory and its drivers
Returns	ROE, ROIC vs cost of capital (value creation)
Balance sheet & cash flow	Leverage, working capital, FCF generation
Management & governance	Track record, capital allocation, alignment
Valuation	Is the price sensible vs intrinsic value?

Qualitative + quantitative. The numbers (financials, ratios) show *what* happened; the qualitative work (moat, management, industry) explains *why* and whether it's sustainable. Great analysis fuses both.

Growth vs value styles. *Value* investors seek cheap stocks relative to fundamentals; *growth* investors pay up for superior growth. Both are fundamental — they weight the drivers differently.

Catalysts. Fundamental value only matters to a trade if something will make the market recognize it — new products, margin inflection, deleveraging, management change, sector re-rating. Identifying catalysts turns a valuation gap into an actionable idea.

Worked examples

Example 1 — top-down chain. Macro: rates are falling and consumption is recovering → favour consumer discretionary. Sector: within it, organized retail is gaining share from unorganized. Stock: pick the best-positioned retailer with a scale/cost moat and clean balance sheet. The macro call, sector selection, and stock pick compound.

Example 2 — bottom-up quality. An analyst ignores the noisy macro and focuses on a company with a durable brand moat, 25% ROIC (well above its ~11% WACC → strong value creation), consistent FCF, and disciplined capital allocation. Even through cycles, a business earning far above its cost of capital compounds value — a classic bottom-up buy.

Example 3 — good company, bad price. A wonderful franchise (high ROIC, wide moat) trades at 60× earnings, pricing in flawless growth for a decade. Fundamentally excellent, but the *price* already reflects it — the fundamental analyst concludes the stock, not the business, is the problem, and passes. Quality ≠ a buy at any price.

How it is tested in interviews

- **"What is fundamental analysis?"** — "Estimating a company's intrinsic value from its underlying economics — the macro, the industry, and the company's own drivers, quality and valuation."

- **"Top-down vs bottom-up?"** — "Top-down starts from the economy and sector and drills to the stock; bottom-up starts with the company's fundamentals. Most research combines both."
- **"How would you analyse a company's competitive position?"** — Porter's five forces / moat sources (brand, cost, switching costs, network, scale, IP) and returns vs cost of capital.
- **"How do you know a company creates value?"** — "ROIC above WACC — it earns more than the cost of the capital it uses."
- **"Why do you need a catalyst?"** — "A valuation gap only pays off if something makes the market recognize it — new products, a margin inflection, deleveraging, a re-rating."

Traps & common mistakes

- Doing only **one level** (company without industry, or macro without the company).
- Confusing a **great business** with a **great stock** — price matters.
- All numbers, no **qualitative** judgment (moat, management, sustainability).
- Ignoring **ROIC vs WACC** — growth without returns above cost of capital destroys value.
- No **catalyst** — a permanently "cheap" stock can stay cheap.

First-principles recap

- Fundamental analysis derives intrinsic value from economy, industry and company.
- **Top-down** (macro → sector → stock) and **bottom-up** (company first) — usually both.
- Fuse quantitative (financials, ROIC) with qualitative (moat, management).
- Value creation = **ROIC > WACC**; a great business isn't a buy at any price.
- Catalysts turn a valuation gap into an actionable idea.

Quick-reference

LEVEL	FOCUS
Macro	Growth, rates, inflation, policy
Sector	Structure, cycle, competition
Company	Moat, drivers, margins, returns, management
Value creation	ROIC > WACC
Catalyst	What makes the market re-rate

Q&A — Fundamental Analysis: Top-Down & Bottom-Up

Theory and worked scenarios on structuring a company/industry/macro analysis.

Q1. Walk through the top-down analysis chain with a concrete example.

Model answer. Start at the macro level (e.g. interest rates are falling and consumption is recovering, favouring rate-sensitive and discretionary sectors), narrow to sector/industry (within consumer discretionary, organised retail is gaining structural share from unorganised players), then to the individual stock (pick the best-positioned retailer within that favoured sector — strongest balance sheet, clearest cost/scale moat). Each layer narrows the opportunity set, and the three calls (macro, sector, stock) compound into the final investment decision.

Q2. How does bottom-up analysis differ philosophically from top-down, and why might a bottom-up investor deliberately ignore a bearish macro call?

Model answer. Bottom-up starts with the individual company's own fundamentals — competitive position, unit economics, management quality, valuation — largely independent of the macro backdrop, on the philosophy that a truly great business bought at a fair price will compound value over time regardless of near-term macro noise, and that macro forecasting itself is notoriously unreliable. A bottom-up investor might deliberately hold a high-quality compounder through a macro downturn they can't predict or time well, betting that business quality matters more over a multi-year holding period than getting the macro call right.

Q3. Worked — is this company creating or destroying value?

Company A has ROIC of 9% and WACC of 11%. Company B has ROIC of 18% and WACC of 12%. Both are growing revenue at 15% per year.

Model answer. Company A is *destroying* value despite growing: it earns 9% on invested capital while that capital costs 11%, so every rupee of growth capital deployed actually erodes value — a classic trap of "growth" that looks impressive on the income statement but destroys shareholder value, since growth in this case simply means investing more capital at a below-cost-of-capital return. Company B is creating substantial value: 18% ROIC against a 12% WACC means each rupee of growth capital earns a 6-point spread above its cost — this is genuine, value-accretive growth. The lesson: revenue growth alone says nothing about value creation; it must be paired with ROIC vs WACC.

Q4. What's the difference between a "great business" and a "great stock," and why does this distinction matter for a recommendation?

Model answer. A great business has strong fundamentals — high and durable ROIC, a real competitive moat, good management. A great stock additionally requires that the *price* doesn't already fully (or more than fully) reflect those strengths. A wonderful business trading at 60x earnings, pricing in a decade of flawless execution, can be a poor investment from the current price even though the underlying business is excellent — the fundamental analyst's job is to separate "is this a good company" from "is this a good price to buy it at," and a recommendation must answer the second question, not just the first.

Q5. Name three sources of competitive moat and give a brief example of each.

Model answer. Cost advantage — a company can produce/deliver at structurally lower cost than competitors (e.g. a scale-driven low-cost manufacturer), letting it either underprice rivals or earn superior margins at the same price. Switching costs — customers face real friction (financial, operational, or data/integration lock-in) in moving to a competitor, common in enterprise software and banking relationships. Network effects — the product/service becomes more valuable as more users join (a payments platform, a marketplace), creating a self-reinforcing advantage that's difficult for a smaller competitor to replicate even with a technically comparable product.

Q6. Why is a "catalyst" necessary even when an analyst has correctly identified that a stock is undervalued?

Model answer. An undervalued stock can, in principle, stay undervalued indefinitely if nothing prompts the broader market to re-examine and re-price it — value alone isn't self-executing. A catalyst (an earnings beat that reveals a hidden margin trend, a management change, a sector re-rating event, a new product launch) is the specific mechanism by which the market's attention is drawn to the mispricing, converting a "this is cheap" observation into an actionable, timed investment thesis. Without an identifiable catalyst, a valuation gap is a real but potentially open-ended and hard-to-time bet.

Q7. A company shows strong revenue growth and margin expansion. What qualitative checks would you still want to run before concluding it's a genuinely strong fundamental story?

Model answer. Check whether growth is broad-based (organic, across products/geographies) or narrowly driven by a single large customer/contract that could be lost; check whether margin expansion is structural (from genuine operating leverage or cost efficiency) or a one-off (a temporary input-cost tailwind, a change in accounting treatment, deferred/cut discretionary spending that will need to resume); assess management's capital-allocation track record (are they reinvesting at good returns, or empire-building); and assess whether the competitive position is actually strengthening or whether the current results reflect a favourable but temporary industry cycle rather than durable company-specific advantage.

Q8. How would you structure an answer to "walk me through how you'd analyse an unfamiliar company in 30 minutes"?

Model answer. State the structure explicitly before diving in: (1) understand the revenue model and unit economics — how does it actually make money; (2) assess the competitive position/moat and industry structure; (3) scan the financial trend — revenue growth, margin trajectory, ROIC vs WACC, balance-sheet health, free-cash-flow generation; (4) form a quick view on management/governance quality from what's available (capital-allocation history, related-party transactions if flagged); (5) sanity-check the current valuation against the quality/growth profile just assessed. Explicitly noting the structure up front, even under time pressure, signals a repeatable process rather than an ad hoc read of whatever numbers happen to be in front of you.

Three-Statement Modeling for Research

The Problem / Why this matters

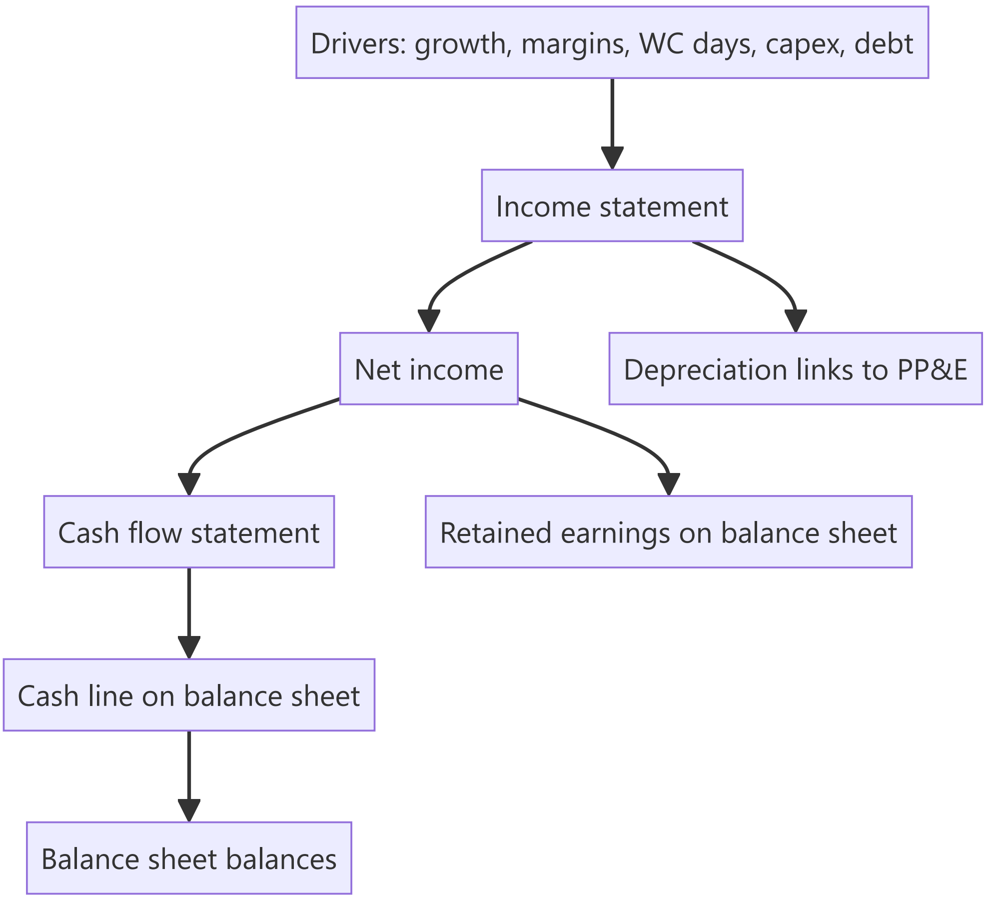
A research view needs numbers: a forecast of what the company will earn and how much cash it will generate. The **three-statement model** — a linked income statement, balance sheet and cash flow driven off a few assumptions — is the workhorse that produces those numbers and feeds valuation. Building and defending one is a core equity-research and IB skill, and "walk me through a 3-statement model" is a standard interview question.

Core Idea

A three-statement model projects the **income statement, balance sheet and cash flow statement**, fully linked so they stay internally consistent, from a set of **operating drivers** (revenue growth, margins, working-capital days, capex, financing). Change an assumption and the whole model — including free cash flow — updates coherently.

Why it works this way

The three statements are connected in reality (net income flows to retained earnings and to cash flow; depreciation links the P&L and the asset base; debt links financing and interest), so a credible forecast must respect those links. Driving everything off a small set of assumptions makes the model transparent, defensible, and easy to sensitize.



Full technical content

Build order (typical): 1. **Revenue** — from drivers (volume × price, or growth rate; segment build). 2. **Costs & margins** — COGS and opex as % of revenue or per-unit; get to EBITDA and EBIT. 3. **Supporting schedules** — depreciation (off PP&E + capex), working capital (off days: DSO, DIO, DPO), debt & interest, equity. 4. **Complete the income statement** — interest (from the debt schedule), taxes, net income. 5. **Build the cash flow** — start from net income, add back D&A, adjust for working-capital changes (from the WC schedule), subtract capex, add/subtract financing. 6. **Complete the balance sheet** — cash (from CF), PP&E (prior + capex − depreciation), working-capital items, debt, retained earnings (prior + NI − dividends). 7. **Check it balances** — assets = liabilities + equity every year; cash ties to the cash-flow statement.

The links that make it consistent: - Net income → retained earnings (BS) and top of cash flow (CF). - Depreciation → reduces PP&E (BS), non-cash add-back (CF), expense (IS). - Capex → increases PP&E (BS), investing outflow (CF). - Working-capital changes → BS balances and CF operating adjustment. - Debt → BS balance, interest on the IS, financing flows on the CF.

Circularity. Interest depends on debt, debt depends on cash, cash depends on interest — a circular reference. Handled with **iterative calculation** (Excel's circular switch) or a **circularity breaker** (average debt / copy-paste). A classic modeling interview question.

Best practices: separate **inputs** (assumptions) from **calculations**; drive off a few clear drivers; keep formulas consistent across columns; build **checks** (balance-sheet balances, cash ties to CF); make scenarios toggle-able; label units. A clean, auditable model beats a clever one.

From model to valuation: the model outputs the **free cash flows** (FCFF/FCFE) that the DCF discounts and the **metrics** (EBITDA, EPS) that multiples apply to — so the model is the engine behind the valuation.

Worked examples

Example 1 — driver-based revenue. A retailer: stores 100 growing to 120, revenue/store ₹5 cr growing 4%/yr. Year-1 revenue = 100 × 5 = ₹500 cr; Year-2 = 110 × 5.2 ≈ ₹572 cr. Costs at 85% of revenue → EBITDA 15% margin. The forecast flows from store count and sales density, not a guessed growth number — defensible and sensitizable.

Example 2 — the balancing check. Net income ₹70 cr, dividends ₹20 cr → retained earnings up ₹50 cr. Cash flow shows net change in cash +₹30 cr. On the balance sheet, cash rises ₹30 cr and retained earnings rise ₹50 cr; other moves (PP&E, debt, working capital) must net so that assets = liabilities + equity. If it doesn't balance, there's a linking error — the check catches it.

Example 3 — the circularity. Adding a revolver whose interest depends on the average debt balance creates a circular reference (interest ↔ debt ↔ cash). Turning on iterative calculation lets Excel converge; alternatively, compute interest on opening (not average) debt to break the loop. Interviewers love this: "where's the circular reference in a 3-statement model?"

Example 4 — full working-capital schedule, driven off days. A company's model needs a working-capital forecast, not a guessed flat number. Given Year-1 revenue ₹800 cr, COGS ₹520 cr, and target days of DSO (Days Sales Outstanding) 45, DIO (Days Inventory Outstanding) 60, DPO (Days Payable Outstanding) 40:

```
Receivables = Revenue × DSO/365 = 800 × 45/365 ≈ ₹98.6 cr
Inventory   = COGS × DIO/365    = 520 × 60/365 ≈ ₹85.5 cr
Payables    = COGS × DPO/365    = 520 × 40/365 ≈ ₹57.0 cr
Net Working Capital = Receivables + Inventory − Payables = 98.6 + 85.5 − 57.0 = ₹127.1 cr
```

If Year-2 revenue grows to ₹880 cr and COGS to ₹560 cr with the *same* days assumptions, NWC becomes 880×45/365 + 560×60/365 − 560×40/365 ≈ 108.5 + 92.1 − 61.4 = ₹139.2 cr — an increase of ₹12.1 cr, which is the **ΔNWC** that flows as a cash *outflow* in the FCFF build (Valuation chapters). This is the concrete mechanism behind the abstract statement "growth costs cash": every rupee of extra revenue tied up in unpaid receivables and unsold inventory (net of the payables the company itself hasn't paid yet) is cash the business doesn't get to keep, and driving it off days — not a flat guess — is what makes the forecast defensible when a PM asks "why did you assume that number."

Example 5 — sanity-checking a model's implied ratios before presenting it. A junior analyst's 5-year forecast shows revenue growing 18% annually while receivable days *fall* from 45 to 30 over the same period, with no stated operational change (no new collections process, no shift in customer mix) driving that improvement. A reviewer should flag this immediately: days ratios don't improve by themselves, and an unexplained, favourable drift in a working-capital assumption is one of the most common ways an overly optimistic forecast quietly inflates projected free cash flow — checking that every year-over-year change in a driver assumption has a stated, defensible reason (not just "the formula extrapolated it") is a core model-review discipline, not a nice-to-have.

How it is tested in interviews

- **"Walk me through a three-statement model."** — Build revenue and costs → schedules (depreciation, working capital, debt) → complete the income statement → cash flow from net income → balance sheet → check it balances. Emphasize the *links*.
- **"How are the three statements linked?"** — Net income → retained earnings and CF; depreciation → PP&E, IS and CF; capex → PP&E and CF; debt → BS, interest and financing.
- **"Where is the circular reference?"** — "Interest ↔ debt ↔ cash. Solved with iterative calculation or by using opening-debt interest."
- **"What checks do you build?"** — "The balance sheet must balance every year, and cash on the balance sheet must equal the cash-flow statement's ending cash."

Traps & common mistakes

- Hard-coding numbers instead of driving off **assumptions**.
- Forgetting a **link** (e.g., depreciation not flowing to PP&E) so the model doesn't balance.
- Not handling **circularity** (or leaving it broken).
- No **checks** — errors go undetected.
- Over-complex models no one (including you) can audit.

First-principles recap

- A 3-statement model links IS, BS and CF, driven off a few operating assumptions.
- Build revenue → costs → schedules → complete IS → CF → BS → **check it balances**.
- Net income, depreciation, capex, working capital and debt are the key links.
- Handle **circularity** (interest ↔ debt ↔ cash) with iteration or a breaker.

- The model outputs the FCF and metrics that feed valuation.

Quick-reference

LINK	FROM → TO
Net income	IS → retained earnings (BS) & CF
Depreciation	IS expense, CF add-back, PP&E ↓
Capex	CF outflow, PP&E ↑
Working capital	CF adjustment, BS items
Debt	BS, interest (IS), financing (CF)
Check	BS balances; CF cash = BS cash

Q&A — Three-Statement Modeling for Research

Theory and worked scenarios on building and troubleshooting a linked financial model.

Q1. Walk me through building a three-statement model from scratch.

Model answer. Start with revenue, driven by explicit operating drivers (volume × price, or a growth rate applied to a segment build) rather than a guessed top-line number. Build costs and margins off revenue to reach EBITDA and EBIT. Build supporting schedules — depreciation (linked to the PP&E and capex build), working capital (driven by DSO/DIO/DPO days), and a debt schedule (linked to financing needs and interest). Complete the income statement with interest (from the debt schedule) and taxes to reach net income. Build the cash flow statement starting from net income, adding back non-cash D&A, adjusting for working-capital changes, subtracting capex, and reflecting financing flows. Complete the balance sheet using the cash flow's ending cash, the PP&E roll-forward, working-capital balances, debt, and retained earnings (prior balance plus net income minus dividends). Finally, check that assets equal liabilities plus equity every single year — if it doesn't balance, there is a linking error somewhere.

Q2. Where exactly does circularity arise in a three-statement model, and name two ways to handle it.

Model answer. Circularity arises because interest expense depends on the debt balance, the debt balance depends on the cash flow (which determines whether the company draws on or pays down a revolver), and cash flow itself depends on net income, which depends on interest expense — a closed loop. Two standard fixes: (1) enable iterative calculation (Excel's circular-reference setting), letting the model converge numerically over repeated recalculation passes; (2) break the circularity structurally by computing interest on the *opening* (prior period) debt balance rather than the average balance, which removes the current period's own cash flow from the interest calculation entirely.

Q3. Worked — trace how a ₹50 cr increase in capex flows through all three statements.

A company increases its planned capex by ₹50 cr in Year 1 versus the prior forecast, funded by drawing on its revolver (debt). Tax rate 25%, no immediate change to depreciation policy in Year 1 itself.

Model answer. Income statement: no immediate Year-1 P&L impact from the capex itself (capex is capitalised, not expensed) — though the higher PP&E base will increase future years' depreciation expense. Balance sheet: PP&E rises by ₹50 cr (gross additions); debt rises by ₹50 cr (the revolver draw funding it) — assuming no cash was used, cash is unaffected by the capex itself in this financing structure. Cash flow statement: investing activities show a ₹50 cr outflow (capex); financing activities show a ₹50 cr inflow (the revolver draw) — the two roughly offset in the net-change-in-cash line, but investing and financing cash flow, viewed separately, both move by ₹50 cr in opposite directions. The check: PP&E is up ₹50 cr and debt is up ₹50 cr on the balance sheet, matching the offsetting ₹50 cr investing outflow and ₹50 cr financing inflow on the cash flow statement — everything ties.

Q4. Why must depreciation be linked to both the PP&E schedule and the income statement, rather than being entered as a flat assumption in each place separately?

Model answer. If depreciation is entered independently in the income statement and in the PP&E roll-forward, the two numbers can drift out of sync (e.g. an analyst updates the capex assumption but forgets to update the resulting depreciation expense to match), silently breaking the model's internal consistency and causing the balance sheet to stop balancing without an obvious error message. Linking depreciation as a single calculated schedule feeding both the income statement (as an expense) and the PP&E roll-forward (as a reduction) is what makes the model a genuinely connected machine rather than three separately-typed documents that happen to sit next to each other.

Q5. What are the two balance-sheet-integrity checks every well-built model should include, and what does a failure of each one tell you?

Model answer. (1) Assets = Liabilities + Equity, checked every forecast year — a failure means there's a broken link somewhere in how an item flows from one statement to another (a classic culprit: net income not fully flowing to retained earnings, or a working-capital change not correctly reflected on both the cash flow and balance sheet). (2) Cash on the balance sheet equals the ending cash balance from the cash flow statement — a failure specifically flags an error in the cash-flow build itself (a financing or investing line not correctly captured) even if the balance sheet appears superficially to balance via some other, unrelated plug.

Q6. Why is "driving off assumptions" (rather than hard-coding forecast numbers) considered a best practice, beyond just being tidier?

Model answer. A driver-based model (e.g. revenue = stores × sales-per-store, rather than a bare "revenue grows 12%" hard-coded number) is transparent about *why* the forecast is what it is, defensible in front of a portfolio manager or interviewer who will ask what's driving the number, and mechanically sensitisable — changing one assumption (e.g. sales-per-store growth from 4% to 6%) automatically ripples correctly through the whole model. A hard-coded forecast number hides the underlying logic, can't be meaningfully stress-tested, and is a strong signal in an interview or work-sample review that the candidate hasn't actually thought through the business drivers.

Q7. A three-statement model outputs FCFF for a DCF. What specific line items from the model does the analyst need to pull, and what must they be careful not to double-count?

Model answer. FCFF = EBIT × (1 − tax rate) + D&A − Capex − ΔNWC, so the analyst pulls EBIT (from the income statement), D&A (from the depreciation schedule), capex (from the investing section of the cash flow build), and the change in net working capital (from the working-capital schedule). The care point: EBIT must be taxed as if unlevered (ignoring the model's actual interest expense) specifically to avoid double-counting the interest tax shield, which is captured separately inside WACC when the DCF discounts these cash flows — pulling the model's actual (post-interest) net income and working backward incorrectly is a common error that inflates or corrupts the FCFF figure.

Q8. Interviewer asks: "Your model isn't balancing — where do you look first?" What's a structured troubleshooting approach?

Model answer. First check the most common culprits in order: (1) does net income correctly flow to both retained earnings on the balance sheet and the top of the cash flow statement — a mismatch here is the single most frequent error; (2) does the depreciation figure match between the income statement expense and the PP&E roll-forward reduction; (3) are all working-capital changes reflected consistently on both the cash flow statement's operating adjustments and the balance sheet's corresponding line items; (4) does the debt schedule's ending balance match what's shown on the balance sheet, and does the associated interest expense match what's in the income statement. Working through these links systematically, rather than randomly changing numbers until the model happens to balance, is both faster and signals real understanding of how the statements connect.

Applied Equity Valuation

The Problem / Why this matters

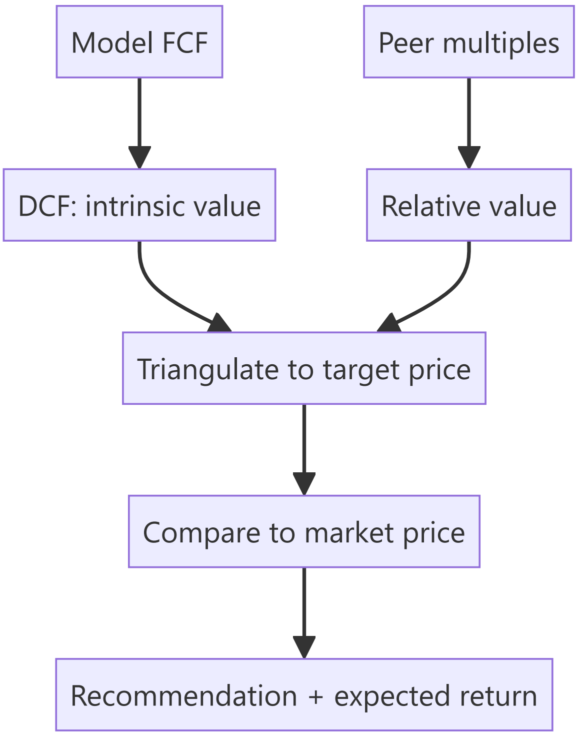
Research turns analysis into a **number** — an estimate of what a share is worth — so it can be compared to the market price. This chapter applies the valuation toolkit (covered deeply in the Valuation book) specifically to **equity research**: how analysts actually use DCF and multiples to set a target price and a recommendation. Interviewers test whether you can value a company end to end and defend the assumptions.

Core Idea

Applied equity valuation triangulates **intrinsic value** (DCF) and **relative value** (multiples vs peers) into a target price. The gap between that target and the market price — plus catalysts — drives the buy/sell/hold recommendation and the expected return.

Why it works this way

No single method is definitive: DCF is theoretically pure but assumption-sensitive; multiples reflect the market's current view but can be collectively wrong. Using both and cross-checking gives a value *range* you can defend, and forces you to understand *why* they differ.



Full technical content

DCF in research: - Project 5–10 years of **free cash flow** (FCFF) from the model. - Discount at **WACC**; add a **terminal value** (Gordon growth or exit multiple). - Sum to **enterprise value**; bridge to **equity value** (subtract net debt, minority interest; add associates); divide by diluted shares → **intrinsic price**. - Run **sensitivity** on WACC and terminal growth (the two swing factors) and present a range.

Relative valuation (multiples): - Pick the right multiple for the sector: **EV/EBITDA** (capital-structure neutral, most sectors), **P/E** (equity-level, mature/stable), **EV/Sales** (loss-making/high-growth), **P/B** (banks/financials), **PEG** (growth-adjusted). - Apply the peer multiple to the company's metric → implied value. - Adjust for differences in growth, margins, returns and risk vs peers.

Setting the target price. Weight the methods (e.g., 60% DCF, 40% comps), or use a football-field range and pick a point. The **12-month target price** implies an expected return vs the current price, which maps to the rating:

EXPECTED RETURN	TYPICAL RATING
> +15%	Buy / Overweight
−10% to +15%	Hold / Neutral
< −10%	Sell / Underweight

Bank/financials valuation uses P/B and ROE (excess-return / residual-income models), and DDM — since FCFF/EV multiples don't work for financials.

High-growth/loss-makers rely on revenue multiples, unit economics, and a path-to-profitability DCF with explicit scenarios.

Sanity checks: does the implied terminal multiple make sense? Is terminal growth below GDP? Does the target imply a reasonable forward P/E? Reconcile DCF and comps — a large gap needs an explanation.

Worked examples

Example 1 — DCF to target. FCFF grows to ₹120 cr in year 5; WACC 11%, terminal growth 4%. Terminal value = 120 × 1.04 / (0.11 − 0.04) = ₹1,783 cr, discounted back; sum of PV of FCFs + PV of TV = enterprise value ₹1,400 cr. Less net debt ₹200 cr = equity value ₹1,200 cr; ÷ 100 mn shares = **₹120/share** intrinsic. Market price ₹100 → ~20% upside → **Buy**.

Example 2 — comps cross-check. Peers trade at 12× EV/EBITDA. The company's EBITDA is ₹130 cr → implied EV ₹1,560 cr; less net debt ₹200 cr = equity ₹1,360 cr ÷ 100 mn = **₹136/share**. The DCF said ₹120, comps say ₹136 — a defensible range of ₹120–136 vs a ₹100 price; the analyst sets a target around ₹128 and rates it Buy.

Example 3 — why DCF and comps differ. DCF (₹120) is below comps (₹136) because the analyst's forecast is more conservative than the growth the market is pricing into peers. Stating *why* they differ — and which you trust — is the analytical value-add. If you think the market's peer growth is too optimistic, lean on the DCF.

How it is tested in interviews

- **"How would you value this company?"** — "DCF for intrinsic value — project FCFF, discount at WACC, add terminal value, bridge EV to equity, per share — cross-checked with peer multiples (EV/EBITDA, P/E), then triangulate to a target price and compare to the market."
- **"Which multiple would you use for a bank?"** — "P/B with ROE, or a residual-income/DDM model — EV/EBITDA and FCFF don't work for financials."
- **"Your DCF says 120, comps say 136 — what do you do?"** — "Present the range, explain the difference (my forecast is more conservative than the growth peers price in), and lean on the method whose assumptions I trust more."

- **"How does the target price map to a rating?"** — "The implied 12-month return vs the current price: strong upside → Buy, roughly fair → Hold, downside → Sell."

Traps & common mistakes

- Relying on **one method** — always triangulate DCF and comps.
- Using the **wrong multiple** for the sector (EV/EBITDA on a bank).
- Terminal value assumptions that imply **absurd** growth or multiples.
- A target price with no **sensitivity** or range.
- Not explaining **why** DCF and comps differ.

First-principles recap

- Applied valuation triangulates **DCF (intrinsic)** and **multiples (relative)** into a target price.
- Bridge DCF enterprise value to equity value to a per-share number.
- Pick the sector-appropriate multiple; financials use P/B/ROE and DDM.
- The target's implied return vs the price sets the **rating**.
- Explain and reconcile any gap between methods.

Quick-reference

METHOD	USE
DCF (FCFF/WACC/TV)	Intrinsic value
EV/EBITDA	Most sectors, capital-neutral
P/E	Mature/stable, equity-level
P/B + ROE / DDM	Banks & financials
EV/Sales	Loss-making/high-growth
Target → rating	Implied return vs price

Q&A — Applied Equity Valuation

Theory plus fully-solved numerical problems on turning a model into a target price and rating.

Q1. How would you value a company for a research recommendation?

Model answer. Triangulate two independent methods: a DCF (project FCFF, discount at WACC, add terminal value, bridge enterprise value to equity value, divide by diluted shares) for intrinsic value, and relative valuation (apply the appropriate peer multiple — EV/EBITDA, P/E, EV/Sales, or P/B depending on sector) for relative value. Weight or range the two into a target price, compare to the market price, and set a rating from the implied return.

Q2. Worked — DCF to target price.

FCFF in year 5 = ₹150 cr, WACC = 10.5%, terminal growth = 4%. Sum of PV of years 1-5 FCFF = ₹480 cr. Net debt = ₹250 cr. Diluted shares = 80 mn.

Model answer.

```
TV = 150 × 1.04 / (0.105 − 0.04) = 156 / 0.065 = ₹2,400 cr (undiscounted, at year 5)
PV(TV) = 2,400 / (1.105)^5 = 2,400 / 1.6763 ≈ ₹1,431.6 cr
Enterprise Value = 480 + 1,431.6 = ₹1,911.6 cr
Equity Value = 1,911.6 − 250 = ₹1,661.6 cr
Value per share = 1,661.6 / 80 = ₹20.77
```

If the market price is ₹17, upside ≈ (20.77–17)/17 ≈ 22% → a Buy on the standard >15% threshold.

Q3. Worked — reconciling DCF with a comps cross-check.

Continuing Q2: peers trade at 9.5× EV/EBITDA; the company's current-year EBITDA is ₹260 cr.

Model answer.

```
Implied EV = 260 × 9.5 = ₹2,470 cr
Implied Equity Value = 2,470 − 250 = ₹2,220 cr
Implied value per share = 2,220 / 80 = ₹27.75
```

DCF says ₹20.77, comps say ₹27.75 — a real gap. The analyst must explain it (e.g. "peers are pricing in faster medium-term growth than my base-case forecast assumes") and state which figure they trust more, rather than silently averaging the two or picking the more favourable number without justification.

Q4. Why doesn't EV/EBITDA work for valuing a bank?

Model answer. EV/EBITDA assumes debt is a financing choice separate from operations — but for a bank, deposits (debt-like liabilities) *are* the raw material of the business, and "EBITDA" is not economically meaningful when interest income/expense is the core operating activity, not a financing afterthought. Banks are instead valued on **P/B linked to ROE** (a bank earning above its cost of equity should trade above 1x book, and vice versa) or a **dividend-discount/residual-income model**.

Q5. Worked — P/B via ROE for a bank.

Bank's sustainable ROE = 15%, cost of equity = 12%, long-run growth = 8%.

Model answer. Using the Gordon-growth-derived P/B relationship:

```
P/B = (ROE − g) / (Ke − g) = (0.15 − 0.08) / (0.12 − 0.08) = 0.07 / 0.04 = 1.75x
```

If current book value per share is ₹200, the implied fair value is ₹200 × 1.75 = **₹350/share**. The intuition: a bank earning ROE above its cost of equity (15% > 12%) deserves to trade above 1x book, and the multiple compresses as the ROE-Ke spread narrows.

Q6. A high-growth, loss-making company can't be DCF'd on near-term FCFF — how do you value it?

Model answer. Use revenue multiples (EV/Sales) benchmarked to comparable growth-stage peers as the primary cross-check, alongside a longer-horizon DCF with an explicit, well-reasoned path to profitability (modelling the margin trajectory year by year rather than assuming near-term steady-state margins), and always stress-test the terminal-margin assumption specifically, since that single assumption typically drives most of the valuation for a pre-profit name.

Q7. What's wrong with presenting a single-point target price with no range?

Model answer. It implies false precision about inherently uncertain inputs (WACC, terminal growth, peer multiples all carry real estimation error) and hides how sensitive the conclusion is to assumptions at the edge of a reasonable range. A credible target price is always accompanied by a sensitivity table (e.g. WACC × terminal growth) or a football-field chart spanning the methods used, so a reader can see how much of the "Buy" case depends on optimistic inputs.

Q8. Your DCF and comps disagree by more than 10%. Is that a red flag on your work?

Model answer. Not automatically — the two methods answer different questions (DCF reflects the analyst's own forecast, comps reflect what the market is currently paying for similar growth/risk) and can legitimately diverge when the analyst's view differs from what's priced into peers. It becomes a red flag only if the analyst can't articulate *why* they diverge — an unexplained gap suggests an error somewhere (a modelling mistake, a wrong peer set, or an unrealistic terminal assumption) rather than a genuine, defensible difference of view.

The Research Note & Investment Thesis

The Problem / Why this matters

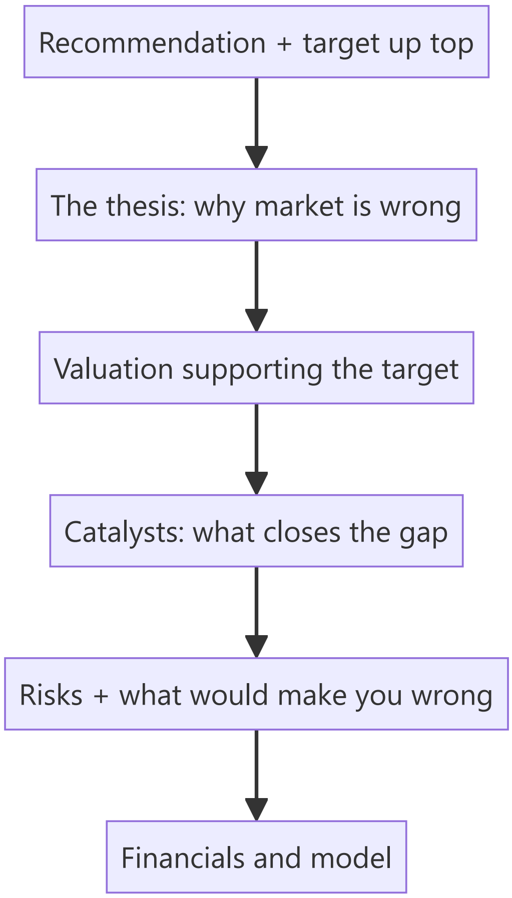
Analysis is worthless if it isn't communicated persuasively. The **research note** is the analyst's product — the written argument that a stock is mispriced, structured so a busy portfolio manager can grasp the thesis, valuation, catalysts and risks in minutes. Writing a tight note and articulating a crisp thesis is exactly what an equity-research interview tests (often via a written case or "pitch me a stock"). This chapter is the writing/communication counterpart to the analysis.

Core Idea

A research note communicates a **differentiated investment thesis** — why the market is wrong about this stock — supported by valuation, backed by catalysts, and honest about risks, ending in a clear **recommendation and target price**. Structure and clarity are as important as the analysis.

Why it works this way

PMs read dozens of notes; they act on the ones that are clear, differentiated, and defensible. A note that buries the thesis, agrees with consensus, or hides the risks gets ignored. So the note leads with the conclusion, states the *variant view*, quantifies the value, and pre-empts objections.



Full technical content

Types of note: initiation (full deep-dive launching coverage), **update/earnings note** (reaction to results or news), **thematic** (sector/idea). All share the core structure.

Anatomy of a research note: 1. **Recommendation & target** — rating (Buy/Hold/Sell), 12-month target price, and expected return — *at the top*. 2. **Investment thesis** — 3–5 crisp points on *why the market is wrong* and you're right (the variant view). 3. **Valuation** — the DCF and/or multiples supporting the target, with the key assumptions. 4. **Catalysts** — the specific events that will close the value gap and their timing. 5. **Risks** — the main downside risks and what would break the thesis. 6. **Business overview & financials** — the model, drivers, and key metrics.

The investment thesis — the heart. A great thesis is: - **Differentiated** — different from consensus (a variant perception); if you agree with the price, there's no trade. - **Specific & supported** — concrete drivers and numbers, not vague optimism. - **Falsifiable** — you can state exactly what would prove you wrong. - **Catalyst-driven** — a reason the market will come around.

The "variant view" test. Ask: *what do I believe that the market doesn't, and why am I right?* If you can't answer, you don't have a thesis — you have a description.

Writing craft: lead with the conclusion (BLUF — bottom line up front); be concise; use specific numbers; separate fact from opinion; make it skimmable (headers, bullets, a chart); and be intellectually honest about risks (it builds credibility).

Position on consensus. Note where you sit vs the Street (above/below consensus estimates and rating) — the value is in the *difference*, and being non-consensus and right is where alpha lives.

Worked examples

Example 1 — a crisp thesis (3 points). *Buy, target ₹1,200 (+20%).* Thesis: (1) The market underrates margin expansion from premiumization — we model 300 bps vs consensus 100 bps. (2) Rural distribution moat is widening, driving volume share gains consensus misses. (3) Valuation at 22× forward P/E is below the 5-year average despite better growth. Catalysts: next two quarters of margin beats. Risks: rural slowdown; raw-material inflation. *Differentiated, specific, catalyst-driven, falsifiable.*

Example 2 — description vs thesis. Weak note: "Company X is a good business with strong brands and steady growth; we rate it Buy." That's a *description* — it says nothing the market doesn't know and gives no reason the stock is mispriced. Strong note: "We are 15% above consensus on FY26 EPS because the Street hasn't modelled the new plant's margin uplift; at the current price that's not in the stock." The second has a variant view.

Example 3 — honest risk section building credibility. A Buy note explicitly says: "If Q1 volume growth comes in below 8% (vs our 15%), our margin thesis breaks and we would downgrade." Stating the falsification point makes the whole note more credible — the PM trusts an analyst who knows what would prove them wrong.

How it is tested in interviews

- **"Pitch me a stock."** — Lead with the recommendation and target, then 2–3 thesis points (your variant view), the valuation, catalysts, and risks — in that order, concisely.
- **"What makes a good research note?"** — "Conclusion up top, a differentiated and falsifiable thesis, valuation that supports the target, specific catalysts, and honest risks."
- **"What's an investment thesis?"** — "A variant view — what you believe that the market doesn't, and why you're right — with catalysts to close the gap."

- **"How is your view different from consensus?"** — Interviewers want to hear you position vs the Street; the value is in the difference.

Traps & common mistakes

- Writing a **description**, not a **thesis** (no variant view).
- Burying the **recommendation** instead of leading with it.
- Omitting **catalysts** (why does the gap close?) or **risks** (looks like cheerleading).
- Agreeing with **consensus/price** — no edge, no trade.
- Not stating what would **falsify** the thesis.

First-principles recap

- The research note is the analyst's product — clarity and structure matter as much as analysis.
- Lead with **recommendation and target**; then thesis, valuation, catalysts, risks.
- A thesis must be **differentiated, specific, falsifiable, and catalyst-driven**.
- Value lives in the **difference from consensus** — the variant view.
- Honest risk disclosure builds credibility.

Quick-reference

SECTION	CONTENT
Top	Rating + target + expected return
Thesis	Variant view (why market is wrong)
Valuation	DCF/multiples supporting target
Catalysts	Events that close the gap + timing
Risks	Downside + falsification point
Thesis test	What do I believe that the market doesn't?

Q&A — The Research Note & Investment Thesis

Theory and worked scenarios on writing a persuasive, defensible research note.

Q1. What does BLUF ("bottom line up front") mean in the context of a research note, and why do PMs specifically demand it?

Model answer. BLUF means the recommendation, target price, and expected return appear at the very top of the note, before the supporting analysis — not as a conclusion the reader has to dig for at the end. Portfolio managers read many notes under real time constraints and need to immediately know what action, if any, is being recommended; a note that builds up to its conclusion risks being abandoned before the PM reaches the actionable part, regardless of how good the underlying analysis is.

Q2. Rewrite this weak thesis statement into a strong one: "The company has good management and a strong brand, so we rate it a Buy."

Model answer. The weak version is a description with no variant view — "good management and a strong brand" are already widely known facts likely reflected in the current price. A strong version names a specific, differentiated view: e.g. "We are 12% above consensus FY26 EBITDA because the Street hasn't yet modelled the margin benefit from the new plant ramping to full utilisation by Q3 — at the current price, that upside isn't reflected." The rewrite adds a concrete, checkable claim about what the market is missing, not just a favourable general description of the company.

Q3. What's the difference between an initiation note and an earnings/update note, and how does the required depth differ?

Model answer. An initiation note is a full deep-dive launching formal coverage — it must establish the complete investment case from scratch (business overview, industry context, full model, valuation, thesis, catalysts, risks) since the reader may have no prior context. An earnings/update note reacts to new information (a results print, management guidance change, a competitive development) against an *existing* thesis — it's shorter and more focused, explicitly stating whether the new information confirms, strengthens, weakens, or breaks the standing thesis, rather than re-deriving the whole investment case each time.

Q4. What is the "variant view" test, and apply it to this scenario: an analyst's DCF and comps both land within 2% of the current market price.

Model answer. The variant view test asks: "what do I believe that the market doesn't, and why am I right?" If an analyst's valuation lands essentially at the market price with no differentiated view on any input, there is no thesis to publish as a rated call — agreeing with the market adds no information the market doesn't already have. In this scenario, the honest output is a Hold/Neutral rating (fairly valued, no edge identified) rather than manufacturing an artificial Buy or Sell case to have something to publish; a credible analyst is comfortable publishing "fairly valued, no strong view" when that's genuinely the finding.

Q5. Why does explicitly stating a falsification point ("if X happens, I'm wrong and will downgrade") make a research note *more* credible rather than less?

Model answer. It demonstrates the analyst has thought rigorously enough about their own thesis to identify what evidence would disprove it, rather than holding a view that can be rationalised indefinitely regardless of new data (a form of confirmation bias that erodes trust over time as a PM watches an analyst explain away every disappointing data point). A PM who sees a clear, specific falsification criterion knows exactly what to watch for and trusts that the analyst will act on that evidence rather than defend a stale call — this is why "what would make you wrong" is one of the most consistently asked follow-up questions after any pitch.

Q6. A research note states the company's own guidance without independent scrutiny, and rates the stock a Buy on that basis. What's the flaw in this approach?

Model answer. Restating management's own guidance and rating the stock accordingly is not independent research — it adds no analytical value beyond what the company itself has already disclosed, and it fails the variant-view test entirely (the market has access to the same guidance). A legitimate research note independently assesses whether that guidance is achievable, conservative, or aggressive (via the analyst's own model, channel checks, or industry comparison) and forms a view that may agree or disagree with management's framing — simply relaying guidance as if it were the analyst's own conclusion is a common junior-analyst failure mode.

Q7. How should a note position itself explicitly relative to Street consensus, and why does this matter for how the thesis is read?

Model answer. A well-written note states directly where its estimates and rating sit versus consensus (e.g. "our FY26 EPS is 8% above consensus; we are the only Buy among 12 covering analysts" or "we are inline with consensus estimates but see a re-rating catalyst the Street hasn't priced"). This matters because it immediately tells the reader where the actual edge (if any) lives — is it a numbers edge (different estimates), a multiple edge (different view on what multiple the market should apply), or a timing edge (agreeing with consensus fundamentals but seeing an unrecognised near-term catalyst) — each requires different supporting evidence and is tested differently by follow-up questions.

Q8. What should a "risks" section actually contain, and what's the difference between a well-written risks section and a legally-defensive boilerplate one?

Model answer. A well-written risks section names the specific, material risks to *this particular thesis* (e.g. "our margin-expansion call breaks if raw-material costs rise faster than the company can pass through in pricing"), ideally with a rough sense of magnitude or probability, and explicitly ties back to the falsification point from the thesis section. A boilerplate risks section lists generic, near-universal risks ("macroeconomic conditions may change," "competition may increase") that apply equally to almost any stock and provide no real information about what's specifically at stake in *this* call — the specificity of the risks section is itself a signal of how rigorously the analyst has actually stress-tested their own thesis.

The Stock Pitch

The Problem / Why this matters

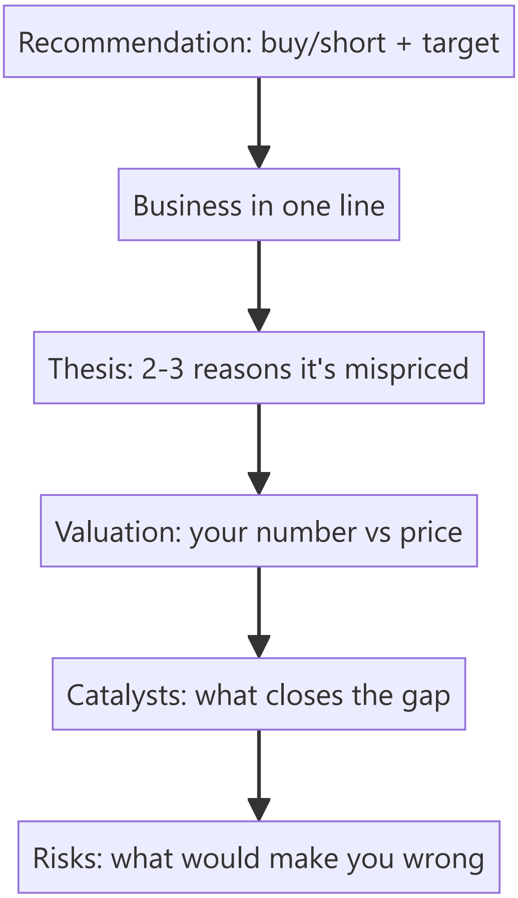
"Pitch me a stock" is the single most common question in equity research, buy-side, and markets interviews — and often the deciding one. It tests everything at once: whether you understand a business, can value it, have a differentiated view, know the catalysts and risks, and can communicate all of it in two minutes. A strong, structured pitch is a skill you can prepare and deliver on demand.

Core Idea

A stock pitch is a **concise, structured argument** for a specific action (buy or short) on a specific stock, delivered in a repeatable format: recommendation → business → thesis → valuation → catalysts → risks. It's the verbal version of the research note, compressed for a live conversation.

Why it works this way

The listener (interviewer or PM) wants to know quickly: *what's the trade, why is it mispriced, what's it worth, what makes it move, and what could go wrong*. Leading with the recommendation and following a clear skeleton signals that you think like an analyst and lets you cover everything without rambling.



Full technical content

The pitch skeleton (in order): 1. **Recommendation** — "I'd buy [X] with a 12-month target of ₹[Y], about [Z]% upside." (Or short.) Lead with it. 2. **Business** — one or two sentences: what the company does and how it makes money. 3. **Thesis** — 2–3 crisp reasons the market is mispricing it (your variant view). This is the core. 4. **Valuation** — your intrinsic value/target and how you got there (DCF and/or a multiple), vs the current price. 5. **Catalysts** — the specific events that will make the market recognize the value, with rough timing. 6. **Risks** — the main risks and what would prove you wrong (and ideally why the risk/reward is still favourable).

Delivery principles: - **Lead with the conclusion**; don't build up to it. - Be **specific** — real numbers (target, upside, key metric), not vague adjectives. - Have a **differentiated view** — say how you differ from consensus. - Keep it **~2 minutes**; then handle follow-up questions. - Show **balance** — acknowledging risks makes you credible.

Long vs short pitches. A **long** pitch argues undervaluation + positive catalysts. A **short** pitch argues overvaluation, deteriorating fundamentals, or a specific negative catalyst (accounting red flags, structural decline) — and must address the asymmetry (shorts have unlimited downside, borrow cost, and squeeze risk).

Preparation. Have **2–3 pitches ready** (ideally one long, one short), including at least one you genuinely follow. Know the numbers cold (valuation, key drivers) because the follow-ups probe them. Be ready for: "What's the market missing?", "What would make you wrong?", "What's it worth and why?", "What's the catalyst?"

Common follow-ups and how to handle them:

FOLLOW-UP	WHAT THEY'RE TESTING
"Why is it mispriced?"	Your variant view / edge
"What's your target and how?"	Valuation rigour
"What's the catalyst?"	Actionability
"What could go wrong?"	Intellectual honesty
"Why hasn't the market seen this?"	Depth of edge

The written initiation-of-coverage note — full structure

A pitch is the spoken compression of a much longer document: the **initiation of coverage (IoC)** note, typically 15-40 pages, that a sell-side analyst publishes when first taking up formal coverage of a stock. Knowing its structure cold is expected of anyone claiming equity-research experience. 1. **Cover page / summary box** — rating, target price, current price, upside/downside %, market cap, key financial snapshot (revenue, EBITDA margin, EPS, P/E) for the forecast years at a glance. 2. **Investment summary** (1 page) — the thesis compressed to 3-5 bullet points, written to be read standalone by a portfolio manager with 30 seconds. 3. **Business overview** — segments, revenue mix, geographic mix, competitive positioning, moat. 4. **Industry overview** — market size, growth drivers, competitive structure (Porter's Five Forces), regulatory backdrop. 5. **Financial analysis** — historical trend analysis (revenue, margins, returns on capital), quality-of-earnings checks (cash conversion, working-capital trends, one-off items flagged and excluded). 6. **Forecast and assumptions** — the driver-based model's key assumptions stated explicitly and defended (this is where a reader checks whether the analyst's numbers are grounded or arbitrary). 7. **Valuation** — DCF (with WACC/terminal-growth sensitivity table) and relative valuation (peer comp table), reconciled to a target price. 8. **Catalysts and risks** — a dedicated section, not an afterthought; risks should be specific and, where possible, quantified (e.g. "each 1% of rupee appreciation compresses EPS by ~0.6%"). 9. **Appendix** — detailed model outputs, historical financial statements, management background, ESG considerations if material.

Sensitivity table (a near-universal request): a DCF target price is never presented as a single number without a sensitivity grid — typically WACC (rows) × terminal growth rate (columns), showing how the target moves across a realistic range of each assumption (e.g. WACC 10.5%-12.5% in 50bp steps, terminal growth 4%-6% in 50bp steps) — this single table is often the first thing a skeptical PM asks to see, since it reveals how much of the "buy" case depends on optimistic assumptions at the edges of the grid versus holding up across the whole range.

Worked examples

Example 1 — a two-minute long pitch. "I'd buy [FMCG co], target ₹1,200, ~20% upside. It's a branded consumer-staples leader with a rural distribution moat. Thesis: the market underrates margin expansion from premiumization — I model 300 bps versus consensus 100 — and volume share gains from distribution the Street isn't crediting; at 22× forward it's below its historical average despite better growth. On a DCF and comps I get ₹1,200 versus ₹1,000 today. Catalysts: the next two quarters of margin beats. Risks: a rural slowdown or raw-material inflation — but at this valuation, I think the risk/reward is favourable." *Every element, two minutes.*

Example 2 — a short pitch. "I'd short [X], target 30% downside. It's a lender growing loans 40% a year while under-provisioning; my thesis is that credit costs are being deferred, not avoided — restated for realistic provisions, earnings are ~40% lower than reported. Valuation at 4× book assumes durable high ROE that I think is illusory. Catalyst: rising NPAs over the next two quarters and a likely provisioning reset. Risks: it can stay expensive and squeeze shorts, so I'd size it carefully and use a stop." *Addresses the short's asymmetry.*

Example 3 — handling "what would make you wrong?" For the FMCG long: "If the next two quarters show margin *contraction* instead of the expansion I expect — say raw-material inflation the company can't pass through — my core thesis breaks and I'd exit." A crisp falsification point signals a real analyst.

Example 4 — a banking-sector long pitch. "I'd buy [private bank], target ₹950, ~18% upside. It's a private-sector bank with a strong deposit franchise — CASA ratio above 45%, well ahead of peers. Thesis: the market is discounting it for near-term NIM compression from deposit re-pricing, but I think that's already 80% priced in at 2.3x book versus a 5-year average of 2.8x, and the market is underrating the credit-cost normalization as legacy stressed-asset provisions roll off over the next four quarters. On a Gordon-growth-based P/B framework (ROE 16%, cost of equity 13%, growth 12% → implied P/B ~2.8x) applied to next year's book value, I get ₹950. Catalysts: NIM bottoming and credit-cost guidance in the next two quarters' results. Risks: a systemic deposit-cost shock if rate cuts don't materialise as expected, and asset-quality slippage in the unsecured retail book — a segment I'm watching closely given sector-wide unsecured-lending stress." *Shows sector-specific valuation framework (P/B via ROE-CoE-growth, not DCF, which is standard for banks given the difficulty of forecasting bank free cash flow directly) and a sector-specific risk (unsecured retail asset quality).*

Example 5 — a technology-sector short pitch. "I'd short [mid-cap SaaS co], target 25% downside. It trades at 12x forward revenue on the strength of a headline 130% net-revenue-retention figure, but digging into the cohort disclosures, that NRR is concentrated in a handful of large accounts renewing at expanded contract value, while the broader customer base shows flat-to-declining logo retention — a classic 'blended metric hides a worsening core' pattern. My thesis: as the large-account expansion cycle normalises over the next 2-3 quarters, blended NRR converges toward the weaker underlying logo-retention trend, and the multiple re-rates toward peer SaaS names growing similarly but without this concentration risk (6-8x forward revenue). Catalyst: the next NRR print by customer-cohort-size disclosure (if the company provides it) or a deceleration in the headline number itself. Risks: SaaS multiples can stay elevated on momentum regardless of fundamentals for longer than a short position can be held comfortably, and a single large new logo win could reverse the narrative — sizing and stop-discipline matter more here than in the long book." *Demonstrates disaggregating a headline metric — the specific analytical move interviewers look for in a "find the flaw in this metric" style question.*

Sector-specific valuation frameworks — a quick reference

Interviewers routinely ask "how would you value a bank / an insurer / a commodity company differently from a normal company" — a DCF built on standard free cash flow to firm breaks down or requires heavy adaptation for several sectors: - **Banks/NBFCs**: valued primarily on **P/B (price-to-book) linked to ROE**, not DCF — a bank's "free cash flow" is not economically meaningful the way it is for an industrial company, since debt (deposits) is the raw material of the business, not just a financing choice. The Gordon-growth P/B framework (Example 4 above) is the standard tool. - **Insurance**: **Embedded Value (EV)** and **Value of New Business (VNB) multiples** are sector-specific metrics — EV captures the discounted value of in-force policies plus net worth, and the market typically prices life insurers on P/EV rather than P/E or P/B. - **Commodity/cyclical companies** (metals, cement, oil & gas): valued through-the-cycle, often on **normalised EV/EBITDA** (using a mid-cycle commodity-price assumption, not the current spot price, to avoid over/under-valuing at cycle extremes) rather than a single-year DCF that's hostage to where the commodity price happens to sit today. - **Real estate**: **Net Asset Value (NAV)** — summing the discounted value of each project/land parcel individually — is the standard framework, since a blended company-wide DCF obscures project-level economics that actually drive value. - **Early-stage/pre-profit tech**: revenue multiples (EV/Revenue) or, for genuinely forecastable unit economics, a DCF built on a longer explicit forecast period with explicit path-to-profitability assumptions, always paired with a clear statement of how sensitive the valuation is to the terminal-margin assumption specifically.

How it is tested in interviews

- **"Pitch me a stock."** — Deliver the skeleton: recommendation + target → business → 2–3 thesis points → valuation → catalysts → risks, in ~2 minutes.
- **"What's the market missing?"** — Your variant view — the single most important thing; have it sharp.
- **"What's it worth and how did you get there?"** — Give a target with a one-line valuation method; know the numbers.
- **"What would make you wrong?"** — A specific falsification point, delivered confidently.
- **"Long or short something you follow"** — Always have real pitches ready, not textbook examples.

Traps & common mistakes

- **Burying the recommendation** — lead with it.
- No **variant view** — just describing a "good company."
- **Vague** — no target, no numbers.
- Ignoring **risks** — sounds naive; acknowledging them builds credibility.
- Not knowing your own numbers when **probed**.
- For shorts, ignoring **borrow cost, squeeze, and unlimited downside**.

First-principles recap

- A pitch = recommendation → business → thesis → valuation → catalysts → risks, in ~2 minutes.

- **Lead with the conclusion**; be specific with numbers.
- The thesis is a **variant view** — what the market is missing.
- Name **catalysts** (actionability) and **risks/falsification** (credibility).
- Prepare 2–3 real pitches and know the numbers cold.

Quick-reference

ELEMENT	ONE LINE
Recommendation	Buy/short + target + upside
Business	What it does, how it earns
Thesis	2–3 reasons it's mispriced (variant view)
Valuation	Your number vs price + method
Catalysts	What closes the gap + when
Risks	Downside + what proves you wrong

Q&A — The Stock Pitch

Theory and scenario drills for the single most-asked live question in markets interviews.

Q1. "Pitch me a stock" — what's the structure, and how long should it take?

Model answer. Recommendation with target and upside first → one-line business description → 2-3 point thesis (the variant view) → valuation (your number vs the price, and how you got there) → catalysts (what closes the gap, roughly when) → risks (what would prove you wrong). Target roughly two minutes before handling follow-ups; lead with the conclusion, never build up to it.

Q2. What's the single most important element of a pitch, and why?

Model answer. The variant view — what the market is missing that you see. A pitch that just describes a good company with no view on *why the market is wrong* isn't research, it's a book report; the interviewer is specifically screening for whether the candidate can articulate a genuine, differentiated edge.

Q3. How does a short pitch differ structurally from a long pitch?

Model answer. The skeleton is the same, but a short must explicitly address the asymmetry of shorting: unlimited theoretical downside (a stock can rise indefinitely, unlike a long where the max loss is the investment), borrow cost, and squeeze risk — and should generally identify a specific negative catalyst (an accounting red flag, a structural decline, a credit-quality deterioration) rather than a vaguer "it's overvalued" argument, since "overvalued" alone can stay overvalued for a long time and cost a short position dearly while waiting.

Q4. Write a full ~90-second pitch for a hypothetical company from these facts: a listed QSR chain trading at 28x forward P/E, same-store sales growth decelerating from 12% to 6% over the last three quarters, but management guiding aggressive new-store additions (25% unit growth) to offset the deceleration.

Model answer. "I'd short [QSR Co], target 20% downside. It trades at 28x forward earnings — a premium multiple that assumes sustained high growth — but same-store sales growth has decelerated from 12% to 6% over three consecutive quarters, a trend management isn't addressing directly, instead leaning on 25% unit growth to mask the core deceleration in the headline numbers. My thesis: new-store economics take 12-18 months to mature, so aggressive unit growth in the near term will actually depress consolidated margins before it helps, and once same-store-sales deceleration becomes impossible to hide behind unit growth, the multiple should compress toward the 18-20x peer average for decelerating QSR names. Catalyst: same-store sales guidance or print showing continued deceleration in the next one to two quarters. Risks: a same-store-sales re-acceleration (a new product launch or marketing push could reverse the trend), and premium-multiple growth names can stay expensive on momentum longer than expected — I'd size and stop-manage this position carefully given that risk."

Q5. An interviewer asks "why hasn't the market already priced this in?" — what's a strong answer structure?

Model answer. Name a specific, credible reason the market is slow or wrong here — common legitimate reasons include: the signal is buried in a disclosure most analysts don't dig into (e.g. a footnote or a segment breakdown), the market is anchored on a backward-looking headline metric that masks the real trend (see Q4's blended same-store-sales/unit-growth example), the stock is under-covered (few analysts, low institutional ownership) so mispricing persists longer, or the market is systematically short-termist about a benefit that plays out over multiple quarters. A weak answer ("I just think I'm right and everyone else is wrong") signals overconfidence rather than a genuine edge.

Q6. What's the purpose of the sensitivity table in a written pitch/note, and what's the most common grid used?

Model answer. It shows how the target price moves across a realistic range of the most uncertain assumptions, so a reader can judge how much of the "Buy" case depends on optimistic inputs versus holding up broadly. The most common grid is WACC (rows) × terminal growth rate (columns) for a DCF-based target, typically spanning roughly ±1-2% around the base case in each dimension.

Q7. Give the sector-appropriate valuation approach for: (a) a bank, (b) a life insurer, (c) a commodity/cyclical company, (d) a real-estate developer.

Model answer. (a) Bank — P/B linked to ROE (Gordon-growth P/B relationship) or a residual-income/DDM model; standard FCFF/DCF breaks down since deposits are the business's raw material, not just financing. (b) Life insurer — Embedded Value (EV) and Value of New Business (VNB) multiples, typically priced on P/EV rather than P/E or P/B. (c) Commodity/cyclical — normalised EV/EBITDA using a mid-cycle commodity-price assumption rather than current spot, to avoid over/under-valuing at cycle extremes. (d) Real-estate developer — Net Asset Value (NAV), summing the discounted value of each project/land parcel individually rather than a single blended company-wide DCF.

Q8. What does "the risk/reward is still favourable" mean at the end of a pitch, and why include it?

Model answer. After naming the real risks (Q3, Q5), a strong pitch explicitly states why the expected value still favours the trade — e.g. "even if the risk materialises, the downside is roughly X% while the base case upside is Y%, an asymmetric setup" — because simply listing risks without this framing can make the pitch sound tentative; naming risks *and* explaining why they don't change the conclusion is what signals genuine conviction backed by analysis, rather than either naive overconfidence or unearned hedging.

Technical Analysis Essentials

The Problem / Why this matters

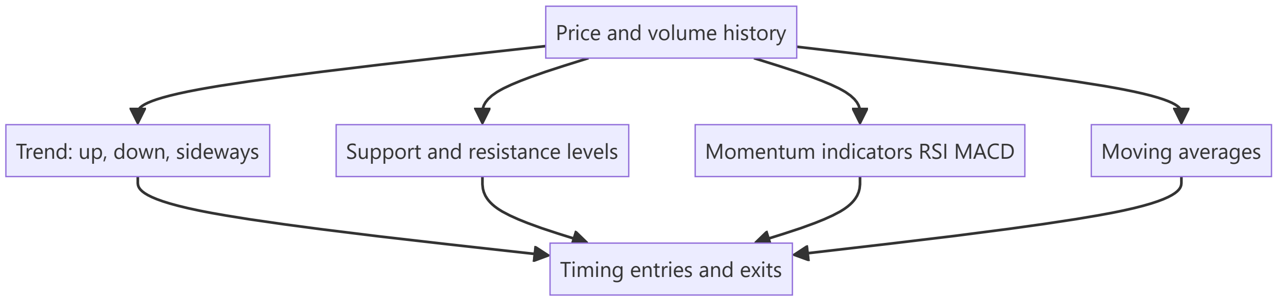
While fundamental analysis asks *what a stock is worth*, **technical analysis** asks *what the price is likely to do next*, from the price and volume history itself. It's the core toolkit for traders and technical research analysts, and it's used alongside fundamentals for timing entries and exits. Even fundamentally-driven roles expect you to understand trends, support/resistance, and the main indicators — and for trading/TRA roles it's central.

Core Idea

Technical analysis studies **price and volume patterns** to forecast future price movement, resting on three assumptions: the price discounts everything, prices move in trends, and history tends to repeat (because human behaviour does). Its tools identify trends, key price levels, momentum, and reversals.

Why it works this way

Price reflects the collective actions of all participants, so patterns in price can reveal the balance of supply and demand and the psychology (fear/greed) driving it. Trends persist because participants react and herd; levels matter because memory and orders cluster at them; patterns repeat because human behaviour under uncertainty is repetitive.



Full technical content

The three assumptions of technical analysis: 1. **The market discounts everything** — all known information is already in the price, so study the price. 2. **Prices move in trends** — once established, a trend is more likely to continue than reverse. 3. **History repeats** — patterns recur because market psychology is consistent over time.

Trend. The primary concept: an **uptrend** = higher highs and higher lows; a **downtrend** = lower highs and lower lows; **sideways/range** = no clear direction. "The trend is your friend" — trade with it until it clearly breaks. **Trendlines** connect the lows (uptrend) or highs (downtrend).

Support & resistance. Support = a price level where buying tends to emerge and halt declines; **resistance** = where selling tends to emerge and cap advances. A broken resistance often becomes new support (and vice versa). These levels are where traders place orders and stops.

Moving averages (MA). Smooth price to reveal the trend. Simple (SMA) or exponential (EMA). A price above its MA is bullish; below is bearish. Crossovers signal trend changes: a **golden cross** (50-day crosses above 200-day) is bullish; a **death cross** (50-day below 200-day) is bearish.

Momentum / oscillators:

INDICATOR	WHAT IT SHOWS
RSI (Relative Strength Index)	Momentum, 0–100; > 70 overbought, < 30 oversold; divergences signal reversals
MACD (Moving Average Convergence Divergence)	Trend + momentum via two EMAs and a signal line; crossovers and histogram
Stochastic	Momentum vs recent range
Bollinger Bands	Volatility bands around an MA

Volume confirms moves — a breakout on high volume is more reliable than on low volume. **Chart patterns** (head-and-shoulders, double top/bottom, triangles, flags) signal continuation or reversal.

Technical vs fundamental — complementary. Fundamentals answer *what* to buy (value); technicals answer *when* (timing). Many analysts use fundamentals for the thesis and technicals for entry/exit and risk (stops).

Worked examples

Example 1 — trend and trendline. A stock makes higher highs and higher lows for months (uptrend); a trendline drawn under the rising lows holds. A trader buys on pullbacks to the trendline with a stop just below it, riding the trend until the line breaks — a break signalling the uptrend may be over.

Example 2 — support turned resistance. A stock repeatedly bounces off ₹100 (support). It finally breaks below to ₹90. On the next rally, ₹100 now acts as *resistance* — sellers who were trapped exit there. The old floor becomes the new ceiling, a classic level flip.

Example 3 — RSI divergence. Price makes a new high but RSI makes a *lower* high (bearish divergence) — momentum is weakening even as price rises, warning of a possible reversal. Combined with resistance, it's a signal to tighten stops or take profit.

Example 4 — golden cross vs a false start. The 50-day MA crosses above the 200-day MA (a golden cross) after a long downtrend, a classically bullish signal. Two weeks later, price stalls and the 50-day MA turns back down without the 200-day MA reversing — a "false" golden cross that didn't lead to a sustained trend. A disciplined technical analyst treats the golden cross as raising the odds of a trend change, not guaranteeing one, and waits for price to also clear a nearby resistance level or hold above the crossed MAs for several sessions before committing full size — precisely the "probabilities, not certainties" discipline this chapter's traps section warns about.

Example 5 — combining fundamentals and technicals on one name. An equity analyst has a fundamental Buy thesis on a stock (undervalued on DCF and comps, per the Applied Equity Valuation chapter) but the stock is in a technical downtrend, trading below its 200-day MA with weak relative strength versus the sector. Rather than buying immediately on the fundamental case alone, the analyst waits for a technical confirmation — a break above the downtrend line with rising volume, or the stock reclaiming its 200-day MA — before initiating the position, using the technical setup purely for entry timing and initial stop placement (just below the reclaimed MA) while the fundamental thesis remains the reason for being long at all. This is the concrete version of "fundamentals for what, technicals for when" that interviewers ask candidates to articulate.

How it is tested in interviews

- **"What is technical analysis and its assumptions?"** — "Studying price and volume to forecast prices, on three assumptions: the price discounts everything, prices move in trends, and history repeats."

- **"Technical vs fundamental analysis?"** — "Fundamentals estimate intrinsic value (what to buy); technicals study price/volume for direction and timing (when). They're complementary."
- **"What is support and resistance?"** — "Levels where buying (support) or selling (resistance) tends to emerge; broken resistance often becomes support."
- **"What do RSI and MACD tell you?"** — "RSI is a 0–100 momentum gauge (>70 overbought, <30 oversold); MACD combines trend and momentum via EMA crossovers. Divergences warn of reversals."
- **"What's a golden cross?"** — "The 50-day MA crossing above the 200-day — a bullish trend signal; the reverse (death cross) is bearish."

Traps & common mistakes

- Treating technicals as **certainty** — they're probabilities, not guarantees.
- Ignoring **volume** confirmation of breakouts.
- Using indicators in **isolation** rather than confluence (trend + level + momentum).
- Fighting the **trend** ("catching a falling knife").
- Forgetting technicals are best **combined** with fundamentals and risk management (stops).

First-principles recap

- Technical analysis forecasts price from price/volume history.
- Three assumptions: price discounts everything, trends persist, history repeats.
- Core tools: **trend**, **support/resistance**, **moving averages**, **RSI/MACD**, **volume**.
- Golden/death crosses and divergences signal trend/momentum shifts.
- Best used **with** fundamentals — value for *what*, technicals for *when*.

Quick-reference

TOOL	SIGNAL
Trend	Higher highs/lows = up; trade with it
Support/resistance	Buy/sell levels; break flips role
Moving average	Above = bullish; golden/death cross
RSI	>70 overbought, <30 oversold, divergence
MACD	EMA crossover, momentum
Volume	Confirms breakouts

Q&A — Technical Analysis Essentials

Theory and worked scenarios on chart-based analysis fundamentals (see the much deeper Technical-Analysis-Complete volume elsewhere in this master for full chart-pattern depth).

Q1. State the three foundational assumptions of technical analysis, and briefly justify each.

Model answer. (1) The market discounts everything — all known information (fundamentals, sentiment, expectations) is already reflected in the current price, so studying price itself captures the net effect of all available information without needing to separately analyse each input. (2) Prices move in trends — once a directional move is established, continuation is statistically more likely than immediate reversal, because participants react to and reinforce established moves (momentum, herding). (3) History repeats — chart patterns recur because they reflect recurring patterns of crowd psychology (fear, greed, FOMO) under uncertainty, which doesn't fundamentally change across market cycles even as the specific assets and eras differ.

Q2. Worked — identify the signal from this scenario.

A stock has been in a clear uptrend for six months. It pulls back to its rising 50-day moving average three separate times, bouncing higher each time. On the fourth pullback, it closes decisively below the 50-day MA on above-average volume.

Model answer. The first three bounces off the rising 50-day MA are consistent with the "trend is your friend" principle — the moving average is acting as dynamic support in an established uptrend, and each successful bounce reinforces that the trend remains intact. The fourth event — a decisive close below the MA on above-average volume — is a meaningfully different signal: volume confirmation matters (a break on light volume is less reliable than one on heavy volume), so this combination suggests the uptrend's support has genuinely failed rather than being another buyable dip, warranting a reassessment of the trend rather than automatically buying the fourth pullback the way the first three were bought.

Q3. What's the difference between what RSI and MACD each measure, and can they disagree with each other?

Model answer. RSI (Relative Strength Index) is a bounded 0-100 momentum oscillator measuring the speed and magnitude of recent price changes, primarily used to flag overbought (>70) or oversold (<30) conditions and divergences. MACD (Moving Average Convergence Divergence) combines trend and momentum information via the relationship between two exponential moving averages and a signal line, primarily used for crossover and histogram signals. Yes, they can disagree — e.g. RSI can show "overbought" while MACD's crossover remains bullish, since RSI is more sensitive to short-term speed of movement while MACD reflects a somewhat longer-term trend-momentum blend; a technical analyst uses multiple indicators in confluence specifically because any single one can give a misleading signal in isolation.

Q4. Explain "support becomes resistance" (or vice versa) with the underlying behavioural logic, not just the rule.

Model answer. When a price level repeatedly holds as support, many market participants place buy orders near it and some who bought elsewhere set mental reference points around it; when the level finally breaks decisively, those who bought at or near the old support level are now underwater and often eager to sell "if I can just get back to even" — creating a cluster of latent sell orders exactly at the old support level, which is now approached from below on any subsequent rally. This behavioural mechanism (trapped buyers becoming eager sellers at their breakeven) is what turns a former support level into new resistance, not some mechanical rule independent of actual participant psychology.

Q5. How should technical and fundamental analysis be combined in a single research process, according to this chapter?

Model answer. Fundamental analysis answers "what to buy" (is the business undervalued, and why) while technical analysis answers "when to buy/sell" (is the current price action and level structure favourable for entry, and where should a stop be placed to manage risk if the thesis is wrong). A common integrated approach: use fundamental research to build the conviction list and set a target price/thesis, then use technical levels (support, trend, moving averages) to time the actual entry and to set objective, rules-based exit/stop points rather than relying purely on discretionary judgment about when to cut a losing position.

Q6. Why is technical analysis theoretically inconsistent with even the weak form of the Efficient Market Hypothesis, and how do practitioners typically respond to that critique?

Model answer. Weak-form EMH holds that all past price and volume information is already reflected in the current price, meaning patterns in historical price data (the entire basis of technical analysis) should carry no predictive power for future prices. Practitioners typically respond that markets are not perfectly efficient in practice — documented behavioural biases (herding, anchoring, disposition effect) create real, if imperfect and hard-to-time, patterns in how information gets incorporated into price, and that technical analysis is better understood as a probabilistic tool reflecting crowd psychology than as a claim that markets are informationally inefficient in a strict academic sense.

Q7. What's the danger of "using indicators in isolation," and how would a disciplined technical analyst avoid it in practice?

Model answer. Any single indicator can and does generate false signals regularly — an RSI reading "oversold" doesn't guarantee a bounce, a single moving-average crossover doesn't guarantee a sustained trend change. A disciplined analyst looks for confluence: multiple independent signals (e.g. a support level, a bullish RSI divergence, and a MACD crossover) aligning at the same point, which meaningfully raises the odds of a signal being genuine versus noise, rather than acting on any single indicator's reading in isolation.

Q8. A stock breaks out above a well-established resistance level, but on noticeably below-average volume. How should this affect confidence in the breakout?

Model answer. Volume is the confirmation mechanism for a breakout's reliability — a genuine breakout driven by real conviction typically attracts increased participation (higher volume) as more buyers step in and short-sellers cover. A breakout on light volume suggests a lack of broad conviction behind the move, raising the probability of a "false breakout" (the price re-entering the prior range shortly after) — a technically disciplined analyst would treat this breakout with reduced confidence and might wait for volume confirmation or a successful retest of the broken level before fully committing, rather than treating the price break alone as sufficient signal.

Market Efficiency & Behavioral Finance

The Problem / Why this matters

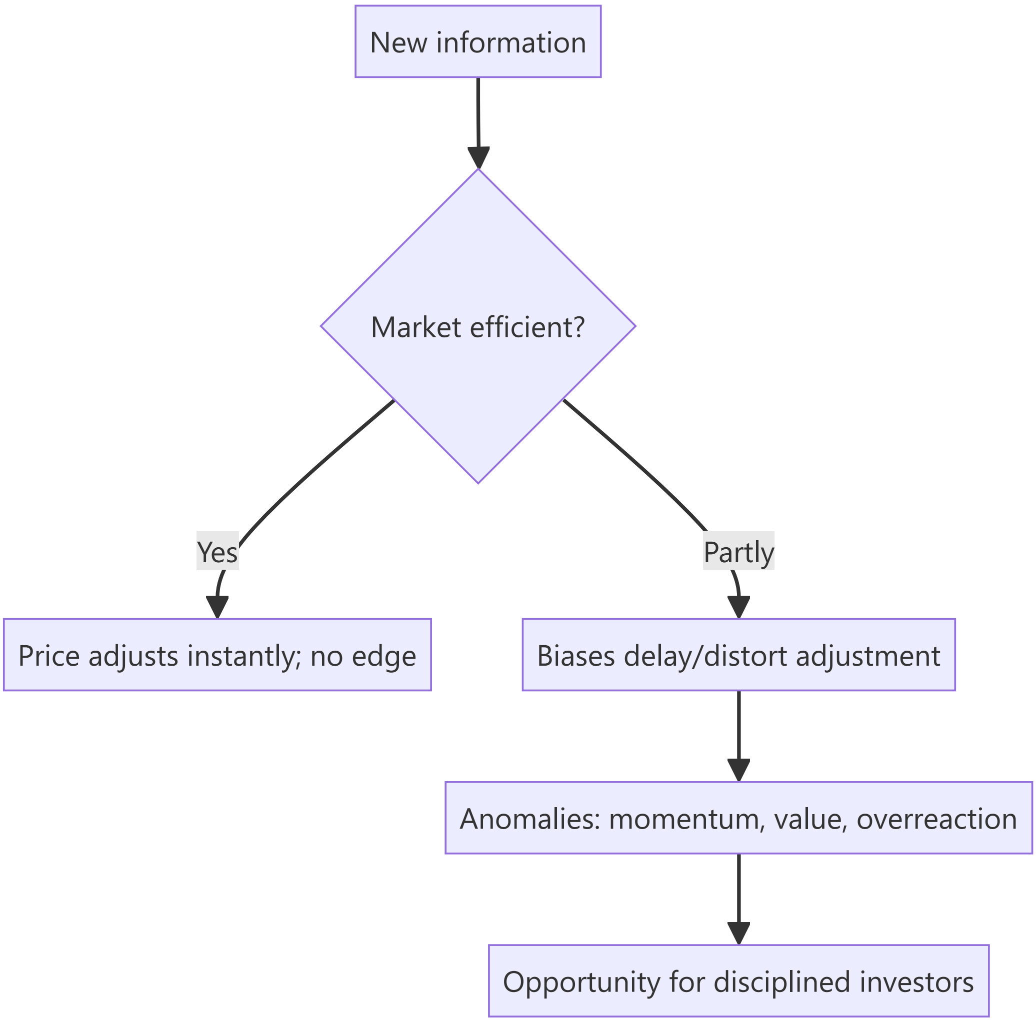
Can you beat the market? The answer shapes everything an investor does. The **Efficient Market Hypothesis (EMH)** says prices already reflect information, so consistently beating the market is nearly impossible — while **behavioral finance** documents the systematic biases that make markets *inefficient* enough to create opportunities. Every research and investing role sits in the tension between these two, and interviewers test whether you understand both.

Core Idea

The **EMH** holds that asset prices reflect available information, so you can't consistently earn risk-adjusted excess returns. **Behavioral finance** counters that investors are systematically irrational — subject to biases that push prices away from fundamentals — creating anomalies a disciplined investor can exploit. Reality lies between: markets are *mostly* efficient but not perfectly.

Why it works this way

If a piece of information predicted higher returns, rational investors would trade on it until the opportunity vanished — that's the efficiency argument. But real investors are human: they overreact, herd, anchor and fear losses, and those biases are *systematic* (not random), so they don't fully cancel out — leaving persistent, if hard-to-capture, mispricings.



Full technical content

The three forms of EMH:

FORM	PRICES REFLECT	IMPLICATION
Weak	All past prices/volume	Technical analysis shouldn't work
Semi-strong	All public information	Fundamental analysis on public data shouldn't give an edge
Strong	Even private/inside information	No one can beat the market, even insiders

Evidence: markets are *broadly* semi-strong efficient (prices react fast to news), but **anomalies** persist that challenge it.

Anomalies (evidence against strict EMH): - **Value effect** — cheap (low P/B, P/E) stocks historically outperform. - **Momentum** — recent winners keep winning over 3–12 months. - **Small-cap effect**, **low-volatility anomaly**, **post-earnings-announcement drift**, calendar effects. These underpin factor investing (Fama-French).

Behavioral biases (why markets misprice):

BIAS	EFFECT
Overconfidence	Excess trading, underestimating risk
Anchoring	Fixating on a reference price
Loss aversion	Losses hurt ~2× gains; holding losers, selling winners (disposition effect)
Herding	Following the crowd → bubbles and crashes
Recency bias	Overweighting recent events
Confirmation bias	Seeking evidence that fits your view
Mental accounting	Treating money differently by source/label
Overreaction / underreaction	To news → reversals and drift

Bubbles and crashes are the macro expression of these biases — herding and overconfidence inflate prices far above value, then fear collapses them.

Implications for investors: - If markets were perfectly efficient → **index passively** (no edge to be had). - Because they're not perfectly efficient → disciplined, unemotional investors can exploit others' biases (the behavioral edge), but it's hard and requires patience. - Know your *own* biases — the first job is not to be the sucker.

Worked examples

Example 1 — semi-strong efficiency in action. A company reports earnings 20% above expectations at 9:00 am; by 9:01 the stock has jumped to reflect it. A retail investor reading the news at noon can't profit from the surprise — it's already priced. That's semi-strong efficiency: public information is incorporated almost instantly.

Example 2 — loss aversion / disposition effect. An investor holds a stock down 30%, refusing to sell "until it comes back," while quickly booking a 10% gain elsewhere. Loss aversion makes them ride losers and cut winners — exactly backwards. Recognizing this bias (and using rules/stops) is the behavioral edge.

Example 3 — herding bubble. During a mania, prices detach from any fundamental value as investors buy because others are buying (herding) and extrapolate recent gains (recency). Fundamentals say "expensive," but the crowd pushes higher — until sentiment turns and it crashes. Bubbles are behavioral finance writ large.

Example 4 — testing weak-form efficiency with a simple rule. A researcher backtests a rule — "buy after two consecutive up-days, sell after two consecutive down-days" — on 10 years of Nifty 50 data and finds it produces a small, statistically insignificant edge after transaction costs, though a raw (pre-cost) backtest shows a modest positive return. This is a textbook weak-form-efficiency result: any exploitable pattern purely in past price sequences is either too small to survive real trading frictions (bid-ask spread, brokerage, slippage) or gets arbitraged away once enough participants exploit it — precisely why weak-form efficiency, though not perfect, holds up well empirically for simple, well-known price-pattern rules on liquid, widely-traded stocks.

Example 5 — anchoring in a real equity-research scenario. An analyst initiated coverage on a stock at ₹500 with a ₹650 target eighteen months ago. New information now suggests fair value is closer to ₹420, but the analyst's revised note sets the target at ₹520 — a number that "splits the difference" between the old target and the new fair-value estimate, rather than moving decisively to ₹420. This is anchoring: the analyst is unconsciously anchoring on their own prior published number rather than updating fully to the new evidence, a well-documented bias among professional forecasters, not just retail investors — one reason many research desks require an explicit, written justification whenever a target price revision is smaller than the underlying model's fair-value change would imply.

How it is tested in interviews

- **"What is the Efficient Market Hypothesis?"** — "Prices reflect available information, so you can't consistently earn risk-adjusted excess returns. Three forms: weak (past prices), semi-strong (public info), strong (private info)."
- **"If markets are semi-strong efficient, what stops working?"** — "Trading on public information/fundamental analysis of public data — it's already in the price. And technical analysis fails under even the weak form."
- **"Do you believe markets are efficient?"** — "Broadly semi-strong efficient — news is priced fast — but not perfectly; behavioral biases and documented anomalies (value, momentum) leave exploitable inefficiencies."
- **"Name a behavioral bias and its effect."** — Loss aversion → holding losers too long (disposition effect); herding → bubbles; anchoring → fixating on a reference price.
- **"If markets were perfectly efficient, how would you invest?"** — "Passively, via low-cost index funds — there'd be no edge to justify active fees."

Traps & common mistakes

- Treating EMH as **all-or-nothing** — it's a spectrum; markets are mostly-but-not-perfectly efficient.
- Confusing the **three forms** (weak/semi-strong/strong).
- Assuming biases **cancel out** — they're systematic, not random.
- Believing anomalies are **free money** — they're real but hard and risky to capture.
- Ignoring your **own** biases.

First-principles recap

- EMH: prices reflect information → hard to beat the market; forms are weak/semi-strong/strong.
- Behavioral finance: systematic biases push prices from value, creating anomalies.
- Reality: mostly efficient, not perfectly — news is priced fast, but inefficiencies persist.
- Anomalies (value, momentum) underpin factor investing.
- The behavioral edge is discipline; the first task is not to be biased yourself.

Quick-reference

CONCEPT	NOTE
Weak EMH	Past prices priced → TA fails
Semi-strong	Public info priced → FA edge hard
Strong	Even private info priced
Anomalies	Value, momentum, small-cap, PEAD
Key biases	Loss aversion, herding, anchoring, overconfidence
If perfectly efficient	Index passively

Q&A — Market Efficiency & Behavioral Finance

Theory and worked scenarios on EMH, anomalies, and behavioural biases.

Q1. State the three forms of the Efficient Market Hypothesis and what each implies for a specific investing strategy.

Model answer. Weak form — prices reflect all past price/volume data, implying technical analysis should not generate consistent excess returns. Semi-strong form — prices reflect all publicly available information, implying fundamental analysis of public data (filings, news, ratios) should not generate a consistent edge, since the information is already priced. Strong form — prices reflect even private/insider information, implying not even insiders could consistently beat the market (empirically the weakest-supported form, and largely why insider trading is illegal and profitable when it occurs — it's real evidence against strong-form efficiency).

Q2. If you believe markets are broadly semi-strong efficient but not perfectly so, how should that shape how you spend research time?

Model answer. Time spent restating or re-deriving already-public, widely-followed information (headline financial ratios, well-known industry trends) adds little value since that's already priced in efficiently. Time is better spent on the sources of genuine edge that survive semi-strong efficiency: deeper, harder-to-replicate analysis of public information (Part 6 of the equity-research-process chapter's "analytical edge"), primary research/channel checks that surface information ahead of wide dissemination, and exploiting behavioural mispricings where other investors' documented biases create persistent, if hard-to-time, opportunities.

Q3. Worked — apply loss aversion and the disposition effect to a portfolio decision.

An investor holds two positions: Stock A is up 25% and Stock B is down 25%, with no new fundamental information on either since purchase. They're considering trimming the portfolio to raise cash for a new idea.

Model answer. The disposition effect predicts the investor will be tempted to sell Stock A (the winner, "locking in a gain feels good") and hold Stock B (the loser, "I'll sell once it comes back to break-even") — even though, absent new information, this is exactly backwards from a purely rational standpoint: there's no fundamental reason the winner is now a worse holding than the loser, and in fact selling winners and holding losers systematically has been shown to hurt realised returns. A disciplined investor would evaluate both positions purely on current expected forward return versus alternatives, ignoring the emotionally-loaded fact of whether each position is currently a paper gain or loss.

Q4. Give a real anomaly that challenges strict EMH, and explain the standard efficient-markets rebuttal to it.

Model answer. Momentum — stocks that have performed well over the past 3-12 months tend to continue outperforming over the subsequent few months, a well-documented empirical anomaly that shouldn't exist under strict EMH (past price data should carry no predictive information). The standard efficient-markets rebuttal is that momentum returns may simply compensate for a real risk factor not being fully understood/priced (a risk-based explanation, consistent with EMH broadly), rather than being unexploited free money — the debate between "momentum is a genuine anomaly from underreaction" (behavioural explanation) and "momentum is compensation for unmeasured risk" (efficient-markets explanation) remains genuinely contested in the academic literature, and a well-prepared candidate should be able to present both sides rather than assert one is obviously correct.

Q5. Distinguish herding from momentum investing — they can look similar from the outside but are conceptually different.

Model answer. Momentum investing is a systematic strategy based on documented historical patterns (recent winners tend to keep winning over a specific time horizon), applied with discipline and typically with defined entry/exit rules. Herding is the behavioural tendency to follow the crowd's actions specifically *because* others are doing it (social proof), often without independent analysis, and is a primary driver of bubbles — herding can push prices far beyond what any momentum-based risk/reward framework would justify, and unlike disciplined momentum investing, herding-driven buying typically has no defined exit discipline, which is exactly what makes bubbles collapse suddenly and violently when sentiment reverses.

Q6. Why does the chapter argue "the first job is not to be biased yourself" before trying to exploit others' biases?

Model answer. Every documented behavioural bias (overconfidence, loss aversion, anchoring, confirmation bias, herding) applies to professional analysts and portfolio managers just as much as to retail investors — an analyst who is unaware of their own susceptibility to, say, confirmation bias (seeking evidence that supports an existing thesis while discounting contrary evidence) will systematically produce worse research regardless of technical skill. Genuine behavioural edge requires first building the discipline and self-awareness to recognise and counteract one's own biases (e.g. deliberately seeking disconfirming evidence, pre-committing to falsification criteria as covered in the research-note chapter) before that discipline can be turned into an edge against less self-aware market participants.

Q7. A stock reports earnings that beat expectations, and the price continues drifting upward gradually over the following weeks rather than jumping immediately to a new level. What phenomenon does this describe, and what does it imply about market efficiency?

Model answer. This describes post-earnings-announcement drift (PEAD), a well-documented anomaly where prices underreact to earnings surprises initially and continue adjusting gradually over subsequent weeks rather than immediately incorporating the full information content of the surprise. It's evidence against strict semi-strong efficiency (which would predict instantaneous, complete adjustment), and is one of the more robust, widely replicated anomalies in the academic literature — though, consistent with the broader theme of this chapter, exploiting it in practice is complicated by transaction costs and the fact that the effect, while statistically real, is not large enough per trade to guarantee reliable profit after costs for every investor who tries to trade it.

Q8. How would you answer "do you believe markets are efficient?" in a way that shows nuanced understanding rather than picking one extreme side?

Model answer. "Markets are broadly semi-strong efficient in the sense that public information gets incorporated into prices quickly — you can't expect to consistently profit just from reading yesterday's news or restating public filings. But they're not perfectly efficient: well-documented behavioural biases are systematic rather than random, meaning they don't fully cancel out in aggregate, and this leaves real, empirically-documented anomalies like value and momentum effects. My view is that genuine edge comes from either deeper analytical work than the market has done on public information, primary research that gets ahead of wide information dissemination, or disciplined exploitation of others' behavioural biases — not from a belief that markets are broadly irrational or easily beaten." This answer explicitly avoids the all-or-nothing trap the chapter warns against.

Corporate Actions

The Problem / Why this matters

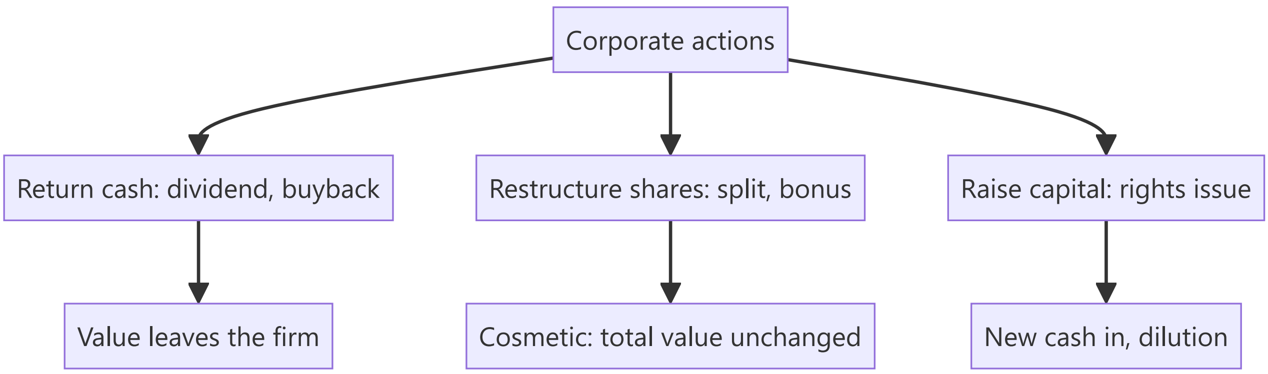
Companies regularly do things that change their shares — pay dividends, split the stock, issue bonus or rights shares, buy back stock. These **corporate actions** change share counts, prices, and per-share metrics, and an analyst must adjust for them correctly (in models, historical price series, and valuation) — and know which ones create value and which merely rearrange it. Interviewers test whether you know that a split or bonus doesn't change value, and how a buyback flows through.

Core Idea

A **corporate action** is an event initiated by a company that affects its securities and shareholders. Some return cash (dividends, buybacks); some change the share structure without changing total value (splits, bonus issues); some raise capital (rights issues). Knowing the mechanics and the value impact of each is essential.

Why it works this way

Per-share numbers depend on the share count, so any action that changes shares changes per-share metrics — but often not total value. A split cuts the price and multiplies shares proportionally (same market cap); a dividend moves cash from the company to shareholders (value leaves the firm). Distinguishing *value-changing* from *cosmetic* actions is the key skill.



Full technical content

Dividends. Cash paid per share from profits/reserves. Reduces cash and retained earnings. Key dates: **declaration, ex-dividend** (buy before ex-date to receive it), **record**, **payment**. The price typically drops by ~the dividend on the ex-date. **Dividend yield** = dividend/price.

Stock split. Divides each share into more shares at a proportionally lower price — e.g., a **1:1 split** (₹100 → two shares of ₹50). Market cap, and your total value, **unchanged**; only the share count and price change. Purpose: improve affordability and liquidity. (A **reverse split** consolidates shares into fewer, higher-priced ones.)

Bonus / scrip issue. Free additional shares to existing holders from reserves (e.g., **1:1 bonus** = one free share per share held). Like a split, it increases shares and cuts the price proportionally — **no change in total value** (reserves just move to share capital). Purely accounting/signaling.

Rights issue. Existing shareholders get the *right* to buy new shares, usually at a **discount**, in proportion to holdings (e.g., 1 new share per 4 held at a discount). Raises capital; those who don't subscribe are **diluted** (though the right itself has value and can be sold). The price adjusts to a **theoretical ex-rights price (TERP)** blending old and new.

Buyback (share repurchase). The company buys its own shares (tender or open-market) and cancels them. Reduces cash and equity, shrinks share count → **boosts EPS**. Returns cash like a dividend but more tax-efficiently/flexibly; signals shares are cheap. Doesn't create value by itself (depends on price paid vs value and alternative uses of cash).

ACTION	SHARES	PRICE/SHARE	TOTAL VALUE	CASH IMPACT
Dividend	Unchanged	Falls ~dividend	Falls (cash out)	Cash out
Split (1:1)	×2	÷2	Unchanged	None
Bonus (1:1)	×2	÷2	Unchanged	None
Rights	Up	TERP (blend)	Up by cash raised	Cash in
Buyback	Down	~Unchanged	Falls (cash out)	Cash out

Analyst adjustments. Adjust **historical prices** for splits/bonus (so charts aren't distorted), adjust **share counts** in models, and use **diluted** shares. Never compare a pre-split price to a post-split price without adjusting.

Worked examples

Example 1 — a split creates no value. A stock at ₹1,000 does a 1:5 split → 5 shares at ₹200 each. An investor who held 10 shares (₹10,000) now holds 50 shares (₹10,000). Nothing changed except affordability. *A 2-for-1 or 5-for-1 split does not make you richer.*

Example 2 — buyback and EPS. A firm has net income ₹100 cr and 100 mn shares → EPS ₹10. It buys back 10 mn shares (cancelled) → 90 mn shares → EPS ₹100 cr / 90 mn = **₹11.1**. EPS rose ~11% with no change in profit — purely from a smaller share count. (Whether it *created value* depends on whether the shares were bought below intrinsic value.)

Example 3 — rights issue TERP. A stock at ₹200; a 1:4 rights issue at ₹100. For every 4 old shares (₹800) you buy 1 new at ₹100 → 5 shares for ₹900 → TERP = ₹180. The price "falls" from ₹200 to ₹180, but that's the dilution of the discounted new shares, not a loss for subscribers.

Example 4 — a full dividend-date sequence, worked with actual dates. A company declares a ₹15/share dividend on 5 June (**declaration date**). It sets 20 June as the **record date** — only shareholders on record that day receive the dividend. Given a T+1 settlement cycle, the **ex-dividend date** is set one business day before the record date, 19 June — an investor must buy the stock by 18 June (so the trade settles by 19 June) to appear on the register by the 20th record date. A trader who buys the stock on 19 June (the ex-date itself) does *not* receive the dividend, and the stock opens on the 19th trading roughly ₹15 lower than the prior close, reflecting the entitlement no longer attaching to a new buyer. **Payment date** — the actual date the ₹15 lands in shareholders' accounts — might be 5 July, several weeks after the ex-date. An analyst modelling total shareholder return must track all four dates correctly: using the wrong date (e.g. the payment date instead of the ex-date) to adjust a historical price series would misalign the mechanical price drop from the actual dividend entitlement.

Example 5 — buyback method comparison: tender offer vs open-market. A company wants to return ₹500 cr to shareholders via buyback and is choosing between two mechanisms. A **tender offer**

buyback sets a fixed price (often at a premium to market, e.g. 20% above the current price) and a fixed quantity, with shareholders tendering shares proportionally if oversubscribed — pros: certainty of price and total outlay, and it signals strong management confidence (paying a visible premium); cons: slower (requires a formal offer process) and the premium paid may not represent good capital allocation if the stock wasn't genuinely undervalued by that much. An **open-market buyback** purchases shares gradually on the exchange over an extended window (up to a regulatory cap, e.g. a maximum % of daily volume) at prevailing market prices — pros: flexible, can be paused if the price rises above what management considers fair value, and avoids overpaying a large explicit premium; cons: slower to complete the full ₹500 cr, and provides a weaker signalling effect than a tender offer's explicit premium commitment. The choice itself is informative to an analyst: a company choosing a tender offer at a large premium is making a stronger public statement about undervaluation than one running a measured open-market programme.

How it is tested in interviews

- **"Does a 2-for-1 stock split create value?"** — "No — twice the shares at half the price; total value and market cap are unchanged. It improves affordability and liquidity."
- **"How does a buyback affect EPS and the statements?"** — "Cash and equity fall; share count shrinks, so EPS rises. It returns cash like a dividend but more flexibly/tax-efficiently. It only creates value if shares are bought below intrinsic value."
- **"Split vs bonus issue?"** — "Both increase shares and cut the price proportionally with no value change; a split changes the face value, a bonus capitalizes reserves into share capital."
- **"What is a rights issue and TERP?"** — "Existing holders can buy new shares at a discount; the price settles at a theoretical ex-rights price blending old and new. Non-subscribers are diluted."
- **"Why adjust historical prices for a split?"** — "So the chart and returns aren't distorted by the mechanical price change."

Traps & common mistakes

- Thinking a **split/bonus** makes shareholders richer — it's cosmetic.
- Confusing a **buyback's EPS boost** with value creation (depends on price paid).
- Forgetting **rights issues dilute** non-subscribers.
- Comparing pre- and post-split prices **without adjusting**.
- Missing the **ex-date** logic for dividends/entitlements.

First-principles recap

- Corporate actions change shares, price, and per-share metrics — often not total value.
- **Splits and bonus issues are cosmetic** (value unchanged).
- **Dividends and buybacks** return cash (value leaves the firm); buybacks lift EPS.
- **Rights issues** raise capital and dilute non-subscribers (price → TERP).
- Always adjust historical prices and share counts for splits/bonus.

Quick-reference

ACTION	VALUE EFFECT
Dividend	Cash out; price drops ex-date
Split / bonus	Cosmetic; total value unchanged
Buyback	Cash out; EPS up; value if bought cheap
Rights issue	Capital in; dilutes non-subscribers; TERP
Adjust for	Splits/bonus in prices & share counts

Q&A — Corporate Actions

Theory and worked numerical scenarios on dividends, splits, bonuses, rights issues, and buybacks.

Q1. Does a 2-for-1 stock split make a shareholder wealthier? Walk through why or why not with numbers.

Model answer. No. If an investor holds 100 shares at ₹500 (₹50,000 total value) and the company executes a 2-for-1 split, they now hold 200 shares at ₹250 (still ₹50,000 total value) — the market capitalisation and the investor's total holding value are mechanically unchanged; only the share count and per-share price have changed. The purpose of a split is to improve affordability (a lower per-share price may attract more retail participation) and liquidity (more shares outstanding can mean tighter spreads and more trading volume), not to create value.

Q2. Worked — buyback impact on EPS.

A company has net income of ₹240 cr and 60 mn shares outstanding (EPS = ₹4.00). It executes a buyback, repurchasing and cancelling 6 mn shares.

Model answer. New share count = 60 − 6 = 54 mn. New EPS = 240 / 54 = ₹4.44. EPS rose from ₹4.00 to ₹4.44 (an 11.1% increase) purely from the reduced share count, with no change in the underlying net income or business performance. Whether this buyback *created value* for remaining shareholders depends entirely on whether the repurchase price was below the shares' intrinsic value — buying back overvalued shares actually destroys value even while mechanically boosting EPS, a distinction the chapter emphasises as a common analytical trap.

Q3. Worked — theoretical ex-rights price (TERP) calculation.

A stock trades at ₹450 before a 1:3 rights issue (one new share for every three held) priced at ₹300.

Model answer. For every 3 existing shares (value 3 × 450 = ₹1,350), the holder can buy 1 new share at ₹300, giving 4 shares for a total cost of 1,350 + 300 = ₹1,650. TERP = 1,650 / 4 = ₹412.50. The price is expected to settle around ₹412.50 post-rights — a fall from ₹450, but this is not a loss for a shareholder who subscribes to their full entitlement: they now hold more shares at a lower average cost, and total portfolio value is preserved. A shareholder who does *not* subscribe is diluted and their proportional ownership falls, though they can potentially recoup some value by selling the rights entitlement itself if it's tradeable.

Q4. Distinguish which corporate actions are "cosmetic" (no total value change) from which are "value-changing," and explain the underlying reason for the distinction.

Model answer. Cosmetic actions — stock splits and bonus issues — merely restructure the existing equity claim into a different number of shares/different face value, drawing from the company's own reserves (bonus) or simply redefining share units (split), with no cash or assets actually leaving or entering the company; total value is mechanically preserved. Value-changing actions — dividends and buybacks — involve actual cash leaving the company to shareholders, directly reducing the company's asset base (and, in the case of a dividend, typically causing an ex-date price adjustment roughly equal to the dividend). Rights issues are value-changing in the other direction — real new cash enters the company, funding growth or debt reduction, at the cost of dilution for non-subscribers.

Q5. Why does a stock's price typically fall by roughly the dividend amount on the ex-dividend date, and is this a "loss" for existing shareholders?

Model answer. On the ex-dividend date, buyers of the stock are no longer entitled to the declared dividend (only holders as of the record date receive it) — so the stock is mechanically worth roughly the dividend amount less than it was the day before, reflecting that upcoming cash payment no longer being part of what a new buyer receives. This is not a loss for the existing shareholder who is entitled to the dividend: they hold a slightly lower-priced stock plus a dividend payment equal (approximately) to the price drop — their total value (stock plus declared dividend receivable) is essentially unchanged by the mechanical ex-date adjustment itself.

Q6. What's the difference between a bonus issue and a rights issue, and why would a company prefer one over the other?

Model answer. A bonus issue gives existing shareholders additional shares for free, funded by capitalising the company's own reserves — no new cash enters the company, and it's purely a cosmetic restructuring (Q4) often used as a signal of management confidence or to improve share liquidity/affordability. A rights issue requires shareholders to pay for the new shares (at a discount to market) — it genuinely raises new capital for the company. A company needing actual growth capital or debt reduction uses a rights issue; a company wanting to signal confidence or improve trading liquidity without needing new cash uses a bonus issue.

Q7. An analyst is building a 10-year historical price chart for a stock that underwent a 1:1 bonus issue five years ago. What adjustment must they make, and what happens if they skip it?

Model answer. They must adjust all historical prices *before* the bonus issue date by the bonus ratio (halving pre-bonus prices, in a 1:1 case) so the entire price series is on a consistent, comparable per-share basis. If they skip this adjustment, the chart will show an artificial, dramatic price "crash" exactly at the bonus-issue date that has nothing to do with the company's actual performance — a classic and easily avoidable analytical error that would badly distort any technical analysis, historical-return calculation, or valuation multiple trend built on the unadjusted series.

Q8. A company announces a large buyback funded entirely by increasing debt (leverage), rather than using existing cash reserves. What should an analyst think about beyond the simple EPS-accretion math?

Model answer. Beyond the mechanical EPS boost (Q2), a debt-funded buyback increases financial leverage and interest expense, raising the company's financial risk — the analyst should assess whether the resulting capital structure remains prudent (debt/EBITDA, interest coverage) and whether management is buying back shares because they're genuinely undervalued (a legitimate capital-allocation decision) or simply to engineer a short-term EPS increase without a corresponding improvement in the underlying business, sometimes to hit management compensation targets tied to EPS — a distinction directly relevant to assessing management quality and capital-allocation discipline (a recurring theme from the fundamental-analysis chapter).

Capital Raising: Follow-ons, Rights, QIPs & Block Deals

The Problem / Why this matters

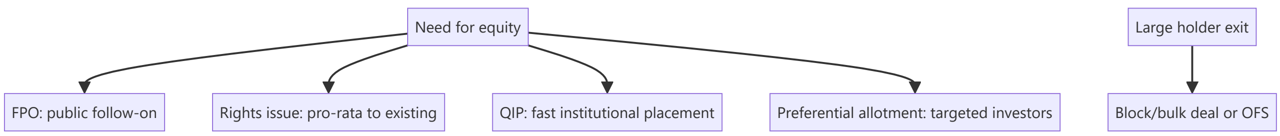
An IPO is a company's first equity raise, but listed companies raise equity again and again — to fund growth, cut debt, or let large holders exit. **Equity capital markets (ECM)** covers all these follow-on ways to issue or place equity: FPOs, rights issues, QIPs, preferential allotments, and block/bulk deals. Knowing each mechanism, when it's used, and its effect on existing shareholders is core to ECM, IB, and research roles.

Core Idea

Beyond the IPO, listed firms raise or place equity through several routes, differing by **who can buy, how fast, and at what dilution**: broad public offers (FPO), pro-rata offers to existing holders (rights), fast institutional placements (QIP), targeted placements (preferential), and secondary sales of existing shares (blocks/OFS). Each balances speed, price, investor base, and dilution.

Why it works this way

Different situations call for different tools: a company wanting a quick institutional raise uses a QIP (fast, no lengthy prospectus); one wanting to treat all shareholders fairly uses a rights issue; a large holder wanting to exit uses a block deal or OFS. The menu exists so issuers can optimize for speed, price, and the desired investor base.



Full technical content

Primary raises (new shares → company gets cash, dilution):

ROUTE	WHO BUYS	SPEED	NOTES
FPO	Public	Slow (prospectus)	A full follow-on public offer
Rights issue	Existing holders, pro-rata	Moderate	Usually at a discount; non-subscribers diluted; price → TERP
QIP	Qualified institutional buyers	**Fast**	India-specific; no lengthy prospectus; institutions only
Preferential allotment	Selected investors	Fast	Targeted (e.g., a strategic/PE investor); priced per SEBI formula

Secondary sales (existing shares → selling holder gets cash, no new capital, no dilution):

ROUTE	WHAT IT IS
OFS (Offer for Sale)	Promoters/large holders sell via an exchange mechanism
Block deal	A large negotiated trade (min size) in a special window at a pre-agreed price
Bulk deal	Large trades executed in the normal market (disclosed)

Key distinctions: - **Fresh issue vs secondary sale** — fresh issue raises capital for the company and dilutes; secondary sale just transfers existing shares (holder gets cash, no dilution). - **Dilution** — new shares reduce existing holders' ownership % and EPS unless the capital earns above the cost of equity. - **Pricing** — placements are usually at a small discount to market to attract buyers; rights at a larger discount; SEBI sets floor-price formulas for QIPs/preferential. - **Signaling** — equity issuance can signal management thinks shares are fairly/over-valued (pecking-order); a raise to cut debt vs to fund high-return growth is read very differently.

ECM's job. Investment banks (ECM desks) advise on the route, timing, size, and pricing, and place the shares with investors — balancing the issuer's proceeds against a successful, well-received deal.

Worked examples

Example 1 — QIP for speed. A company needs ₹1,500 cr quickly to fund expansion. Rather than a months-long FPO, it does a **QIP**: places new shares with institutions at a ~3% discount to the market over a couple of days. Fast, institutional, and dilutive — the company gets the cash without a full prospectus process.

Example 2 — rights issue to deleverage. A leveraged firm does a **1:2 rights issue** at a 30% discount to cut debt. Existing holders can maintain their stake by subscribing; those who don't are diluted but can sell their rights. The stock re-rates to TERP; the balance sheet strengthens.

Example 3 — block deal exit (no dilution). A PE fund holding 8% wants to exit. It sells the entire stake in a single **block deal** to institutions at a small discount in the block window. The *fund* receives the cash; the company issues no new shares, so there's **no dilution** — just a change of ownership. (Contrast with a QIP, which is new shares and dilutive.)

How it is tested in interviews

- **"Ways a listed company can raise equity?"** — FPO, rights issue, QIP, preferential allotment (primary/dilutive); plus OFS/block deals for existing holders to sell (secondary, non-dilutive).
- **"What is a QIP and why use it?"** — "A Qualified Institutional Placement — a fast raise from institutions without a full prospectus; used when a company wants speed and an institutional base."
- **"Rights issue vs QIP?"** — "Rights is pro-rata to all existing holders (fair, at a discount, non-subscribers diluted); QIP is a quick placement to institutions only."
- **"Block deal vs QIP — is it dilutive?"** — "A block deal is a secondary sale of existing shares — the seller gets the cash, no dilution. A QIP is new shares — the company gets the cash, dilutive."
- **"Why might a raise move the stock down?"** — "Dilution, plus the signal that management may think shares are fully valued; the use of proceeds (debt reduction vs high-return growth) matters."

Traps & common mistakes

- Confusing **primary** (new shares, company cash, dilution) with **secondary** (existing shares, holder cash, no dilution).

- Thinking a **block deal/OFS** raises capital for the company — it doesn't.
- Forgetting **rights issues dilute** non-subscribers and reset the price to TERP.
- Ignoring the **signaling** and use-of-proceeds read of a raise.

First-principles recap

- Listed firms raise equity via FPO, rights, QIP, preferential (primary, dilutive) and sell via OFS/blocks (secondary, non-dilutive).
- **QIP** = fast institutional raise; **rights** = pro-rata to existing holders at a discount.
- Fresh issue → company cash + dilution; secondary sale → holder cash, no dilution.
- Placements price at a discount; SEBI sets floors.
- Read every raise for **dilution, signaling, and use of proceeds**.

Quick-reference

ROUTE	TYPE	CASH TO	DILUTION
FPO	Primary	Company	Yes
Rights	Primary, pro-rata	Company	Yes (non-subscribers)
QIP	Primary, institutions	Company	Yes
Preferential	Primary, targeted	Company	Yes
OFS / block	Secondary	Selling holder	No

Q&A — Capital Raising: Follow-ons, Rights, QIPs & Block Deals

Theory and worked scenarios on ECM mechanisms beyond the IPO.

Q1. List the primary (dilutive, company receives cash) capital-raising routes and the secondary (non-dilutive, existing holder receives cash) routes, and state the key distinguishing question to classify any deal you encounter.

Model answer. Primary routes: FPO, rights issue, QIP, preferential allotment — all involve newly issued shares, with proceeds going to the company, and all dilute existing non-participating shareholders. Secondary routes: OFS, block deals, bulk deals — all involve a transfer of *existing* shares, with proceeds going to the selling shareholder, and no dilution or new capital for the company. The key distinguishing question for any deal: "are new shares being created (primary), or are existing shares just changing hands (secondary)?" — this single question resolves the classification and its capital-structure/dilution implications immediately.

Q2. Why would a company choose a QIP over a rights issue when both raise capital from investors, and what does the company give up by choosing QIP?

Model answer. A QIP is dramatically faster (days, versus the weeks-to-months a rights issue with its formal offer document and subscription period requires) and doesn't require the extended prospectus/disclosure process, since it's placed only with qualified institutional buyers. What the company gives up: fairness/inclusivity to retail and smaller existing shareholders (who get no opportunity to participate and maintain their proportional stake, unlike in a rights issue), and potentially a broader/more diversified new-investor base — a QIP concentrates the new shares among a smaller number of institutional holders. Companies needing capital urgently, or comfortable diluting retail without offering them participation, favour QIP; companies wanting to treat all existing shareholders equitably favour rights issues despite the slower timeline.

Q3. Worked — is this transaction dilutive? A private equity fund holding 12% of a listed company sells its entire stake to a group of mutual funds via a block deal at a 2% discount to the prevailing market price.

Model answer. This is a secondary transaction — the PE fund is selling *existing* shares it already owns to other investors; no new shares are created, so there is zero dilution to any other shareholder, and the company itself receives no cash from this transaction (only the exiting PE fund receives proceeds). The only effect on the company's shareholder register is a change in *who* holds that 12% stake (institutional buyers replacing the PE fund), not a change in total shares outstanding or the company's capital position. A common interview trap is assuming any large equity transaction involving a listed company is dilutive — always check whether shares are newly issued (primary) or merely transferred (secondary) first.

Q4. What does it mean that "SEBI sets floor-price formulas for QIPs and preferential allotments," and why does this regulation exist?

Model answer. Rather than letting a company negotiate an arbitrarily low placement price with select institutional or preferential investors, SEBI mandates a minimum ("floor") price, typically calculated from a formula based on recent average trading prices over a specified look-back window. This exists to protect existing (particularly retail) shareholders from being diluted at an unfairly cheap price that would transfer value from existing holders to the new placement investors — without a floor-price rule, a company (or its promoters, in a preferential-allotment scenario benefiting an insider) could place new shares to favoured parties at a steep, unjustified discount, effectively expropriating value from other shareholders.

Q5. How should an analyst interpret the signal when a company announces a large equity raise specifically to fund debt repayment, versus one raising equity to fund high-return growth capex?

Model answer. Equity raised to repay debt is often read as a defensive, deleveraging move — potentially signalling the company (or its bankers) judge the current leverage level as risky, and it dilutes existing shareholders without adding new earning assets, generally a less positive signal for near-term EPS/growth. Equity raised to fund high-return growth capex can be read more positively if the analyst believes the incremental capital will earn well above its cost (ROIC > WACC, per the fundamental-analysis chapter) — dilution is a real cost, but it's justified if it funds genuinely value-accretive investment. The "why" behind a raise (stated use of proceeds) is therefore as important to the analyst's read as the raise's size and pricing.

Q6. What is the "pecking order" signalling concern raised in this chapter regarding equity issuance, and how does it apply to interpreting a surprise equity raise?

Model answer. Pecking-order theory suggests companies prefer to fund needs with internal cash first, then debt, and equity issuance last — because managers, who typically have better information about the company's prospects than outside investors, are more likely to issue equity when they believe the stock is fairly valued or overvalued (since issuing undervalued equity would unnecessarily give away value to new shareholders at existing holders' expense). This means a surprise equity raise can itself be read by the market as a mild negative signal about management's view of the current valuation — a nuance an analyst should be alert to beyond just the raise's stated purpose, since the *choice* to raise equity at all (versus debt) carries information.

Q7. Explain the difference between a block deal and a bulk deal, and why the distinction (special window vs normal market) matters for how each affects the stock's observed price.

Model answer. A block deal is executed in a special, separate trading window with a minimum transaction size, at a pre-agreed price negotiated directly between buyer and seller — because it happens outside the regular continuous order book, a large block deal doesn't directly "walk the book" and cause the market-impact price movement that an equivalent-sized order would in normal trading (see the secondary-markets chapter's market-impact discussion). A bulk deal is a large trade executed within the normal market during regular trading hours (just disclosed due to its size) — it does interact with and can move the regular order book, similar to any other large market order. The special block window exists specifically to let large holders transact size without unduly disrupting the continuous market price.

Q8. A company does a preferential allotment to a single strategic investor at a price close to the SEBI floor price, shortly after a period of share-price weakness. What should an analyst check before concluding this is either clearly positive or clearly negative for existing shareholders?

Model answer. Check: (1) who the strategic investor is and whether their involvement brings genuine strategic value (technology, market access, credibility) beyond just capital — a well-regarded strategic investor committing capital can be a positive signal even at a discount; (2) the actual discount to the pre-announcement price relative to the SEBI-mandated floor — a price right at the regulatory floor (the minimum legally allowed) versus a smaller discount matters for how much value existing shareholders are giving up; (3) the stated use of proceeds and whether it addresses a genuine, previously-flagged need (e.g. deleveraging, funding a specific growth initiative) versus appearing opportunistic; (4) whether the preferential allottee has any pre-existing relationship with promoters/management that could raise related-party or governance concerns. No single factor alone determines whether the deal is good or bad for existing shareholders — it requires weighing all of these together.

Sell-Side vs Buy-Side

The Problem / Why this matters

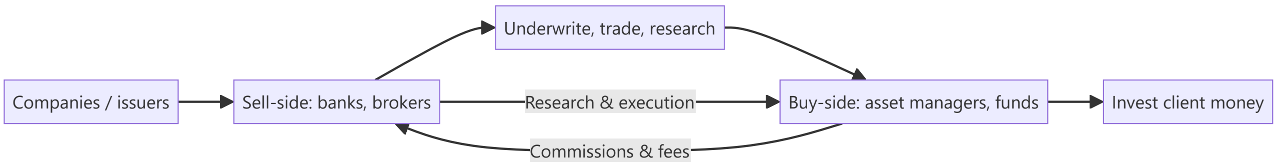
The finance industry splits into two worlds — the **sell-side** (banks and brokers who create, sell, and research products) and the **buy-side** (asset managers who invest client money). Every markets role sits on one side, they interact constantly, and interviewers frequently ask which you want and why. Understanding the distinction — who does what, how each makes money, and how research flows between them — is basic industry literacy.

Core Idea

The **sell-side** creates and sells financial products and services (underwriting, trading, research) to clients and earns fees/commissions. The **buy-side** buys and holds investments on behalf of clients (funds) and earns management/performance fees on assets. Sell-side research is *published* to win business; buy-side research is *private* and drives the fund's own decisions.

Why it works this way

The two exist because issuers and investors need different services. The sell-side intermediates — helping companies raise capital and helping investors execute and understand — monetized through transactions. The buy-side is the end investor of capital — monetized through investment performance on assets under management. Research flows from sell-side (marketing/service) to buy-side (decision input).



Full technical content

Sell-side (investment banks, brokerages):

FUNCTION	WHAT IT DOES
Investment banking (ECM/DCM/M&A)	Raise capital and advise on deals for companies
Sales & trading	Execute trades, make markets, provide liquidity
Sell-side research	Publish analysis and ratings on stocks/sectors to clients
Earns: **underwriting fees, advisory fees, trading commissions, spreads.** Sell-side research is *distributed* to institutional clients to win their trading/banking business — it's a service, not a profit centre itself.	

Buy-side (asset managers):

TYPE	DESCRIPTION
Mutual funds / AMCs	Pooled long-only funds, benchmark-relative
Hedge funds	Flexible, absolute-return, can short/leverage
Pension & insurance funds	Long-term, liability-driven
PE / VC	Private investing
PMS / AIFs	Portfolio management for HNIs
Earns: **management fees (% of AUM)** and often **performance fees** (e.g., 2-and-20 for hedge funds). Buy-side research is *internal* and confidential — it directly drives what the fund buys.	

How they interact: - Sell-side provides **execution** (trading), **capital-raising access** (IPO allocations), and **research** to the buy-side. - Buy-side pays via **commissions** and consumes sell-side research as one input (increasingly unbundled post-MiFID II). - The buy-side ultimately makes the investment decision; the sell-side facilitates.

Research differences:

	SELL-SIDE RESEARCH	BUY-SIDE RESEARCH
Audience	Published to many clients	Internal to the fund
Purpose	Win business/commissions	Make the fund's own decisions
Coverage	Broad, many names	Focused on holdings/ideas
Incentive	Volume, access, relationships	Being *right* (performance)

Career framing. Sell-side research/banking: broad coverage, client-facing, deal/idea flow. Buy-side: fewer names, deeper conviction, direct accountability for returns. Many move sell-side → buy-side over a career.

Worked examples

Example 1 — a research idea's journey. A sell-side analyst publishes a Buy on Stock X with a note. It's distributed to hundreds of institutional clients. A buy-side PM reads it, does her *own* work, disagrees on one assumption, but likes the idea — and buys through the same bank's trading desk (paying commission). The sell-side earned the commission and strengthened the relationship; the buy-side made the actual decision.

Example 2 — how each makes money. A brokerage (sell-side) earns ₹5 cr in commissions from a fund's trading this year. The fund (buy-side) manages ₹5,000 cr and earns a 1% management fee = ₹50 cr, plus performance fees if it beats its benchmark. Different economics: transaction fees vs fees on assets and performance.

Example 3 — incentive difference. A sell-side analyst is rewarded for coverage, client access, and generating trading interest; a buy-side analyst is rewarded purely for whether the calls make money. This is why buy-side research is more focused and conviction-driven — being *right* is the only thing that pays.

Example 4 — a worked management-fee-plus-performance-fee calculation. A hedge fund manages ₹2,000 cr, charges a 1.5% management fee and a 20% performance fee above a hurdle rate of 8%. In a year the fund returns 22%. Management fee = 1.5% × 2,000 cr = ₹30 cr, charged regardless of performance. Performance fee applies only to the return *above* the 8% hurdle: excess return = 22% − 8% = 14%, on which the fund earns 20% = 14% × 20% = 2.8% of AUM = ₹56 cr. Total fund revenue this year: ₹30 cr + ₹56 cr = **₹86 cr**, versus an investor's net return after fees of roughly 22% − 1.5% − 2.8% ≈ **17.7%**. This concrete "2-and-20-with-a-hurdle" calculation is a standard technical question for anyone claiming buy-side/hedge-fund interest, and the hurdle-rate mechanic specifically (performance fees only above a minimum threshold, not on the whole return) is the detail most candidates get wrong by omitting it.

Example 5 — how MiFID-II-style research unbundling changed sell-side economics in practice. Before unbundling, a buy-side fund's trading commissions implicitly paid for both execution *and* the sell-side research bundled alongside it — a fund couldn't easily tell how much it was really paying for research specifically. Post-unbundling, funds must pay for research explicitly and separately from execution costs, forcing them to actively budget and justify research spend against its measured value. The practical effect on sell-side desks: consolidation toward fewer, higher-conviction analysts covering fewer names with more differentiated views, since broad, generic coverage that used to be "free" (bundled into commissions) no longer automatically earns its keep once a fund has to consciously decide whether a specific research relationship is worth paying for directly.

How it is tested in interviews

- **"What's the difference between sell-side and buy-side?"** — "Sell-side (banks/brokers) creates and sells products — underwriting, trading, research — for fees and commissions. Buy-side (asset managers) invests client money for management and performance fees. Sell-side research is published to win business; buy-side research is private and drives the fund's decisions."
- **"How does each make money?"** — Sell-side: underwriting/advisory fees, trading commissions/spreads. Buy-side: % of AUM management fees plus performance fees.
- **"Do you want sell-side or buy-side, and why?"** — Have a genuine answer: sell-side for broad coverage and client/deal exposure; buy-side for deep conviction and direct accountability for returns.
- **"How do sell-side and buy-side research differ?"** — Audience (published vs internal), purpose (win business vs make money), and incentive (volume/access vs being right).

Traps & common mistakes

- Mixing up which side **invests** (buy-side) vs **intermediates** (sell-side).
- Thinking sell-side research is the sell-side's **profit centre** — it's a service to win trading/banking.
- Not having a **genuine preference** with reasons when asked.
- Forgetting the buy-side does its **own** work and makes the final call.

First-principles recap

- **Sell-side** intermediates (underwrite, trade, research) for fees/commissions.
- **Buy-side** invests client money for fees on assets and performance.
- Research flows sell-side → buy-side; the buy-side decides.
- Incentives differ: sell-side rewards access/volume; buy-side rewards being right.
- Careers often move sell-side → buy-side.

Quick-reference

	SELL-SIDE	BUY-SIDE
Role	Create/sell/execute/research	Invest client money
Examples	IBs, brokerages	Mutual/hedge/pension funds, PMS
Revenue	Fees, commissions, spreads	% AUM + performance fees
Research	Published, wins business	Internal, drives decisions
Reward	Access/volume	Performance (being right)

Q&A — Sell-Side vs Buy-Side

Theory and scenario drills on industry structure, economics, and career framing.

Q1. Explain precisely why sell-side research is described as "not a profit centre itself" — if it doesn't directly make money, why do banks maintain large research departments?

Model answer. Sell-side research is typically distributed to institutional clients without a direct per-report fee (especially pre-unbundling regulatory regimes) — its function is to win and retain the trading commissions, IPO allocation access, and banking relationships that *are* the actual revenue sources. A highly-regarded analyst who consistently produces valuable research attracts institutional client relationships and trading flow to the bank's desks; research is best understood as a marketing and relationship function that indirectly drives commission and banking revenue, not a standalone profit line measured against research-department costs alone.

Q2. How do sell-side and buy-side analysts' incentives differ, and how does that difference shape the character of their research output?

Model answer. A sell-side analyst is rewarded substantially for breadth of coverage, client access, and generating trading interest/relationships — incentives that favour maintaining ratings on many names, being responsive to client questions, and sometimes avoiding overly controversial calls that could damage banking relationships with covered companies. A buy-side analyst is rewarded purely on whether their calls make money for the fund — incentives that favour deep conviction on a smaller number of ideas and genuine intellectual honesty about being wrong, since there's no relationship-management layer diluting the accountability. This difference is why buy-side research is typically described as more focused and conviction-driven.

Q3. Worked scenario — a sell-side analyst's Buy rating and a buy-side fund's actual position disagree. What does this reveal, and is it a problem?

Model answer. A sell-side Buy rating represents one analyst's published view, distributed as one input among many to potentially hundreds of buy-side clients, each of whom does their own independent analysis. It is entirely normal and healthy for a buy-side fund to read the sell-side note, disagree with a specific assumption, and reach a different conclusion (or even the opposite conclusion) — the sell-side analyst's job is to provide a well-reasoned, differentiated view and supporting analysis, not to dictate the buy-side's actual investment decision. If every buy-side fund simply followed every sell-side rating without independent work, there would be no genuine price discovery or alpha generation on the buy-side at all.

Q4. Describe the economics difference between a sell-side brokerage and a buy-side asset manager with a worked comparison.

A brokerage earns ₹8 cr in annual commissions from a specific fund's trading activity. That same fund manages ₹4,000 cr in assets, charging a 1.25% management fee plus a 15% performance fee on returns above its benchmark, and beats its benchmark by 3% this year.

Model answer. Brokerage revenue from this relationship = ₹8 cr (pure commission on transaction volume, independent of whether the fund's trades were ultimately profitable). Fund's management fee = 1.25% × 4,000 cr = ₹50 cr (charged regardless of performance). If outperformance is measured on, say, a ₹4,000 cr base, a 3% outperformance generates ₹120 cr of excess return, of which the fund captures 15% = ₹18 cr in performance fees. Total buy-side economics (₹50 cr + ₹18 cr = ₹68 cr) dwarfs the sell-side's ₹8 cr commission from this single relationship — illustrating why buy-side economics scale with assets under management and performance, while sell-side economics scale with transaction volume, a structurally different (and, at meaningful AUM, typically larger) revenue model.

Q5. What genuine reasons might a candidate give for preferring buy-side over sell-side, and vice versa, beyond generic statements like "I want to make investment decisions"?

Model answer. For buy-side: direct accountability (the P&L reflects your own calls, not a published rating diluted across many clients' independent decisions), typically fewer, higher-conviction names to cover allowing real depth, and a career path more directly tied to demonstrated investing skill. For sell-side: broader exposure across many companies/sectors early in a career (useful for building pattern recognition before specialising), more client-facing and deal-flow exposure (useful for those interested in the transactional/advisory side or building a wide professional network), and a well-trodden training-ground role that many buy-side firms explicitly like to hire from. A strong interview answer states a genuine, specific reason tied to the candidate's actual background and goals rather than a generic preference.

Q6. Why might a company or its investment bank prefer that sell-side research remains largely aligned with, rather than sharply critical of, that company — and what tension does this create for research objectivity?

Model answer. A bank's research on a company it also does (or hopes to do) investment banking business with (underwriting, M&A advisory) creates a structural conflict of interest — overly negative research could jeopardise that banking relationship, while research too closely aligned with management's own narrative undermines the research's actual value and credibility to buy-side clients. This is precisely why regulatory "Chinese wall" separations between research and banking departments exist, and why analysts are formally required to disclose banking relationships alongside their ratings — a well-prepared candidate should be able to name this conflict-of-interest tension directly rather than pretend sell-side research is fully independent of commercial pressures.

Q7. What does it mean that sell-side research economics have shifted "post-MiFID II" toward unbundling, and why does this matter to how research value is now assessed?

Model answer. MiFID II (a European regulatory regime with global ripple effects on institutional research practices) required buy-side firms to pay explicitly for research rather than have it bundled implicitly into trading commissions — this forced buy-side firms to directly justify and budget research spend against its actual, measurable value, rather than treating it as a "free" byproduct of trading relationships. The practical effect has been increased scrutiny on which sell-side research genuinely adds value (leading some firms to consolidate research spend on fewer, higher-conviction providers) — a structural shift a candidate discussing the sell-side research business model should be aware of.

Q8. Describe a realistic career path that moves from sell-side to buy-side, and what specifically makes an analyst attractive for that transition.

Model answer. A common path: several years as a sell-side research associate/analyst building deep sector expertise and a track record of well-reasoned, differentiated calls (not just accurate but well-argued and risk-aware, since buy-side hiring managers scrutinise the *reasoning* behind calls, not just whether they happened to be right), followed by a move to a buy-side fund as a research analyst or associate portfolio manager. What makes a sell-side analyst attractive for this move: demonstrated ability to form genuinely differentiated (not consensus-hugging) views, a track record where calls can be checked against actual subsequent stock performance, deep sector/company knowledge built over the coverage years, and often personal relationships built with buy-side clients during the sell-side tenure who can vouch for the analyst's judgment.

Portfolio Construction & Risk

The Problem / Why this matters

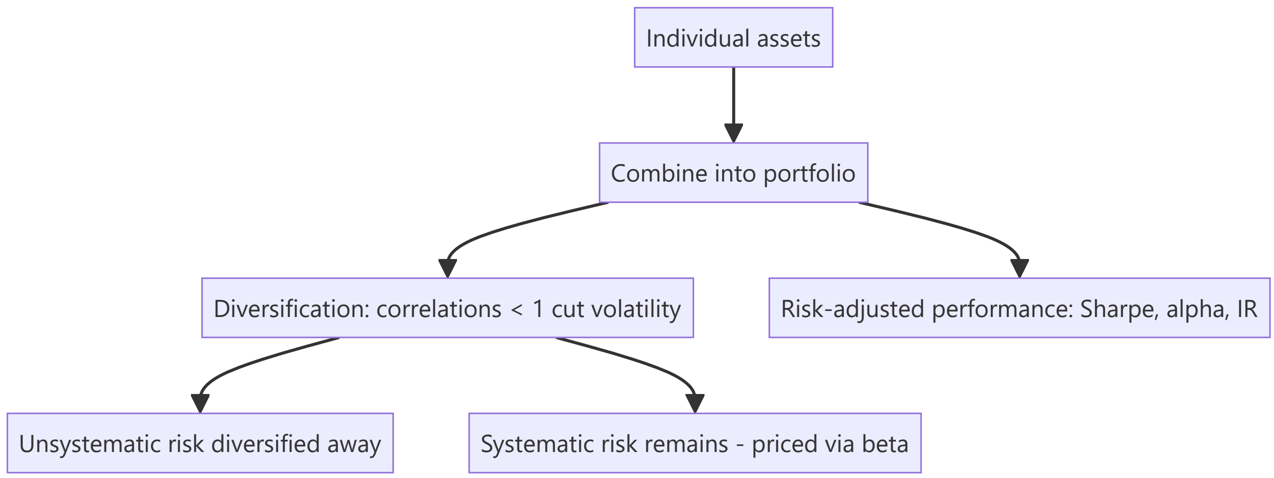
Owning a single great stock is risky; investing is really about building a **portfolio** — combining positions so the whole is better (higher return per unit of risk) than the parts. How you size positions, diversify, benchmark, and measure risk-adjusted performance is the core of asset management and any investing role. Interviewers test diversification, systematic vs unsystematic risk, beta, and the performance ratios.

Core Idea

Portfolio construction combines assets to maximize expected return for a given level of risk, exploiting **diversification** (imperfectly correlated assets reduce portfolio volatility). Risk splits into **systematic** (market, undiversifiable) and **unsystematic** (specific, diversifiable), and performance is judged on a **risk-adjusted** basis (Sharpe, alpha, information ratio), not raw return.

Why it works this way

Because asset returns aren't perfectly correlated, combining them means their ups and downs partly offset, lowering portfolio volatility without proportionally lowering return — the "only free lunch in finance." But the shared exposure to the economy (systematic risk) can't be diversified away, so it's what investors are ultimately compensated for (via beta and the risk premium).

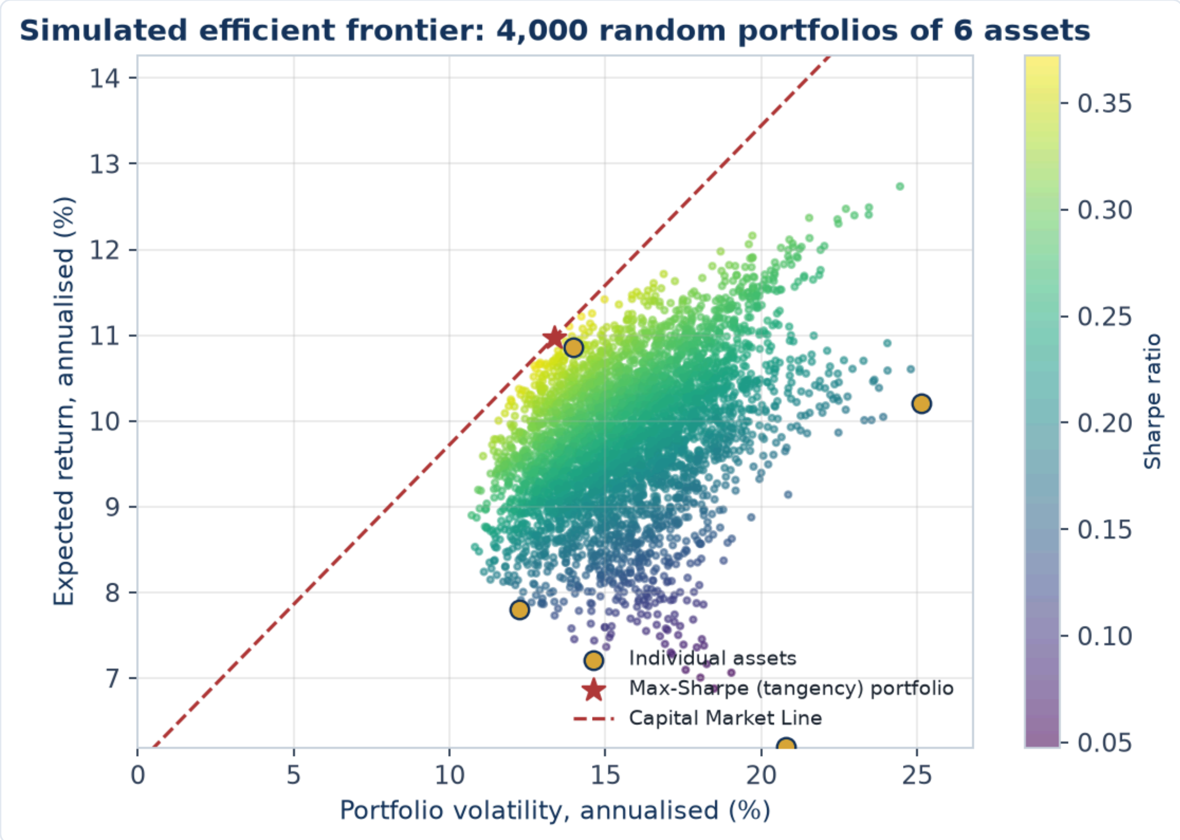


Full technical content

Risk decomposition: - Unsystematic (specific/idiosyncratic) risk — company/sector-specific; **diversifiable** by holding many uncorrelated names. - **Systematic (market) risk** — common exposure to the economy; **not diversifiable**; measured by **beta**; it's what earns the risk premium.

Diversification. Combining assets with correlation < 1 reduces portfolio standard deviation for a given return. The benefit is largest when correlations are low/negative; most idiosyncratic risk is removed with ~20–30 well-chosen names, leaving mainly systematic risk.

Modern Portfolio Theory (Markowitz). For a given return, there's a minimum-variance portfolio; the set of best risk-return combinations is the **efficient frontier**. Adding a risk-free asset gives the **Capital Market Line**; the best risky portfolio is the market portfolio (→ CAPM).



The chart above simulates 4,000 random-weight portfolios of 6 assets, coloured by Sharpe ratio. The individual assets (gold dots) all sit *inside* the cloud of possible portfolios — no single asset is as efficient as a well-diversified combination, the visual proof of Example 1's diversification math. The star marks the max-Sharpe (tangency) portfolio, and the dashed Capital Market Line shows the best achievable risk-return trade-off once a risk-free asset is available to blend with it — exactly the CAPM intuition this section describes.

Position sizing & construction approaches: - **Concentration vs diversification** — high-conviction concentrated books (fewer, larger positions) vs broadly diversified books; a trade-off between idiosyncratic risk and edge. - **Active vs passive** — active tries to beat a benchmark (higher fees, tracking error); passive replicates it cheaply. - **Benchmark-relative** — most funds are measured vs an index; **active weights** (overweight/underweight vs the benchmark) express views; **tracking error** measures deviation. - **Constraints** — position limits, sector caps, liquidity, mandate.

Risk-adjusted performance metrics:

METRIC	FORMULA	MEASURES
Sharpe ratio	(Rp − Rf) / σp	Excess return per unit of total risk
Treynor ratio	(Rp − Rf) / βp	Excess return per unit of market risk
Alpha (Jensen's)	Rp − [Rf + β(Rm − Rf)]	Return above CAPM (skill)
Information ratio	Active return / tracking error	Consistency of outperformance vs benchmark
Beta	Cov(Rp,Rm)/Var(Rm)	Sensitivity to the market

Sharpe uses **total** risk (σ), Treynor uses **systematic** risk (β) — use Treynor when the portfolio is one part of a diversified whole. **Alpha** is the holy grail: return not explained by market exposure.

Rebalancing. Periodically returning to target weights (as prices drift) enforces "sell high, buy low" and controls risk.

Worked examples

Example 1 — diversification cuts risk. Two stocks each with 20% volatility and a correlation of 0.3. A 50/50 portfolio has volatility = $\sqrt{(0.5^2 \cdot 0.2^2 + 0.5^2 \cdot 0.2^2 + 2 \cdot 0.5 \cdot 0.5 \cdot 0.3 \cdot 0.2 \cdot 0.2)} = \sqrt{(0.01 + 0.006)} \approx \sqrt{0.016}$ $\approx$ **12.6%** — well below the 20% of either stock alone, for the average return. That's the diversification benefit from correlation < 1.

Example 2 — Sharpe ratio comparison. Portfolio A returns 15% with 10% volatility; Portfolio B returns 20% with 25% volatility; risk-free 5%. Sharpe A = (15−5)/10 = **1.0**; Sharpe B = (20−5)/25 = **0.6**. B has the higher return but A is the better *risk-adjusted* performer — you'd prefer A (and could lever it to match B's return at lower risk).

Example 3 — alpha vs beta. A fund returned 18% when the market (beta 1.2 exposure) returned 12%, risk-free 5%. CAPM-expected = 5 + 1.2(12−5) = 13.4%. **Alpha = 18 − 13.4 = +4.6%** — genuine outperformance beyond what its market exposure explains. Distinguishing this skill (alpha) from just taking market risk (beta) is the whole game.

Example 4 — building a 4-stock portfolio from scratch, with a concentration/diversification trade-off. A PM has ₹10 cr to deploy across a high-conviction book. Four candidate holdings have expected returns and volatilities of: Stock A (18%, 22%), Stock B (14%, 15%), Stock C (22%, 30%), Stock D (11%, 12%), with an average pairwise correlation of 0.35 among them (a realistic same-market, imperfectly correlated basket). An equal-weight (25% each) portfolio's expected return is simply the weighted average: 0.25×(18+14+22+11) = **16.25%**. Portfolio volatility, using the full correlation structure, comes out meaningfully below the weighted-average volatility of 19.75% — illustrating the same diversification math as Example 1, but now across four holdings instead of two, where the benefit compounds as more imperfectly-correlated positions are added. **The concentration trade-off**: if the PM instead conviction-weights toward the highest-expected-return, highest-volatility names (say 40% A, 10% B, 40% C, 10% D), expected return rises to 0.4×18+0.1×14+0.4×22+0.1×11 = **18.5%**, but so does portfolio volatility, since the weight is now concentrated in the two highest-volatility, still-correlated names — a direct, numeric illustration of Part 18's "concentration vs diversification" trade-off (7.1's position-sizing discussion generalised to a full portfolio): more conviction-weighting raises both expected return *and* risk, and the PM must judge whether the extra ~2.25 points of expected return justifies the higher realised volatility and larger single-name drawdown exposure, given the fund's actual risk mandate and client risk tolerance.

Example 5 — factor investing in practice. A fund constructs a "value + momentum" combined strategy: it ranks its investable universe by a value score (low P/B, low P/E) and a momentum score (12-month trailing return, per Part 14's anomaly discussion), buying stocks that rank favourably on *both* factors simultaneously rather than either alone. The logic: value and momentum have historically shown low or even negative correlation as standalone factors (value tends to do best when momentum struggles, and vice versa, since they capture different market inefficiencies — value bets on mean-reversion in cheap stocks, momentum bets on trend continuation in recent winners) — combining them in a single portfolio can produce a smoother overall return stream than either factor run alone, the same diversification logic from Example 1 and Example 4, applied across *factors* rather than across individual stock-specific risk. This is why many quantitative equity strategies deliberately blend multiple, historically-uncorrelated factors (value, momentum, quality, low-volatility) rather than betting on a single factor.

How it is tested in interviews

- **"Why diversify?"** — "Because assets aren't perfectly correlated, combining them lowers portfolio volatility for a given return — it removes diversifiable, idiosyncratic risk, leaving mostly systematic risk."
- **"Systematic vs unsystematic risk?"** — "Unsystematic is company-specific and diversifiable; systematic is market-wide, undiversifiable, and measured by beta — it's what earns the risk premium."
- **"What does the Sharpe ratio measure?"** — "Excess return per unit of total risk (volatility) — a risk-adjusted performance measure; higher is better."
- **"Sharpe vs Treynor?"** — "Sharpe uses total risk (σ); Treynor uses systematic risk (β). Use Treynor when the portfolio is part of a broader diversified portfolio."
- **"What is alpha?"** — "Return above what CAPM/beta predicts — the manager's skill, not explained by market exposure."

Traps & common mistakes

- Judging on **raw return** instead of **risk-adjusted** return.
- Thinking diversification removes **all** risk — systematic risk remains.
- Confusing **Sharpe (total risk)** and **Treynor (systematic risk)**.
- Confusing **alpha** (skill) with **beta** (market exposure) — high returns from leverage/beta aren't alpha.
- Over-diversifying to closet-indexing (paying active fees for index-like returns).

First-principles recap

- Portfolios exploit **diversification** (correlations < 1) to raise return per unit of risk.
- Risk = **systematic** (market, priced via beta) + **unsystematic** (specific, diversifiable).
- Judge performance **risk-adjusted**: Sharpe, Treynor, alpha, information ratio.
- **Alpha** = skill beyond market exposure; **beta** = market exposure.
- Rebalance to targets; manage tracking error vs the benchmark.

Quick-reference

METRIC	FORMULA
Beta	Cov(Rp,Rm)/Var(Rm)
Sharpe	(Rp − Rf)/σp
Treynor	(Rp − Rf)/βp
Alpha	Rp − [Rf + β(Rm − Rf)]
Information ratio	Active return / tracking error
Diversifiable	Unsystematic only

Q&A — Portfolio Construction & Risk

Theory and worked numerical problems on diversification, risk decomposition, and risk-adjusted performance.

Q1. Worked — two-asset portfolio volatility.

Stock A: 25% volatility. Stock B: 18% volatility. Correlation between A and B: 0.2. Portfolio is 60% A, 40% B.

Model answer.

```
σp² = w_A² σ_A² + w_B² σ_B² + 2 w_A w_B ρ σ_A σ_B
    = 0.6² × 0.25² + 0.4² × 0.18² + 2 × 0.6 × 0.4 × 0.2 × 0.25 × 0.18
    = 0.36 × 0.0625 + 0.16 × 0.0324 + 0.48 × 0.2 × 0.045
    = 0.0225 + 0.005184 + 0.00432
    = 0.032004
σp = √0.032004 ≈ 17.89%
```

The portfolio's volatility (≈17.9%) is below the simple weighted-average volatility of the two holdings (0.6×25 + 0.4×18 = 22.2%) — the gap between 22.2% and 17.9% is the diversification benefit, arising directly from the correlation being well below 1.

Q2. Why can't systematic risk be diversified away, no matter how many additional uncorrelated stocks are added to a portfolio?

Model answer. Systematic risk represents exposure to broad market/economic factors (interest rates, GDP growth, systemic shocks) that affect essentially all stocks simultaneously to some degree — since virtually every stock has some positive correlation to the overall market, adding more holdings can eliminate the *company-specific* (unsystematic) portion of risk that's genuinely uncorrelated across names, but it cannot eliminate the shared, market-wide component that every stock carries. This is precisely why systematic risk (measured by beta) is the risk that's compensated with a risk premium — it's the risk investors are stuck bearing collectively no matter how they diversify within the equity asset class.

Q3. Worked — Sharpe ratio decision.

Fund X: 22% return, 18% volatility. Fund Y: 16% return, 9% volatility. Risk-free rate: 6%.

Model answer.

```
Sharpe X = (22 − 6) / 18 = 16/18 ≈ 0.89
Sharpe Y = (16 − 6) / 9 = 10/9 ≈ 1.11
```

Fund Y has the higher Sharpe ratio despite the lower raw return — it delivers more excess return per unit of total risk taken. An investor could, in principle, lever Fund Y (e.g. via borrowing) to match Fund X's volatility level and would expect to end up with a higher return than Fund X at that same risk level, which is the practical implication of comparing Sharpe ratios rather than raw returns alone.

Q4. Distinguish alpha from beta with a worked numerical example, and explain why conflating the two is a serious analytical error.

A fund returns 19% in a year when the market returned 10%, risk-free rate is 5%, and the fund's beta is 1.5.

Model answer.

```
CAPM-expected return = Rf + β(Rm − Rf) = 5% + 1.5 × (10% − 5%) = 5% + 7.5% = 12.5%
Alpha = Actual return − CAPM-expected return = 19% − 12.5% = +6.5%
```

The fund's 19% return looks impressive, but 12.5 percentage points of it are simply explained by taking on 1.5x the market's systematic risk (beta exposure) — anyone could have replicated most of that 19% by leveraging a passive index fund to a 1.5 beta, at much lower fees. Only the 6.5% alpha represents genuine skill (return not explained by market exposure). Conflating a high raw return achieved mostly through high beta with genuine investment skill is a serious error — it leads to overpaying active-management fees for what is, economically, just leveraged market exposure available far more cheaply.

Q5. When would an analyst use the Treynor ratio instead of the Sharpe ratio to evaluate a fund's risk-adjusted performance, and why does the choice matter?

Model answer. Sharpe ratio uses total risk (standard deviation, σ), making it the right choice when evaluating a portfolio as a standalone investment (all of an investor's risk exposure). Treynor ratio uses only systematic risk (beta), making it the right choice when evaluating a portfolio as *one component* of a broader, diversified overall portfolio — since in that context, the fund's unsystematic/idiosyncratic risk may already be diversified away by the investor's other holdings, and what matters is how efficiently the fund generates excess return per unit of the market risk it does contribute. Using Sharpe when Treynor is appropriate (or vice versa) can lead to incorrectly favouring a fund with low idiosyncratic risk but poor systematic-risk-adjusted returns, or the reverse.

Q6. Worked — information ratio and what it signals about consistency.

Fund A: active return (vs benchmark) averaging 4% annually, with a tracking error of 8%. Fund B: active return averaging 2.5% annually, with a tracking error of 2%.

Model answer.

```
IR_A = 4 / 8 = 0.50
IR_B = 2.5 / 2 = 1.25
```

Despite Fund A's higher average active return (4% vs 2.5%), Fund B has the much higher information ratio — its outperformance is far more *consistent* relative to how much it deviates from the benchmark. Fund A's higher average return comes with much more volatile/inconsistent relative performance (a high tracking error), meaning an investor experiences a bumpier ride of over- and under-performance to earn that average, whereas Fund B delivers steadier, more reliable outperformance per unit of active risk taken — a meaningfully different (and often more prized by institutional allocators) quality than Fund A's higher but less consistent alpha.

Q7. What's the trade-off in choosing a concentrated, high-conviction portfolio (e.g. 15-20 positions) versus a broadly diversified one (e.g. 100+ positions)?

Model answer. A concentrated portfolio allows each position to genuinely reflect the manager's highest-conviction ideas and lets real skill (analytical edge) show up meaningfully in returns if the manager's calls are good — but it also carries more idiosyncratic risk per position, since fewer holdings mean any single stock-specific negative surprise has a larger portfolio-level impact, and the overall portfolio's realised volatility can deviate more sharply from the benchmark. A broadly diversified portfolio diversifies away most idiosyncratic risk (per the diversification math in Q1-Q2), producing smoother, more benchmark-like returns — but if a manager is genuinely skilled at stock selection, over-diversifying dilutes that skill's impact toward benchmark-like ("closet indexing") returns while still charging active-management fees, itself a well-known problem the chapter flags.

Q8. Why does rebalancing a portfolio back to target weights function as a disciplined "sell high, buy low" mechanism, even without any explicit market-timing view?

Model answer. As asset prices move, a position that has appreciated becomes an overweight relative to its original target allocation, and a position that has declined becomes an underweight. Rebalancing back to target weights mechanically requires trimming the now-overweight (appreciated, "expensive" relative to target) position and adding to the now-underweight (declined, "cheap" relative to target) position — enforcing a disciplined sell-high/buy-low pattern purely through the rebalancing rule itself, without requiring the manager to have any specific forecasting view on which asset will outperform next. This also controls risk by preventing any single position's appreciation from silently growing the portfolio's concentration and risk profile beyond its intended target.

ESG in Equity Analysis

The Problem / Why this matters

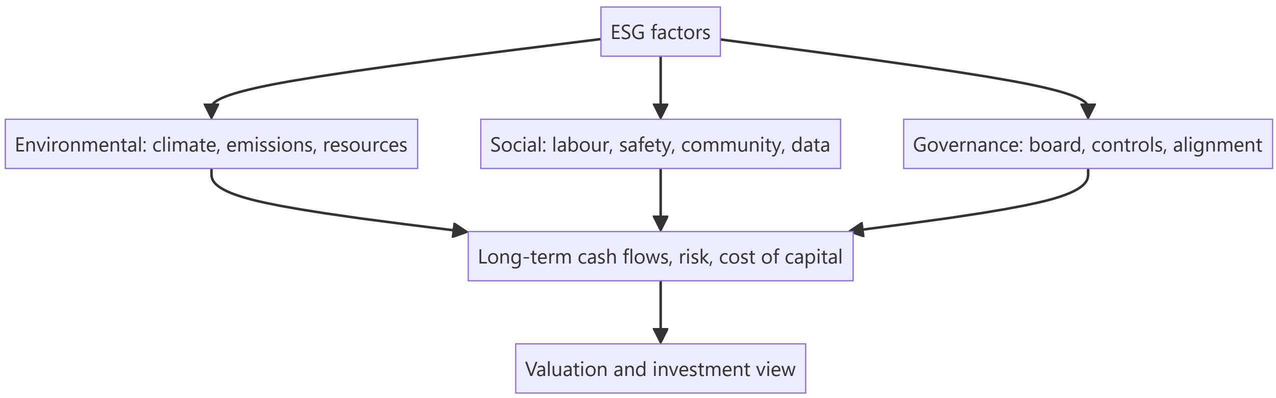
Investors increasingly judge companies not just on financials but on **Environmental, Social and Governance (ESG)** factors — because these non-financial issues create real financial risks and opportunities (regulation, reputation, stranded assets, talent). ESG has moved from niche to mainstream, is increasingly mandated in disclosure (BRSR in India), and comes up in interviews as "why do investors care about ESG?" Understanding how to integrate it into equity analysis is now part of the job.

Core Idea

ESG analysis assesses a company's exposure to and management of **environmental, social, and governance** risks and opportunities, integrating them into the investment view. The premise: material ESG factors affect long-term cash flows, risk, and therefore valuation — so ignoring them means missing risks that standard financials don't capture.

Why it works this way

Many ESG issues are **financially material but slow-moving** — a carbon-intensive business faces future regulation and stranded-asset risk; poor governance invites fraud and value destruction; weak labour or safety practices bring litigation and reputational damage. These show up in cash flows and cost of capital eventually, so a forward-looking analyst prices them in before they hit the financials.



Full technical content

The three pillars:

PILLAR	EXAMPLE FACTORS
Environmental	Carbon emissions, energy/water use, waste, climate transition risk, biodiversity
Social	Labour practices, health & safety, diversity, product safety, data privacy, community relations
Governance	Board independence/quality, executive pay alignment, shareholder rights, audit quality, related-party dealings, anti-corruption

Governance is often seen as the most immediately financially material — weak governance (dominant promoters, poor controls) is a leading cause of fraud and value destruction, especially in emerging markets.

How investors use ESG: - Integration — embed material ESG factors into fundamental analysis and valuation (adjust growth, margins, cost of capital, or apply a risk discount). - **Screening** — exclude (negative screening: tobacco, weapons) or tilt toward leaders (positive/best-in-class). - **Thematic / impact** — invest in a theme (clean energy) or for measurable impact. - **Stewardship** — engage and vote to improve company behaviour.

Materiality. The key discipline: focus on ESG factors that are **financially material** to *that* industry (e.g., emissions for a utility, data privacy for a tech firm, safety for a miner) — not a generic checklist. Frameworks like SASB map materiality by sector.

Instruments & disclosure: green bonds, **sustainability-linked loans**, ESG funds/ETFs; disclosure via GRI, TCFD, and in India the **BRSR (Business Responsibility and Sustainability Report)** mandated for top listed firms by SEBI.

Debates & challenges: - Greenwashing — overstating ESG credentials. - **Inconsistent ratings** — ESG scores from different providers often disagree (different methodologies). - **Performance debate** — whether ESG helps or hurts returns is contested; the strongest case is **risk management** (avoiding blow-ups) rather than guaranteed outperformance.

Effect on cost of capital. Strong ESG can lower cost of capital (broader investor base, lower risk perception); poor ESG can raise it (exclusion, higher risk), feeding directly into valuation.

Worked examples

Example 1 — governance red flag. An analyst finds a company with a promoter-dominated board, frequent related-party transactions, and an auditor change after a qualification. Governance risk is high — historically a precursor to value destruction. The analyst applies a valuation discount (or avoids the stock), a call standard financials alone wouldn't trigger.

Example 2 — environmental transition risk. A thermal-coal power producer looks cheap on current earnings, but tightening emissions regulation and the energy transition threaten **stranded assets** and rising compliance costs over the next decade. Integrating this, the analyst lowers long-term cash flows and raises the discount rate — the stock is less cheap than it looks.

Example 3 — materiality focus. For a software company, carbon emissions are minor but **data privacy and talent retention** are highly material (a breach or attrition could hit revenue and margins). A good ESG analyst weights the *material* factors for that sector, not a generic environmental score.

Example 4 — quantifying a governance discount in a valuation, not just flagging it qualitatively. An analyst values a company at ₹450/share on a clean DCF, but the company has two governance red flags: 35% of revenue runs through related-party transactions with promoter-owned entities (raising round-tripping/value-leakage concerns), and the board has no independent directors on the audit committee. Rather than simply writing "governance risk — be cautious" in the note, a rigorous approach quantifies it: apply a discount rate premium (e.g. +150 basis points to the cost of equity, reflecting the additional risk premium investors should demand for weaker minority-shareholder protection) and re-run the DCF. If the +150bp premium takes WACC from 11.0% to 12.5%, the same cash flows might value the stock at ₹390/share instead of ₹450 — a concrete, defensible ₹60/share governance discount an investment committee can actually act on, rather than a vague qualitative caution that doesn't change the number anyone actually trades on.

Example 5 — BRSR disclosure as a research input, not just a compliance checkbox. India's SEBI-mandated Business Responsibility and Sustainability Report (BRSR, required for the top listed companies by market cap) discloses standardised metrics — GHG emissions (Scope 1 and 2), water/waste intensity, employee attrition and diversity data, and related-party transaction details. An equity analyst covering a manufacturing company can use the BRSR's disclosed emissions-per-unit-of-revenue trend across several years as a genuine research input: a rising emissions-intensity trend at a company operating in a sector facing tightening carbon-pricing regulation is a quantifiable, forward-looking risk signal — directly usable in the Example 2 stranded-asset-style analysis — rather than treating BRSR filing as a purely regulatory box-ticking exercise disconnected from the investment thesis.

How it is tested in interviews

- **"Why do investors care about ESG?"** — "Because material ESG factors create real financial risks and opportunities — regulation, stranded assets, reputation, governance failures — that affect long-term cash flows and cost of capital but aren't captured in standard financials."
- **"What are the three pillars?"** — Environmental (climate, emissions, resources), Social (labour, safety, privacy, community), Governance (board, controls, alignment).
- **"Which is most financially material?"** — "Often governance — weak governance is a leading cause of fraud and value destruction, especially in emerging markets — but materiality is industry-specific."
- **"What are the criticisms of ESG?"** — Greenwashing, inconsistent ratings across providers, and a contested performance record (the strongest case is risk management).
- **"How would you integrate ESG into a valuation?"** — "Focus on material factors for the sector; adjust growth, margins, or the discount rate, or apply a governance/risk discount."

Traps & common mistakes

- Treating ESG as a **generic checklist** rather than **industry-material** factors.
- Ignoring **governance** — the most immediately material pillar.
- Taking a single ESG **rating** at face value (providers disagree).
- Assuming ESG **guarantees** outperformance — the robust case is risk management.
- Missing the **cost-of-capital** channel through which ESG hits valuation.

First-principles recap

- ESG assesses environmental, social and governance risks/opportunities that are financially material.
- **Materiality is industry-specific** — focus on what matters for that sector.
- Governance is often the most immediately material pillar.
- Integrate via adjusted cash flows/discount rate, screening, or stewardship.
- Watch greenwashing and rating inconsistency; the strongest case is risk management and cost of capital.

Quick-reference

PILLAR	MATERIAL EXAMPLES
Environmental	Emissions, transition risk, resources
Social	Labour, safety, data privacy
Governance	Board, controls, related-party, alignment
Uses	Integration, screening, thematic, stewardship
India disclosure	BRSR (SEBI)
Key discipline	Focus on financially material factors

Q&A — ESG in Equity Analysis

Theory and worked scenarios on integrating ESG factors into fundamental equity research.

Q1. Why is "materiality" described as the key discipline in ESG analysis, and what's the error a generic ESG checklist commits?

Model answer. Materiality means focusing specifically on the ESG factors that are genuinely financially relevant to a *particular* company's industry and business model — carbon emissions are highly material for a utility or heavy manufacturer but nearly irrelevant to a software company, while data privacy and talent retention are highly material for a tech firm but not for a cement producer. A generic ESG checklist that scores every company against the same broad list of environmental, social, and governance factors regardless of industry dilutes the signal with irrelevant noise and can produce misleading comparisons (e.g. penalising a software company for a low environmental score on factors that don't actually bear on its business risk) — frameworks like SASB exist specifically to map which factors are material by sector.

Q2. Why does the chapter identify governance as "often the most immediately financially material" ESG pillar, particularly in emerging markets?

Model answer. Weak governance — a dominant, unchecked promoter, frequent undisclosed related-party transactions, weak board independence, poor audit quality — is a well-documented, historically recurring precursor to fraud, capital misallocation, and outright value destruction, and these risks can materialise suddenly and severely (a governance blow-up can wipe out a large fraction of equity value in a short period), unlike environmental or social risks which more often play out gradually over years. In emerging markets specifically, weaker regulatory enforcement and more concentrated ownership structures make governance risk both more prevalent and harder to detect from standard financial disclosures alone, making independent governance assessment a genuinely high-value, differentiated piece of analysis.

Q3. Worked scenario — apply ESG integration (not exclusion) to a valuation. A thermal-power generation company trades at 6x forward earnings, well below the sector average of 9x, with strong current cash flows. What ESG-related analysis should inform whether this "cheap" valuation is a genuine opportunity or a value trap?

Model answer. The low current multiple may already partially reflect market awareness of transition risk, but a rigorous ESG-integrated analysis goes further: model the specific regulatory trajectory for emissions/carbon pricing in the relevant jurisdiction and its likely impact on the plant's economics over the next 10-15 years; assess stranded-asset risk — whether the plant may need to be retired before the end of its natural economic life due to policy or financing-access changes (increasingly, lenders and insurers are reducing exposure to thermal assets, raising the cost and availability of future capital); and explicitly model these effects into extended-horizon cash flows and/or a higher discount rate rather than treating current-year earnings as representative of the long-run value. The conclusion may still be a legitimate contrarian buy if the market is overstating transition risk, or a genuine value trap if the market hasn't yet fully priced a realistic regulatory trajectory — the ESG-integration approach earns that conclusion through explicit analysis rather than either ignoring the risk or reflexively avoiding the sector.

Q4. Distinguish ESG integration from negative screening, and explain why a fund might use one, the other, or both.

Model answer. Integration embeds material ESG factors directly into fundamental analysis and valuation (adjusting growth, margin, or discount-rate assumptions based on ESG-related risk/opportunity) while still potentially owning any company if the valuation is attractive after that adjustment. Negative screening excludes entire categories of companies outright (e.g. tobacco, weapons manufacturers) regardless of valuation, typically driven by a mandate's ethical/values constraints rather than a purely financial materiality judgment. A fund might use integration alone (financially-driven ESG-aware investing with no exclusions), screening alone (values-driven exclusions with otherwise standard financial analysis), or both together (a values-constrained universe, analysed with ESG-integrated fundamental work within that universe) — the choice reflects the fund's mandate and client base, not a single "correct" approach.

Q5. What is "greenwashing," and what should an equity analyst specifically check to avoid being misled by it?

Model answer. Greenwashing is a company overstating or misrepresenting its actual ESG credentials/performance, often through selective disclosure, vague sustainability marketing claims, or ESG-labelled initiatives that are small relative to the company's overall footprint. An analyst should check: whether sustainability claims are backed by third-party-verified, quantified data (not just marketing language) with clear year-over-year metrics; whether disclosed ESG initiatives are material relative to the company's overall scale of operations (a large polluter's small solar-power pilot project doesn't offset its core emissions profile); and whether the company's governance track record (Q2) itself is credible enough to trust its self-reported ESG data in the first place — governance quality is itself a useful filter for how much to trust a company's other ESG disclosures.

Q6. Why do ESG ratings from different data providers often disagree significantly on the same company, and what does this imply for how an analyst should use third-party ESG scores?

Model answer. Different ESG rating providers use different methodologies — different weightings across the E/S/G pillars, different underlying data sources, and different judgments about which specific factors matter for a given industry — meaning it's genuinely common for the same company to receive meaningfully different ESG scores from different providers, sometimes even contradictory qualitative assessments. This implies an analyst should treat any single third-party ESG score as one data point rather than ground truth, and should understand the specific methodology behind a score before relying on it for an investment decision — the chapter's broader point about materiality (Q1) applies here too: an analyst's own judgment about what's genuinely material to *this* company, rather than a black-box aggregate score, is the more defensible basis for investment analysis.

Q7. What is the "cost of capital channel" through which ESG factors affect valuation, distinct from the direct cash-flow channel?

Model answer. Beyond directly affecting projected cash flows (e.g. lower future revenue from reputational damage, higher costs from regulatory compliance), a company's ESG profile can affect the discount rate applied to those cash flows: strong ESG performance can broaden the pool of investors willing to hold the stock (some large institutional mandates now explicitly exclude poor-ESG names or require minimum ESG scores) and can be perceived as lowering certain tail risks, both of which can modestly lower the required return (cost of equity) investors demand. Conversely, poor ESG performance can shrink the addressable investor base (exclusion from ESG-mandated funds) and raise perceived risk, pushing the required return up — meaning two companies with identical projected cash flows could reasonably receive different valuations purely through this cost-of-capital channel if their ESG profiles differ meaningfully.

Q8. How would you respond to the claim "ESG investing is proven to boost returns" in an interview, given the chapter's framing of the performance debate?

Model answer. "The empirical evidence on whether ESG investing boosts returns is genuinely contested and methodology-dependent — I wouldn't claim it's proven to boost returns. The more defensible and robust case for ESG integration is risk management: material ESG factors, when ignored, have been repeatedly shown to precede real financial blow-ups (governance failures leading to fraud, environmental liabilities becoming stranded-asset write-downs, social/labour issues triggering costly litigation or reputational damage) — so ESG-aware analysis is better framed as helping avoid left-tail risk and value-destroying surprises, rather than as a reliable source of additional upside. That's a more defensible, evidence-consistent position than claiming guaranteed outperformance." This response demonstrates nuanced understanding rather than either dismissing ESG or overselling it.

Indian Equity Markets (Capstone)

The Problem / Why this matters

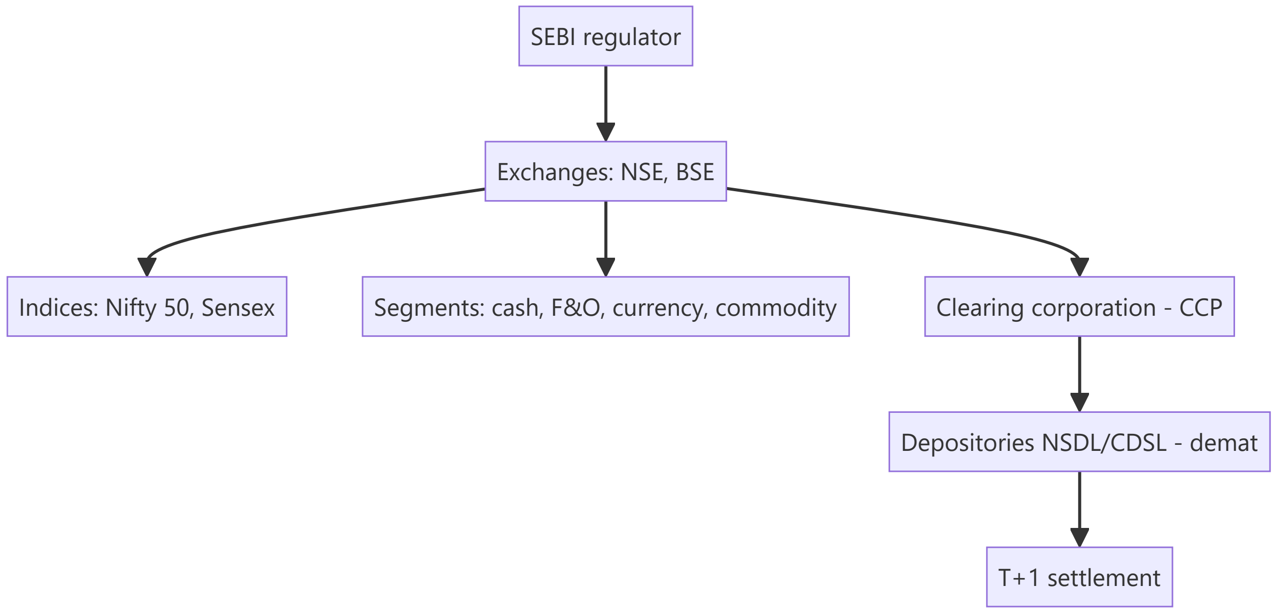
For any finance role in India, you must know your home market cold — the regulator, the exchanges, the indices, how shares are held and settled, the segments, and the participants. It's the most common "home turf" interview check, and it ties together everything in this book in the specific institutional context you'll actually work in. This capstone maps the Indian equity ecosystem end to end.

Core Idea

The Indian equity market is a **modern, electronic, well-regulated** market: **SEBI** regulates it; **NSE and BSE** are the exchanges; **Nifty 50 and Sensex** are the benchmarks; shares are held in **demat** via NSDL/CDSL and settle at **T+1**; and it spans cash, F&O, currency and commodity segments with deep participation from retail, domestic institutions, and foreign investors.

Why it works this way

India built a fully dematerialized, exchange-traded, centrally-cleared market with strong disclosure and one of the world's fastest settlement cycles precisely to ensure trust, liquidity and investor protection — the conditions that let a large, diverse investor base participate and companies raise capital efficiently.



Full technical content

Regulator: SEBI (Securities and Exchange Board of India) — regulates issuers (disclosure), intermediaries (registration/conduct), and markets (surveillance, insider-trading and manipulation enforcement); protects investors. **RBI** oversees monetary policy and banking; **IRDAI** insurance.

Exchanges: NSE (National Stock Exchange — the larger by volume, especially derivatives) and **BSE** (Bombay Stock Exchange — Asia's oldest). Both are fully electronic, order-driven markets.

Benchmark indices: Nifty 50 (NSE, 50 large-caps) and **Sensex** (BSE, 30 large-caps), both **free-float market-cap weighted**. Broader: Nifty Next 50, Nifty 100/500, Midcap/Smallcap, and sectoral indices

(**Bank Nifty**, Nifty IT, etc.).

Holding & settlement: shares held in **dematerialized (demat)** form via depositories **NSDL** and **CDSL** (through Depository Participants); trades cleared by the **clearing corporation** (central counterparty) and settled on **T+1** (India moved to T+1 fully in 2023 and is piloting T+0/instant) — among the fastest globally.

Market segments:

SEGMENT	WHAT TRADES
Cash / equity	Delivery-based and intraday shares
F&O (derivatives)	Index and stock futures & options (Nifty, Bank Nifty are among the world's most traded)
Currency derivatives	USD/INR and other pairs
Commodity	Via MCX/NCDEX (and NSE/BSE segments)

Participants: retail (large and growing, via demat/UPI-based investing), **domestic institutions (DIIs)** — mutual funds, insurers (LIC), pension (EPFO/NPS) — and **foreign portfolio investors (FPIs)**, whose flows often drive index moves. Promoters hold large stakes in many firms.

Trading mechanics: order-driven, price-time priority; **circuit breakers** (index-level halts) and **price bands** (stock-level) curb extreme moves; **UPI/ASBA** streamline IPO applications and payments.

Recent evolution: surging retail participation (record demat accounts), the rise of discount brokers and index/SIP investing, T+1 (moving to instant) settlement, and world-leading F&O volumes (prompting SEBI curbs on retail options speculation).

Worked examples

Example 1 — the full trade lifecycle. A retail investor places a market buy for 100 shares of an NSE-listed company via a broker app. The order matches on the NSE by price-time priority; the **clearing corporation** guarantees it; on **T+1**, shares are credited to the investor's **CDSL/NSDL demat** account and cash debited. Regulated end-to-end by SEBI. Every institution in this chapter touched one small trade.

Example 2 — FPI flows move the Nifty. Foreign investors turn net sellers of ₹15,000 cr in a week on global risk-off. Because FPIs hold large weights in Nifty heavyweights and trade fast, the index falls sharply even without India-specific news — illustrating why FPI flows are watched as a key driver.

Example 3 — F&O depth. Bank Nifty and Nifty options are among the most actively traded derivatives in the world, giving Indian markets deep hedging and speculation venues — but also prompting SEBI to tighten rules to protect retail traders from outsized options losses. India's derivatives market is globally significant.

Example 4 — DII flows offsetting FPI selling. Continuing Example 2: the same week FPIs sell ₹15,000 cr, DIIs (domestic mutual funds, insurers) net-buy ₹11,000 cr, cushioning roughly three-quarters of the FPI outflow's index impact. This DII-as-stabiliser pattern has become a structural feature of Indian markets over the past decade, driven substantially by rising domestic SIP (Systematic Investment Plan) mutual-fund inflows — a monthly, relatively flow-insensitive-to-sentiment source of buying that didn't exist at the same scale two decades ago. An analyst explaining "why didn't the market fall as much as the FPI selling would suggest" should be able to name this DII/SIP-flow offset explicitly, not just cite "buying support" vaguely.

Example 5 — a circuit breaker in action. A stock-specific circuit filter (e.g. 10% for a mid-cap) halts trading in a single name after unexpected, severely negative news (a fraud allegation, a regulatory action) causes a rush of sell orders. The halt gives the market a cooling-off period — allowing information to be digested and genuine two-sided price discovery to resume — rather than letting a single wave of panic-selling execute against a thin, one-sided order book at an arbitrarily bad price. Index-level circuit breakers (triggered by a large move in Nifty/Sensex itself) work the same way at the market-wide level during systemic shocks. Knowing that circuit filters differ by stock category (larger, more liquid stocks typically have wider bands than smaller, more volatile ones) is the kind of granular, "home turf" detail this capstone chapter flags as a common interview differentiator.

How it is tested in interviews

- **"Walk me through the Indian equity market structure."** — SEBI regulates; NSE/BSE are the exchanges; Nifty 50/Sensex are the benchmarks (free-float market-cap weighted); shares held in demat via NSDL/CDSL; cleared by the clearing corporation; settled T+1; segments cash/F&O/currency/commodity.
- **"Who regulates the Indian securities market?"** — "SEBI — issuers, intermediaries, and markets, with investor protection; RBI covers banking/monetary, IRDAI insurance."
- **"What are the main indices and how are they built?"** — "Nifty 50 (NSE) and Sensex (BSE), free-float market-cap weighted large-cap baskets."
- **"How are trades settled in India?"** — "T+1 rolling settlement, shares in demat via NSDL/CDSL, guaranteed by the clearing corporation as central counterparty — among the fastest cycles globally."
- **"What drives Indian market moves?"** — "FPI and DII flows, global cues, rates/inflation and RBI policy, earnings, and sentiment; FPI flows especially move the index."

Traps & common mistakes

- Confusing **SEBI** (securities) with **RBI** (banking/monetary) roles.
- Not knowing India settles at **T+1** (and is going faster).
- Forgetting Nifty/Sensex are **free-float** market-cap weighted.
- Mixing up the **exchange** (NSE/BSE), **depository** (NSDL/CDSL), and **clearing corporation** roles.
- Underrating **FPI flows** and India's globally-large **F&O** volumes.

First-principles recap

- **SEBI** regulates; **NSE/BSE** are the exchanges; **Nifty 50/Sensex** the free-float benchmarks.
- Shares are held in **demat** (NSDL/CDSL) and settle **T+1** (moving to instant).
- Segments: cash, F&O (globally large), currency, commodity.
- Participants: retail (surging), DIIs, FPIs (flow-drivers), promoters.
- A modern, electronic, centrally-cleared, well-regulated market — the context for every Indian finance role.

Quick-reference

ELEMENT	INDIA
Regulator	SEBI (RBI banking, IRDAI insurance)
Exchanges	NSE, BSE
Indices	Nifty 50, Sensex (free-float mcap)
Depositories	NSDL, CDSL (demat)
Settlement	T+1 (→ instant), CCP-guaranteed
Segments	Cash, F&O, currency, commodity
Flow drivers	FPIs, DIIs

Q&A — Indian Equity Markets (Capstone)

The "home turf" questions every India-based markets interview checks for.

Q1. Who regulates Indian equity markets, and what are the two main exchanges?

Model answer. SEBI (Securities and Exchange Board of India) is the primary regulator, covering market conduct, disclosure, intermediary registration, and investor protection. The two main exchanges are NSE (National Stock Exchange) and BSE (Bombay Stock Exchange); NSE is the larger by traded volume in cash equities and derivatives, BSE is India's oldest exchange (founded 1875).

Q2. What are India's two main benchmark indices, and how do they differ in construction?

Model answer. Nifty 50 (NSE, 50 large-cap stocks) and Sensex (BSE, 30 large-cap stocks) — both free-float market-cap-weighted, meaning weights reflect only the shares actually available to trade (excluding promoter/locked-in holdings), not total shares outstanding. This free-float methodology matters because it means an index's composition reflects genuinely tradable liquidity, not raw company size.

Q3. Explain India's settlement cycle and why it matters for an analyst.

Model answer. Indian equities settle on a T+1 basis (trade date plus one business day) — among the fastest cycles globally, reduced from T+2 in a phased rollout completed in 2023. For an analyst, faster settlement reduces counterparty risk in the system and means trading strategies involving quick turnaround face less operational lag than in slower-settling markets.

Q4. What is "demat," and what are the two depositories?

Model answer. Dematerialisation (demat) is holding securities in electronic form rather than physical share certificates — mandatory for trading in India. The two depositories are NSDL (National Securities Depository Limited) and CDSL (Central Depository Services Limited), which maintain electronic records of ownership; an investor's shares sit in a demat account held via a registered depository participant (typically a broker).

Q5. What are the main market segments on Indian exchanges?

Model answer. Cash/equity segment (spot trading in listed shares), F&O (futures & options — index and stock derivatives, one of the largest derivatives markets globally by contract volume), currency derivatives (USD/INR and other pairs), and commodity derivatives (via MCX/NCDEX, and NSE for select commodities) — each with its own participant base, margin rules, and settlement mechanics.

Q6. Who are the main categories of participants in Indian equity markets, and how does their behaviour differ?

Model answer. Retail investors (a rapidly growing base via discount brokers and direct market access), Domestic Institutional Investors — DIIs (mutual funds, insurance companies, pension funds — generally longer-horizon, often a stabilising counterweight to foreign flows), and Foreign Portfolio Investors — FPIs (foreign institutional money, registered and regulated by SEBI, historically a major swing factor in short-term market direction, especially around global risk-sentiment shifts and currency moves). Analysts routinely track monthly FPI/DII net flow data as a sentiment indicator, since large, sustained FPI selling offset by DII buying (or vice versa) is itself informative about market positioning.

Q7. What's the STT (Securities Transaction Tax), and why does it matter to an options trader/analyst specifically?

Model answer. STT is a tax levied on transactions in listed securities and derivatives on Indian exchanges, with different rates for cash-market trades versus options versus futures. It matters specifically at options expiry, where STT on the notional value of a physically/cash-settled in-the-money option that isn't closed out before expiry can be disproportionately large relative to the option's premium — a well-known trap ("the STT trap") that active options traders and anyone modelling realistic F&O trading costs must account for explicitly.

Q8. Why is "knowing your home market cold" specifically emphasised as an interview screen in India?

Model answer. It's a fast, low-effort way for an interviewer to separate candidates who have genuinely engaged with markets (reading regulatory changes, tracking flow data, understanding settlement mechanics) from those with only textbook/global theoretical knowledge — since every India-based markets role, regardless of specific coverage sector, operates within this specific regulatory and market-structure context daily, a candidate who can't answer basic structural questions about SEBI, NSE/BSE, or settlement raises doubts about how seriously they've engaged with the actual job, independent of their technical modelling skill.